Internship

2025 - 2026

Computer Networks and Telecommunications Engineering

Classification of HPV Status in Head and Neck Cancer Using CT/PET Radiomics and Clinical Data

Prepared by EL ANIS EL DIRANY Mohamad Ali

from 01/04/2026 till 31/07/2026

at



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Introduction

- Context

As part of my Bachelor's degree in **Computer Networks and Telecommunications Engineering** at the **Lebanese University, Faculty of Technology**, I completed an internship at **TOGETHERFORCHEIM**. The internship was conducted in the field of **Artificial Intelligence (AI)** and focused on the application of machine learning techniques to the medical domain.

The project addressed the classification of **Human Papillomavirus (HPV) status** in patients with **Head and Neck Squamous Cell Carcinoma (HNSCC)** using **Computed Tomography (CT)** and **Positron Emission Tomography (PET)** radiomic features combined with clinical data. HPV status is an important prognostic and predictive biomarker in head and neck cancer, as it significantly influences treatment strategies, patient prognosis, and overall survival. However, determining HPV status traditionally relies on laboratory testing and pathological analysis, which may not always be immediately available.

Recent advances in artificial intelligence and medical imaging have enabled the development of predictive models capable of extracting quantitative information from medical images. By combining radiomic features with clinical information, machine learning algorithms can assist clinicians in predicting HPV status in a non-invasive manner, supporting decision-making and contributing to the advancement of precision medicine.

The primary objective of this internship was to design, implement, and evaluate machine learning models for HPV classification using CT, PET, and combined clinical-radiomic datasets. This work involved data preprocessing, feature organization, model development, performance evaluation using multiple classification metrics, and model interpretability through feature importance and SHAP analysis. Throughout this internship, I strengthened my knowledge of machine learning, medical image analysis, data science, and predictive modeling while gaining valuable experience in applying AI techniques to solve real-world healthcare challenges.

- Problem Statement and Motivation

Head and Neck Squamous Cell Carcinoma (HNSCC) is one of the most common types of cancer worldwide. Among the factors that influence its prognosis, **Human Papillomavirus (HPV) status** plays a crucial role. Patients with HPV-positive tumors generally have a better response to treatment and improved survival outcomes compared to HPV-negative patients. Therefore, accurately determining HPV status is essential for selecting appropriate treatment strategies and improving patient management.

Conventional methods for determining HPV status rely on pathological and laboratory tests, which may require additional procedures, specialized equipment, and processing time. These limitations have motivated researchers to investigate non-invasive alternatives that can assist clinicians in making faster and more informed decisions.

Recent advances in Artificial Intelligence (AI), radiomics, and medical imaging have demonstrated that quantitative features extracted from CT and PET scans, combined with clinical data, can provide valuable information for disease characterization. Machine learning techniques are capable of identifying complex patterns within these data that may not be easily recognized by human observation.

The motivation behind this project is to develop an AI-based classification model capable of predicting HPV status using CT radiomic features, PET radiomic features, and clinical information. Such a model has the potential to support clinical decision-making, reduce reliance on invasive diagnostic procedures, and contribute to more personalized treatment planning. Ultimately, this work aims to demonstrate how artificial intelligence can enhance healthcare by providing accurate, efficient, and non-invasive predictive tools.

- Objectives to Achieve

The primary objective of this internship was to develop and evaluate Artificial Intelligence (AI) models capable of predicting the **Human Papillomavirus (HPV) status** of patients with **Head and Neck Squamous Cell Carcinoma (HNSCC)** using a combination of **CT radiomic features**, **PET radiomic features**, and **clinical data**.

To achieve this objective, the following goals were established:

- Develop predictive models capable of accurately classifying patients as **HPV-positive** or **HPV-negative**.
- Investigate the predictive performance of three different data sources: **clinical features only**, **CT/PET radiomic features only**, and the **combination of radiomic and clinical features**.
- Train and compare multiple machine learning and deep learning models, including **LightGBM**, **Multi-Layer Perceptron (MLP)**, and **FT-Transformer**, to determine the most effective approach for HPV status prediction.
- Evaluate the performance of each model using appropriate classification metrics, including Accuracy, ROC-AUC, Precision, Recall, F1-score, Balanced Accuracy, Specificity, PR-AUC, Brier Score, and calibration metrics.
- Analyze model interpretability through feature importance and SHAP analysis to identify the variables that contribute most to the prediction.
- Compare the strengths and limitations of the different models and datasets to identify the most reliable and clinically applicable solution.
- Demonstrate the potential of Artificial Intelligence and radiomics as non-invasive tools to support clinicians in the diagnosis and management of head and neck cancer.

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1 Presentation of Collaborative Research and Projects Committee (CRPC): Vision, Expertise, and Impact

Collaborative Research and Projects Committee (CRPC), established in October 2023 under the non-profit association Ensemble pour Chehim/ Together For Chehim (founded in 2015), is a multidisciplinary academic and research platform committed to fostering scientific excellence, collaborative innovation, and societal advancement. Based in Chehim, Lebanon, the CRPC provides structured academic programs, promotes applied research, and builds strategic international partnerships to support students, early-career researchers, and professionals.

The CRPC is led by Dr. Nassib Abdallah (President of the Committee) and Dr. Aya Hage Chehade (Vice-President of the Committee), and brings together a multidisciplinary network of PhD holders, engineers, and other qualified professionals affiliated with research laboratories, industry, and academic institutions across Lebanon and France. This diverse expertise strengthens the committee's ability to achieve its strategic objectives. Together, they oversee the development and implementation of the CRPC's academic programs, research initiatives, and community outreach activities.

Strategic Vision and Mission

The CRPC's activities are guided by four fundamental pillars:

- Academic training and supervision: More than 370 hours of free, in-person training have been delivered locally, tailored to the latest scientific and technological needs.
- International collaboration: Strategic agreements with universities, industries, and research institutions support knowledge exchange and joint project development.
- Widening access to higher education: Special focus is given to underprivileged and geographically remote populations to enhance educational equity.
- Continuous scientific innovation: CRPC actively addresses emerging technological and societal challenges through interdisciplinary research and agile innovation strategies.

This comprehensive mission includes a strong commitment to mentorship, capacity building, community outreach, and research leadership.

Domains of Expertise

CRPC engages in high-impact scientific and technical fields through both education and applied research:

1. Artificial Intelligence and Data Science
    - Machine learning, deep learning, and intelligent systems
    - Applications in healthcare, cybersecurity, and meteorological modeling
2. Computer Science and Software Development
    - Algorithm design, programming foundations, and software engineering
    - Emphasis on project-based learning and collaborative development environments
3. Electronics and Robotics
    - Embedded systems, Arduino-based robotics, automation
    - Prototyping, electronics integration, and practical system development
4. Industrialization and Systems Engineering
    - Bridging innovation and real-world implementation
    - Emphasis on smart systems, industrial automation, and applied technologies
5. Foreign Language Training and Soft Skills Development
    - English language training, including SAT preparation
    - Communication skills, teamwork, and leadership development

These domains are addressed through structured academic curricula, hands-on workshops, and collaborative research activities.

2 Internship Overview

2.1 Functional Analysis

2.1.1 Objective Definition (QQOQCP Method

Who ?	TOGETHERFORCHEIM, Mr. Mohamad Hammoush, Dr. Aya Elhajj Chehade, and myself as the intern.
What ?	Develop AI models to classify HPV status in Head and Neck Squamous Cell Carcinoma using CT/PET radiomic features and clinical data.
Where ?	TOGETHERFORCHEIM.
When ?	During my internship period
How ?	By preprocessing the datasets, extracting radiomic and clinical features, and training models such as LightGBM, MLP, and FT-Transformer.
Why ?	To provide a non-invasive method for predicting HPV status and support clinical decision-making using Artificial Intelligence.

2.1.2 Definition of Requirements

The objective of the project was to develop an Artificial Intelligence system capable of predicting the Human Papillomavirus (HPV) status of patients diagnosed with Head and Neck Squamous Cell Carcinoma (HNSCC) using radiomic features extracted from CT and PET medical images together with clinical information.

To achieve this objective, the following requirements were identified.

Functional Requirements

- Collect and organize the CT radiomic, PET radiomic, and clinical datasets.
- Preprocess the data by cleaning, formatting, and preparing it for machine learning.
- Train classification models using different AI approaches, including **LightGBM**, **Multi-Layer Perceptron (MLP)**, and **FT-Transformer**.
- Predict whether a patient is **HPV-positive** or **HPV-negative**.
- Evaluate the models using appropriate performance metrics such as Accuracy, ROC-AUC, Precision, Recall, F1-score, Balanced Accuracy, Specificity, PR-AUC, and Brier Score.
- Compare the performance of the different models and identify the most effective approach.
- Interpret the predictions using feature importance and SHAP analysis.

Non-Functional Requirements

- Ensure reliable and reproducible experiments.
- Maintain high prediction accuracy and model robustness.
- Provide interpretable results suitable for clinical research.
- Keep the implementation modular and scalable for future improvements.
- Ensure efficient training and testing using Python and machine learning libraries.

2.1.2.1 Horn Diagram

2.1.2.2 Octopus Diagram

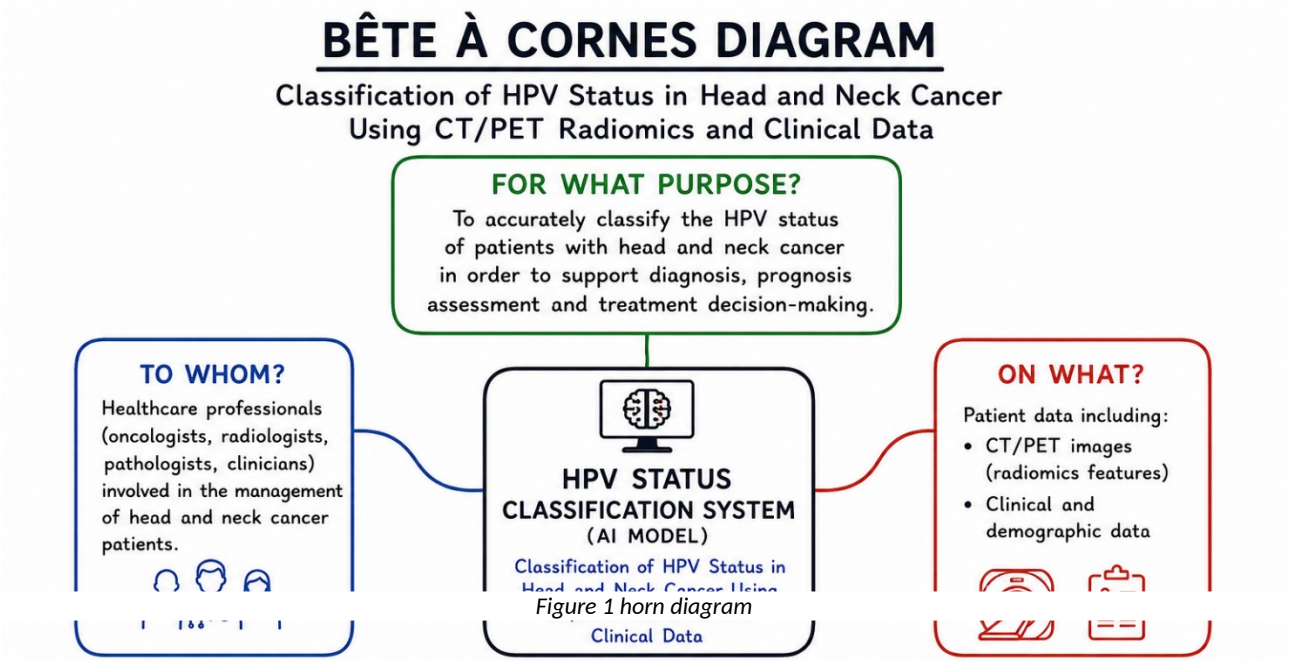


Figure 1 horn diagram

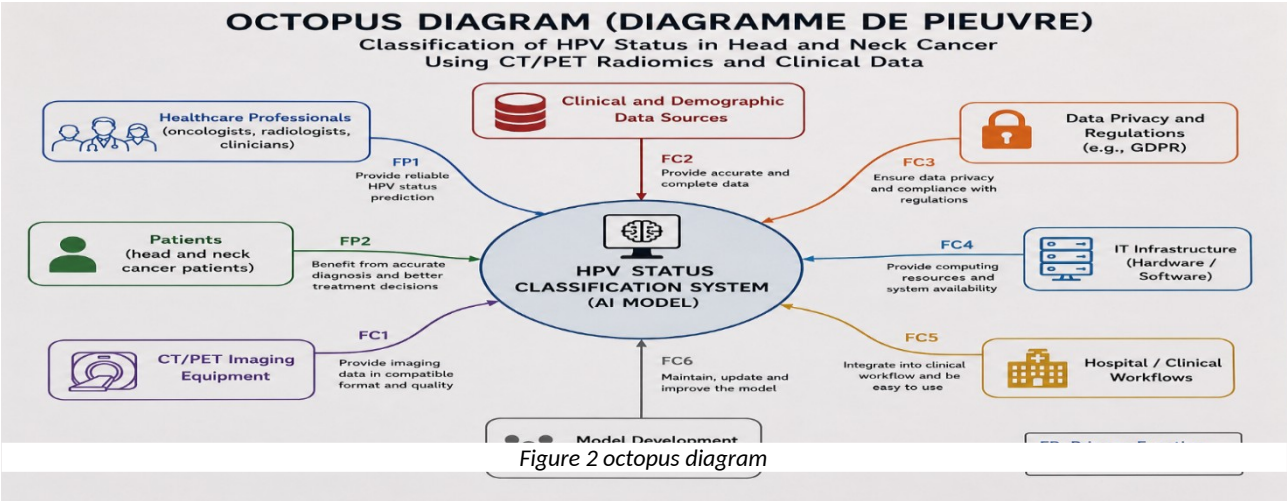


Figure 2 octopus diagram

2.2 Cahier des charges

2.2.1 Diagramme de Gantt

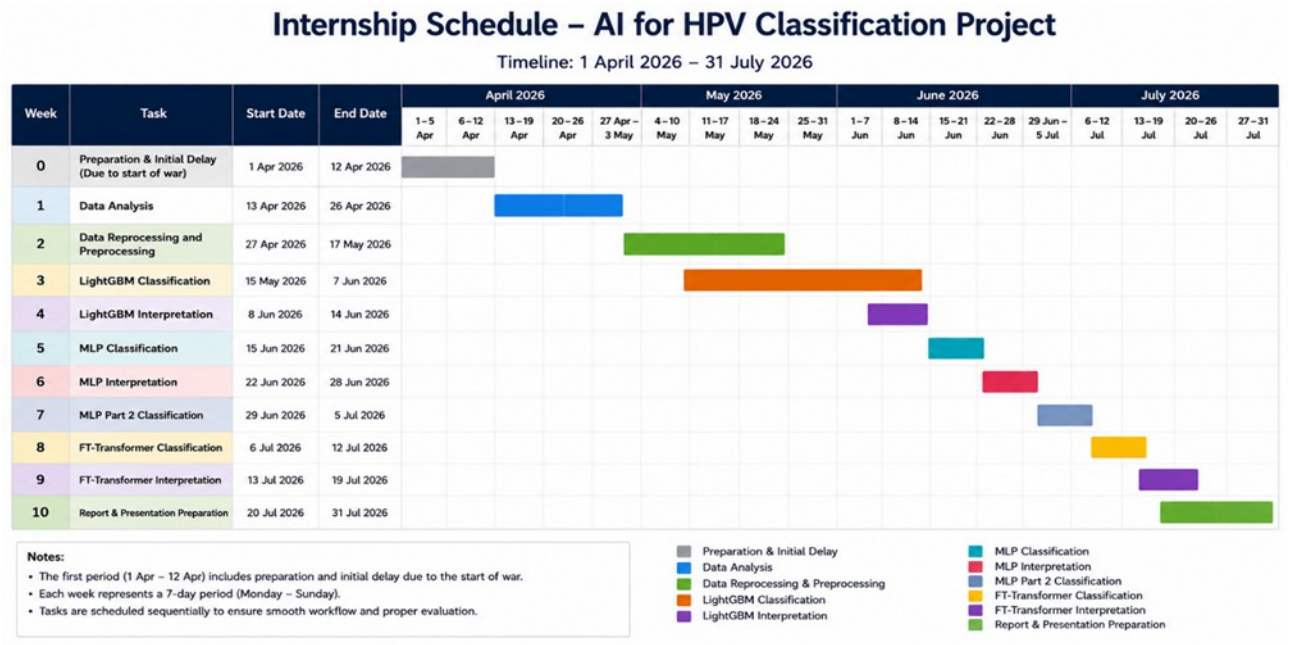
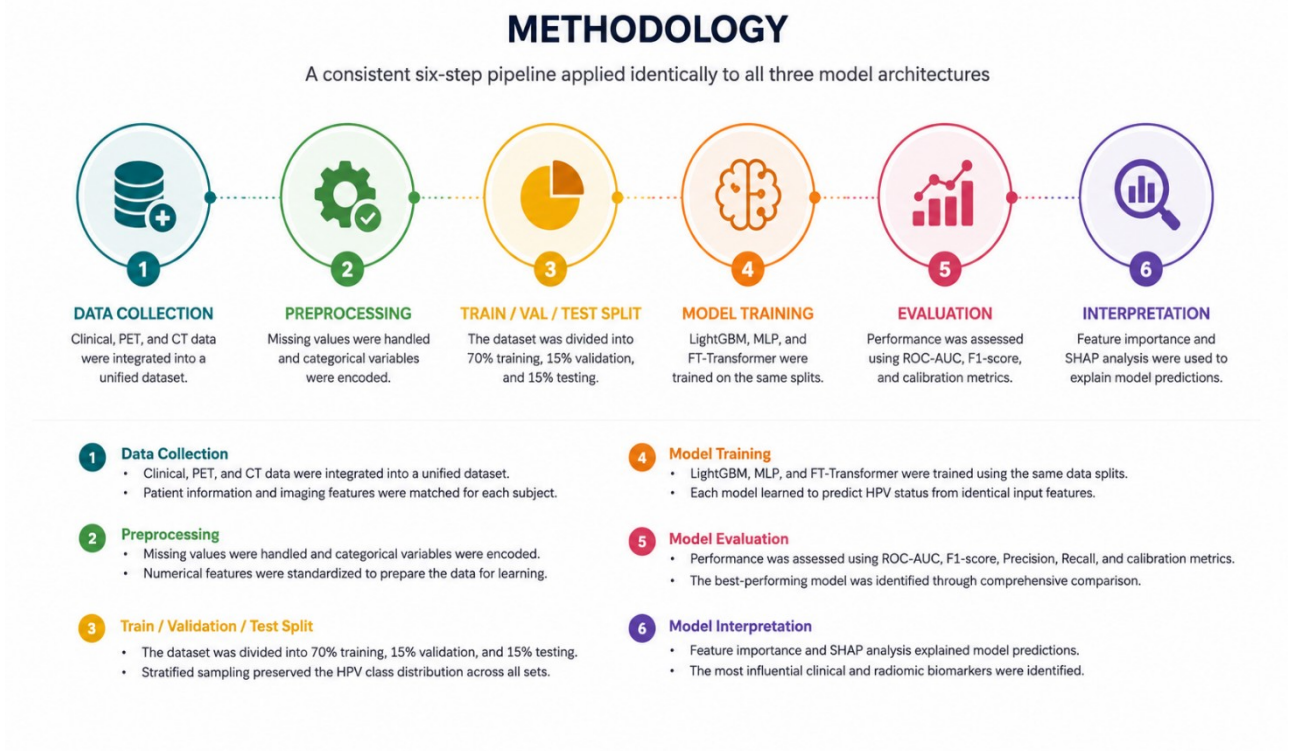


Figure 3 gantt diagram

2.2.2 Methodology



3 State of art

1) Context

Human papillomavirus (HPV) is a group of DNA viruses that infect epithelial cells and is one of the most common viral infections worldwide. Persistent infection with high-risk HPV types, particularly HPV-16, is a major etiological factor for several cancers, including cervical cancer and an increasing proportion of head and neck squamous cell carcinomas (HNSCC), especially oropharyngeal cancer. HPV-positive (HPV+) tumors differ from HPV-negative (HPV−) tumors in terms of tumor biology, prognosis, and response to therapy, making HPV status an essential biomarker for treatment planning and outcome prediction. Positron Emission Tomography (PET) provides functional information by measuring the metabolic activity of tumors, while Computed Tomography (CT) offers detailed anatomical visualization of tumor size, shape, and surrounding structures. The complementary information obtained from PET and CT enables artificial intelligence and radiomics models to extract imaging biomarkers associated with HPV status, allowing non-invasive prediction of HPV positivity and supporting precision medicine in head and neck cancer.

2) The HECKTOR Challenge

The **HEad and neCK TumOR (HECKTOR)** Challenge is an international benchmark competition organized in conjunction with the **MICCAI (Medical Image Computing and Computer-Assisted Intervention)** conference. Since its first edition in **2020**, the challenge has promoted the development and objective evaluation of artificial intelligence methods for analyzing **FDG-PET/CT images** of patients with head and neck cancer. Over the years, HECKTOR has expanded from tumor segmentation to clinically relevant prediction tasks, including prognosis and disease characterization.

In its **2025 edition**, HECKTOR introduced a dedicated task for **HPV status prediction**, where participants were asked to classify tumors as **HPV-positive (HPV+)** or **HPV-negative (HPV−)** using multimodal **PET/CT images** together with basic clinical information such as **age, sex, tobacco use, and alcohol consumption**. This task aims to encourage the development of AI models capable of providing a non-invasive prediction of HPV status to support clinical decision-making. The HECKTOR 2025 dataset contains **1,123 patients** collected from **10 international medical centers**, making it one of the largest publicly available multicenter datasets for head and neck cancer research. Each patient includes co-registered **FDG-PET**

and **CT** scans, expert tumor annotations, and comprehensive clinical metadata, enabling robust evaluation of AI models across diverse clinical settings.

3) Existing AI Approaches for HPV Status Prediction

Most researchers have approached HPV status prediction using a multimodal deep learning model. A common baseline architecture consists of a Convolutional Neural Network (CNN), typically based on a 3D ResNet, to automatically extract features from PET and CT images, while a small fully connected neural network (MLP) processes the patient's clinical information, such as age, sex, tobacco use, and alcohol consumption. The features extracted from both branches are then combined to predict whether the tumor is HPV-positive (HPV+) or HPV-negative (HPV−).

Although these models generally achieve good performance, researchers have consistently observed that they identify HPV-positive tumors more accurately than HPV-negative tumors. This limitation is mainly caused by class imbalance, since the HECKTOR dataset contains substantially more HPV-positive than HPV-negative patients. As a result, the models tend to learn the characteristics of the majority class better, leading to reduced specificity and more misclassification of HPV-negative cases

4) Key Takeaway

Despite the significant progress achieved by recent artificial intelligence models, HPV status prediction remains a challenging task. One of the main difficulties is that the available data are collected from multiple hospitals, where differences in PET/CT scanners, imaging protocols, patient populations, and clinical practices introduce variability that reduces model generalization to unseen data. The HECKTOR challenge was specifically designed to evaluate whether AI models can remain robust under these real-world multicenter conditions. Consequently, improving the generalizability and reliability of AI models across different medical centers remains an important research objective. This challenge provides the motivation for the present project, which aims to investigate and improve automatic HPV status prediction from multimodal PET/CT imaging and clinical data.

4 Work carried out

PHASE 1 : DATA SETS & ANALYSIS

The first two weeks of the internship were dedicated to an in-depth analysis of the available dataset and represented one of the most critical phases of the entire project. During this period, both the clinical data and the CT/PET radiomic features were carefully examined to understand their structure, quality, and relevance to the HPV status classification task. This included reviewing the variables, identifying feature types, checking data consistency, understanding the relationships between clinical and imaging features, and becoming familiar with the overall dataset organization. Establishing a thorough understanding of the data at this early stage was essential before any preprocessing, modeling, or machine learning experiments could be performed. This phase provided the foundation for all subsequent work and ensured that later analyses were based on reliable and well-understood data.

Importance of Data Analysis Before Model Development

Analyzing the data before starting model development is a fundamental step in any machine learning project. A comprehensive exploration of the dataset allows researchers to understand the characteristics of the available information, identify potential issues such as missing values, inconsistencies, or outliers, and evaluate the distribution and quality of the features. It also helps determine the suitability of each variable for the prediction task and provides insight into possible relationships between clinical and radiomic features. Performing this analysis before preprocessing and model training reduces the risk of introducing errors, improves the reliability of the experimental pipeline, and supports informed decisions throughout the development process. Ultimately, a well-executed data analysis phase contributes to building more accurate, robust, and clinically meaningful predictive models

To better understand the available data, the first step was to analyze and explore the three datasets used in this project:

- **HECKTOR file:** Contains the clinical features of the patients.
- **PET file:** Contains the radiomic features extracted from PET images.
- **CT file:** Contains the radiomic features extracted from CT images.

Exploring these datasets was a crucial step before any preprocessing or model development. It helped in understanding the available features, the structure of the data, and identifying any potential issues that could affect the machine learning workflow.

Work applied:

File 1 :HECKTOR 2025 Training Task 3

HECKTOR Clinical Dataset

This dataset contains the clinical information of head and neck cancer patients, including:

- Patient characteristics: age and gender.
- Risk factors: tobacco and alcohol consumption.
- Clinical status: performance status and TNM staging.
- Treatment information: details of the treatment received.
- Outcome: HPV status (target variable for classification).

Analyzing the file:

- PatientID → unique identifier for each patient
- CenterID → hospital/center where patient was treated
- Age → patient age
- Gender → likely binary (0 / 1)
- Tobacco Consumption → smoking history
- Alcohol Consumption → alcohol use
- Performance status (how well a patient can perform daily activities)
- Treatment → type of treatment (numeric coding)
- M-stage → metastasis stage (e.g., M0)
- HPV → Whether the cancer is linked to HPV (Human Papillomavirus)

Feature	Type	Missing Count	Missing %
PatientID	object	0	0%
CenterID	int64	0	0%
Age	int64	0	0%
Gender	int64	0	0%
Tobacco Consumption	float64	95	16.16% ⚠️
Alcohol Consumption	float64	96	16.33% ⚠️
Performance Status	float64	141	23.98% 🔴
Treatment	float64	19	3.23%
M-stage	object	59	10.03% ⚠️
HPV Status	int64	0	0%

Table 1 clinical data analysis

STEP 1:

In this first step, we performed an initial exploratory analysis of the dataset without applying any preprocessing or modifications. The objective was to gain a general understanding of the data structure and the characteristics of the patient population. For example, visualizing the distribution of features such as age

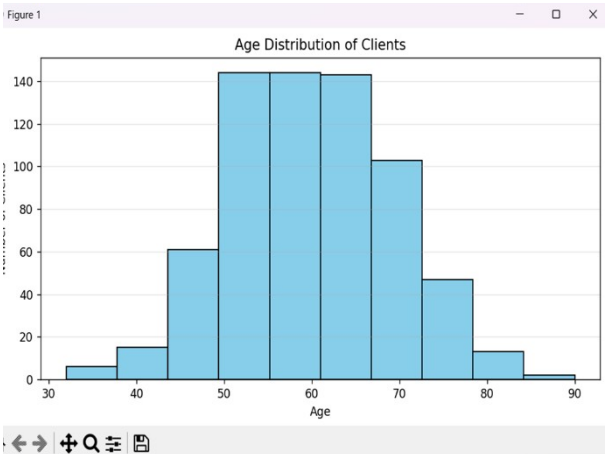


Figure 4 age distribution of clients

STEP 2:

- Age Distribution
- Most patients (both HPV-positive and HPV-negative) are between 50 and 70 years old.
- Few patients are younger than 40 years or older than 80 years.
- HPV-positive patients tend to be younger.
- HPV-negative patients tend to be older.

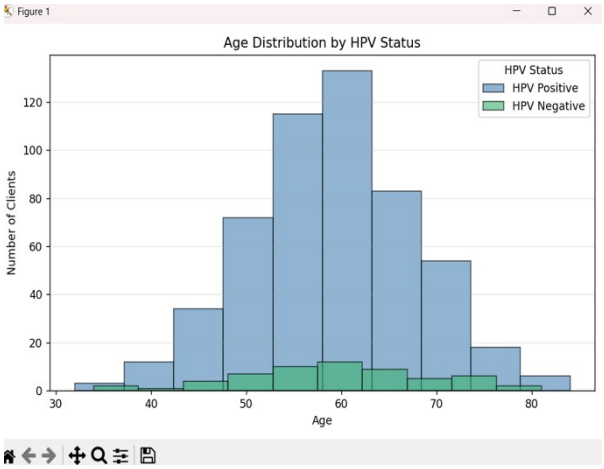


Figure 5 age distribution by hpv status

The age distribution of both HPV-positive and HPV-negative patients is approximately normal, with a considerable overlap between the two groups. However, HPV-negative patients are, on average, slightly older than HPV-positive patients, suggesting a potential association between increasing age and HPV-negative status.

STEP 3:

We are also able to plot the gender count of clients

- TOTAL = 588 clients
- 510 gender 1
- 78 gender 0

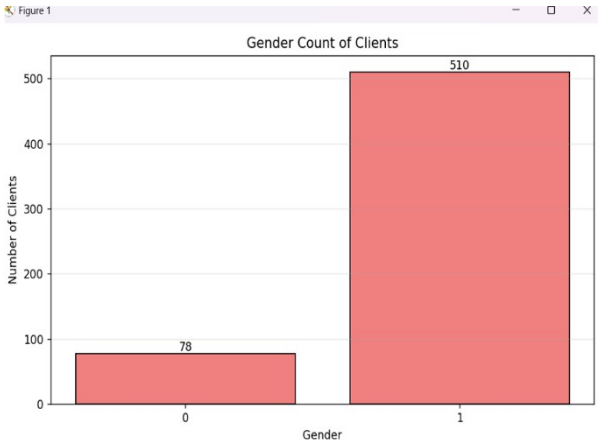


Figure 6 gender counts of clients

STEP 4:

- Treatment Distribution by Gender
- Treatment 0: 100 patients
  - Gender 1: 85 patients (85%)
  - Gender 0: 15 patients (15%)
- Treatment 1: 469 patients
  - Gender 1: 408 patients (87.0%)
  - Gender 0: 61 patients (13.0%)
- Unknown treatment: 19 patients
  - Gender 1: 17 patients (89.5%)
  - Gender 0: 2 patients (10.5%)

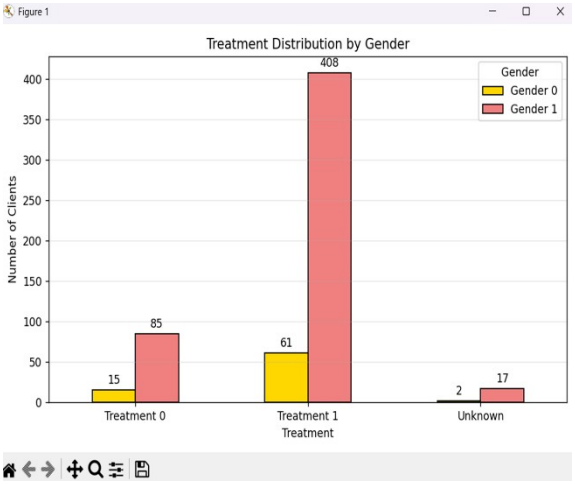
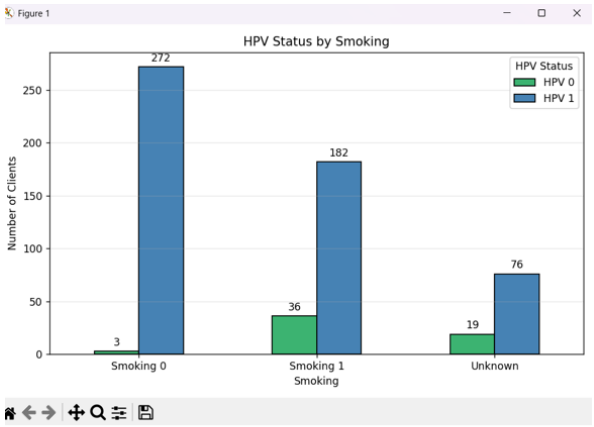


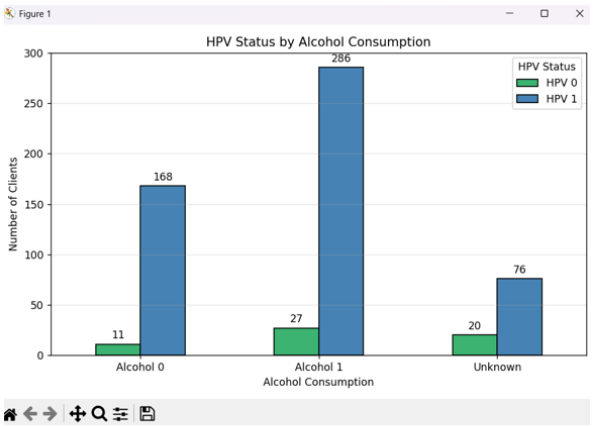
Figure 7 treatment distribution by gender

The treatment distribution is highly imbalanced across genders, with Gender 1 representing the majority of patients in all treatment categories (approximately 85–90%). Furthermore, Treatment 1 is the most frequently administered treatment, accounting for the vast majority of cases, while Treatment 0 and unknown treatments are relatively uncommon. These imbalances should be considered during model development, as they may introduce bias and affect the predictive performance of the classification model.

STEP 5:



The analysis of HPV status in relation to smoking behavior reveals a strong association between the two variables. Non-smokers are predominantly HPV-positive= 272 cases , while smokers show a higher proportion of HPV-negative cases= 36 cases. This pattern aligns with known clinical observations, where HPV-negative cancers are more commonly associated with tobacco use.



The relationship between HPV status and alcohol consumption does not show a strong pattern, as HPV-positive cases dominate across all groups. Unlike tobacco consumption, alcohol does not appear to significantly differentiate between HPV-positive and HPV-negative patients in this dataset.

Figure 8 hpv status by smoking/alcohol consumption

STEP 6:

The analysis of HPV status across M-stage shows that the dataset is largely composed of non-metastatic (M0) patients, with a strong predominance of HPV-positive cases. Very few patients present with metastatic disease (M1), limiting the ability to draw meaningful conclusions regarding the relationship between HPV status and metastasis

M0 stage = 525
506 for HPV positive
M1 stage = 3
3 for HPV positive

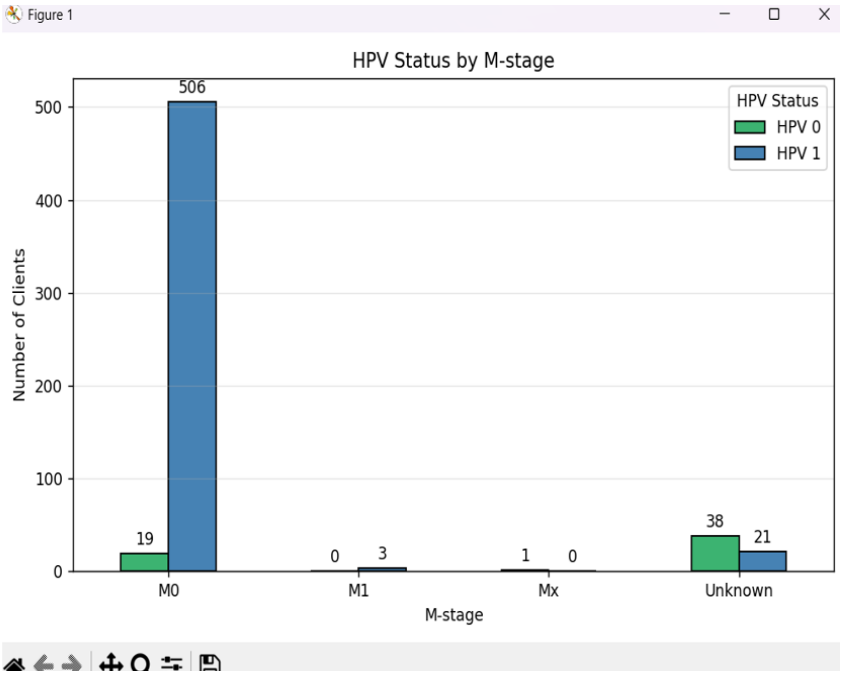


Figure 9 hpv status by M-stage

STEP 7:

The distribution of M-stage across treatment and gender shows that the dataset is predominantly composed of non-metastatic (M0) patients. Treatment 1 is the most frequently administered therapy, particularly among one gender, which is overrepresented in the dataset. The very small number of metastatic cases limits the ability to analyze advanced disease patterns.

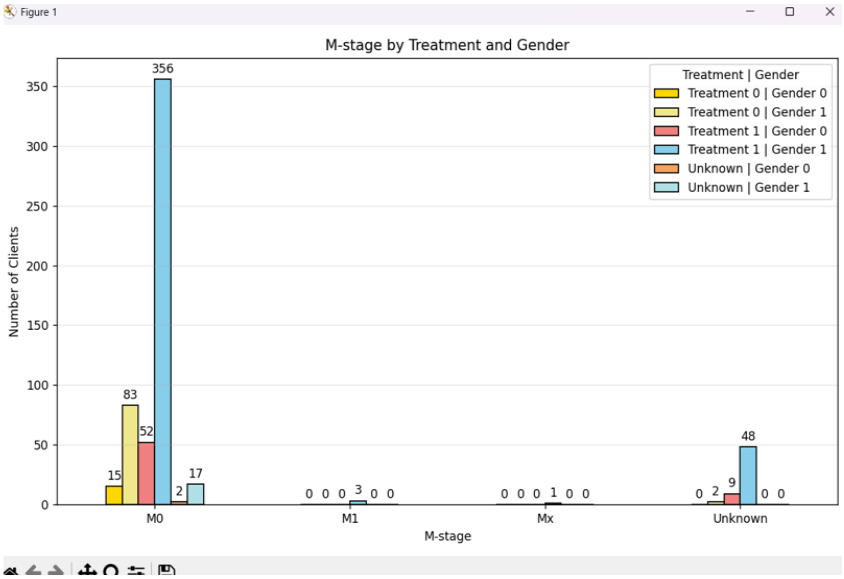


Figure 10 Mstage by treatement and gender

FILE 2 : CT FEATURES

This dataset contains radiomic features extracted from CT images. These features quantitatively describe the tumor by capturing its intensity, texture, and shape characteristics, providing a comprehensive numerical representation of tumor morphology and heterogeneity.

The radiomic features are grouped into three main categories:

- First-Order Features (Intensity)
    - Describe the distribution of voxel (pixel) intensity values.
    - Examples: Mean, Median, Variance, Entropy.
    - Provide information about tumor density and intensity heterogeneity.
- Texture Features
    - Describe the spatial relationships and patterns between neighboring voxels.
    - Include features derived from:
        - GLCM (Gray Level Co-occurrence Matrix): relationships between neighboring pixels.
        - GLRLM (Gray Level Run Length Matrix): lengths of consecutive pixels with the same intensity.
        - GLSZM (Gray Level Size Zone Matrix): sizes of homogeneous intensity regions.
        - GLDM (Gray Level Dependence Matrix): dependence between neighboring voxels.
        - NGTDM (Neighborhood Gray Tone Difference Matrix): differences between a voxel and its surrounding neighborhood.
    - These features capture tumor texture, structural complexity, and heterogeneity.
- Shape Features
    - Describe the three-dimensional geometry of the tumor.
    - Examples: Volume, Surface Area, Sphericity.
    - Provide information about tumor size and physical shape.

STEP 1:

•**X-axis** → mean intensity values
•**Y-axis** → number of patients

The data looks roughly:
centered
symmetric (slightly skewed)
This is good because:
the feature is well-behaved

Some outliers ⚠️
We have:
very low values (negative side)
very high values (~100–150)

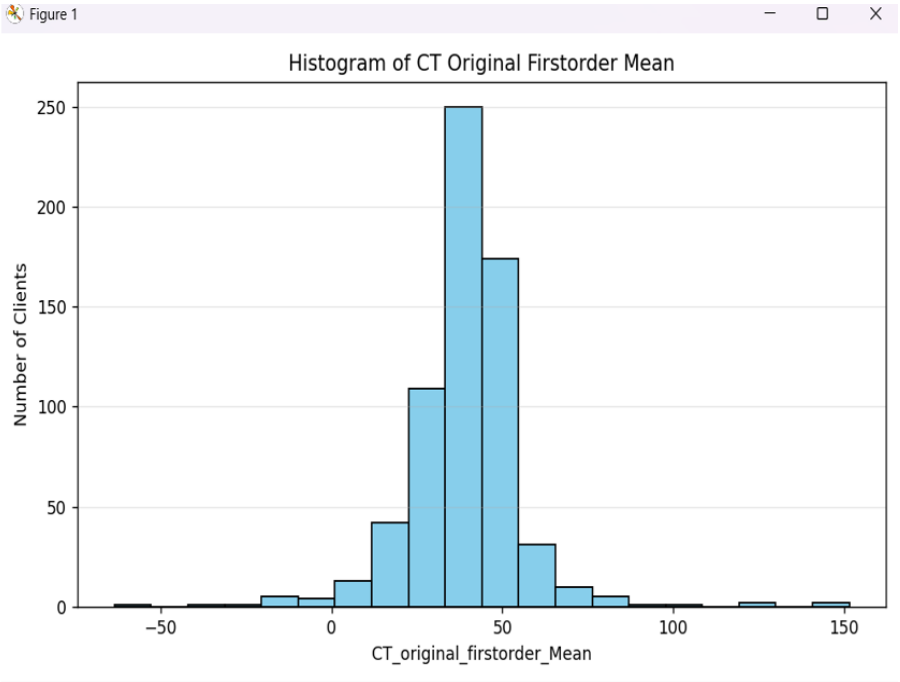


Figure 11 Histogram of ct original firstorder mean

STEP 2:

•**Low entropy** → smooth, uniform tumor
•**High entropy** → irregular, complex tumor

Entropy shows **strong variability between patients**
This makes it:
a more powerful feature

- Most patients (~70–80%) fall within: 👉 1.2 – 2.5
- Low entropy values (<1.0): very few cases (rare)
- High entropy values (>2.5): smaller group of patients (~10–20%)

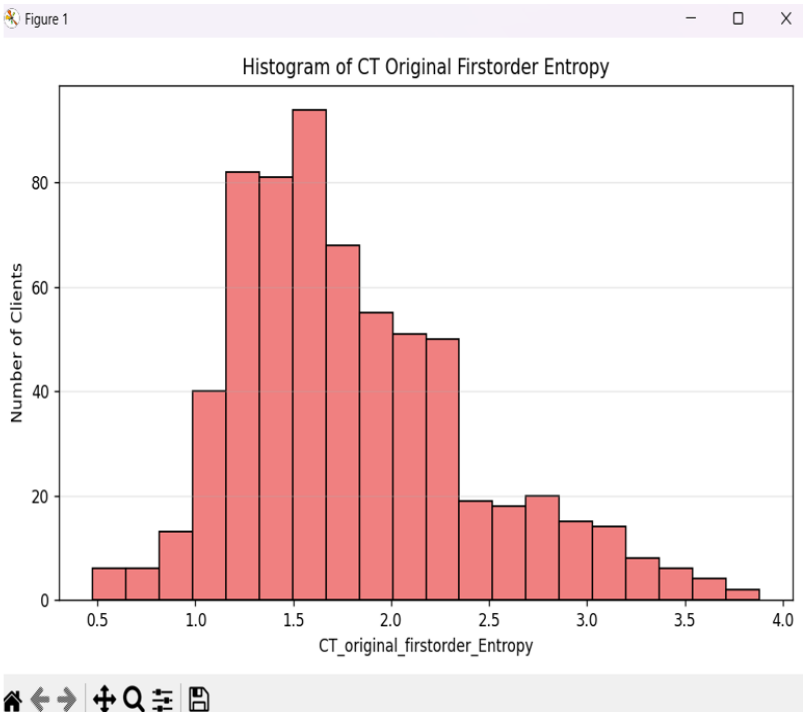


Figure 12 Histogram of ct original firstorder entropy

STEP 3:

Variance refers to:
how much pixel intensities vary inside the tumor

Variance values range from near 0 to over 90,000, with most patients concentrated below 5,000. The distribution is highly right-skewed, indicating the presence of extreme outliers and large variability in tumor intensity for a small subset of patients.

- **Majority of patients (~85–90%)** fall within: **0 – 5,000**
- **Moderate values (5,000 – 20,000):** small portion of patients
- **Very high values (>20,000):** very rare (few extreme cases)

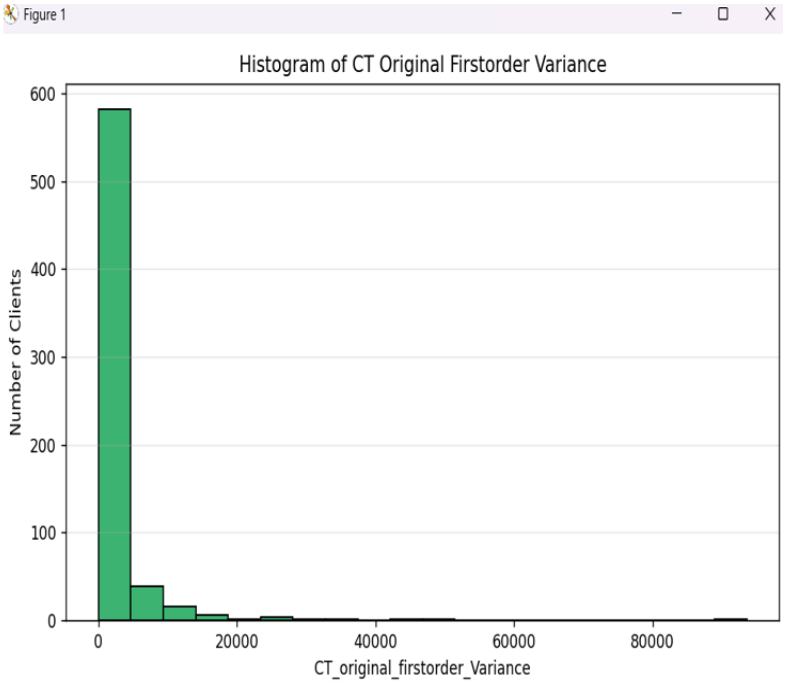


Figure 13 Histogram of ct original Firstorder variance

STEP 4:

- ~70–80% of tumors : small volumes (< 20,000)
- ~15–25%: medium volumes (20,000 – 60,000)
- <5%: very large tumors (> 60,000)

The distribution of tumor volume is highly right-skewed, with most patients having relatively small tumor volumes below 20,000. A small number of cases exhibit very large volumes, creating a long tail in the distribution. This suggests significant variability in tumor size across the dataset.

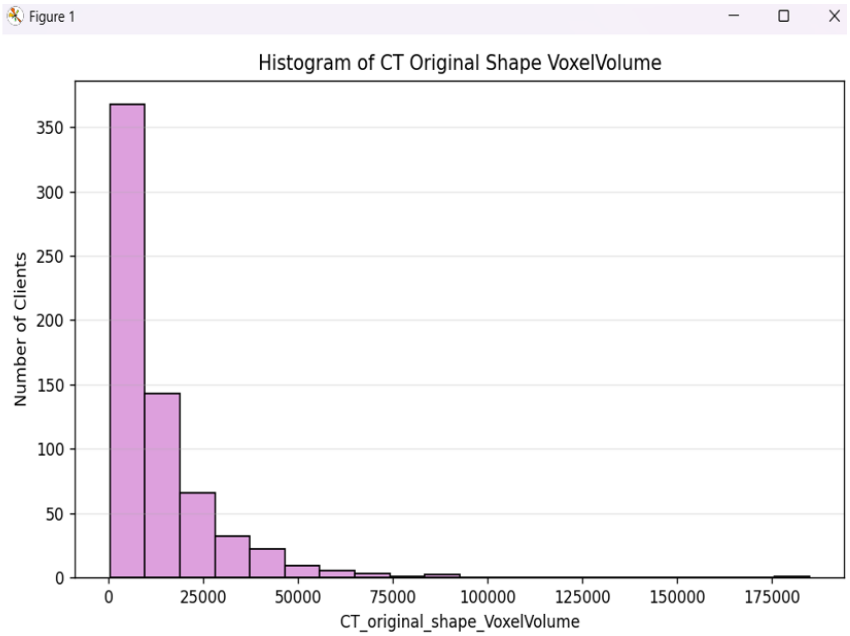


Figure 14 Histogram of ct original shape voxelvolume

STEP 5:

- ~70–80% of tumors: surface area < 5,000
- ~15–25%: between 5,000 – 10,000
- <5%: very high values (>10,000

The surface area distribution is right-skewed, with most tumors exhibiting moderate surface values between 1,000 and 5,000. A small number of cases show very high surface areas, indicating either larger tumor size or increased shape irregularity.

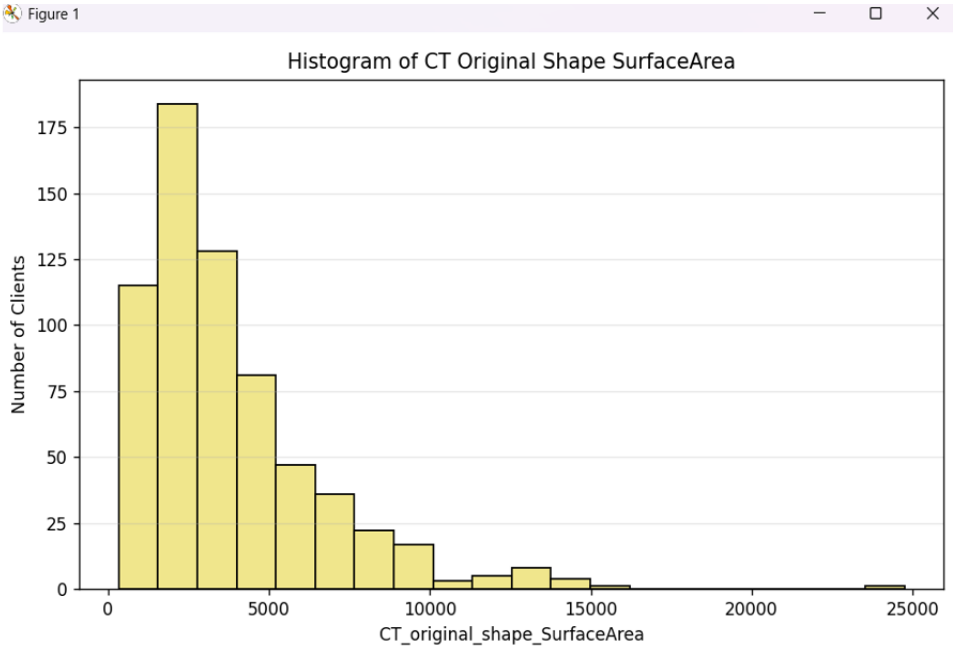


Figure 15 Histogram of ct original shape surface area

STEP 6:

- ~70–80% of tumors: between 0.6 – 0.75
- Small number: below 0.5 (more irregular)
- Very few: above 0.8 (very spherical)

The sphericity distribution is approximately normal, with most tumors having values between 0.6 and 0.75. This indicates that the majority of tumors exhibit moderately regular shapes, while highly irregular or perfectly spherical tumors are relatively rare.

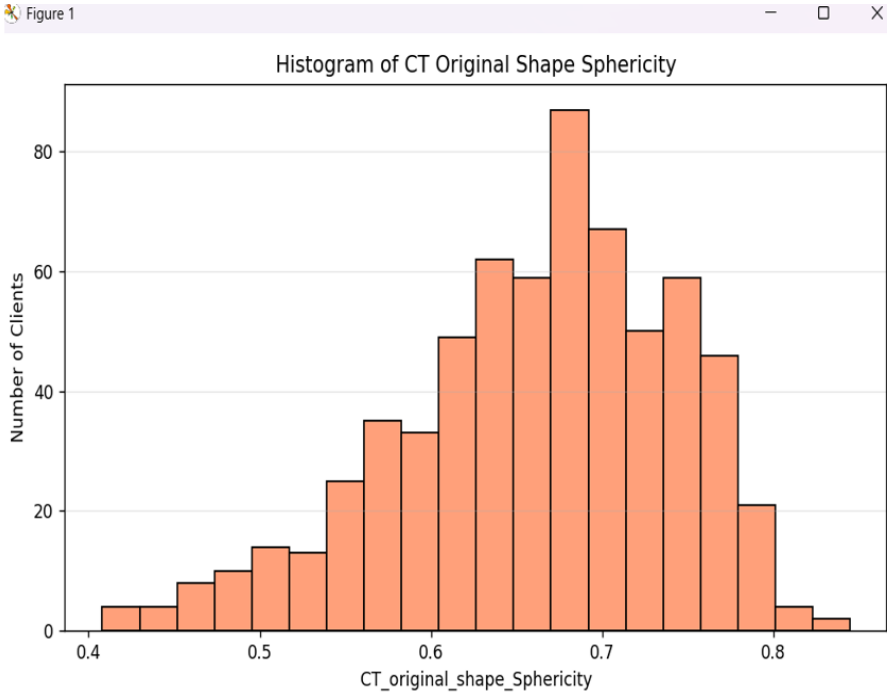


Figure 16 Histogram of ct original shape sphericity

Features overview:

Texture Feature Families

GLCM (Gray Level Co-occurrence Matrix)

- Describes the spatial relationships between neighboring pixels.
- Measures texture properties such as contrast, correlation, homogeneity, and entropy.
- Captures local texture patterns and heterogeneity.

GLRLM (Gray Level Run Length Matrix)

- Measures the length of consecutive pixels with the same gray level.
- Describes texture continuity and run-length patterns.
- Captures the distribution of short and long homogeneous regions.

GLSZM (Gray Level Size Zone Matrix)

- Measures the size of homogeneous gray-level regions.
- Describes the distribution of small and large texture zones.
- Provides information about tumor heterogeneity and regional organization.

NGTDM (Neighborhood Gray Tone Difference Matrix)

- Measures the difference between a voxel and its neighboring voxels.
- Captures local texture characteristics such as coarseness, contrast, complexity, busyness, and strength.
- Describes the overall texture complexity and irregularity of the tumor.

PHASE 2: DATA RE/PREPROCESSING

Data preparation and preprocessing are fundamental steps in any machine learning project, as the quality of the input data directly influences the performance and reliability of the developed models. Proper preprocessing ensures that the data are consistent, complete, and suitable for analysis while minimizing the risk of introducing bias or errors during model training.

In this project, the first step consisted of **merging the clinical dataset with the CT and PET radiomics datasets**, resulting in a unified dataset containing all the information required for HPV status classification. The merged dataset was then **split into training, validation, and testing sets** to ensure an unbiased evaluation of the machine learning models. Performing the split before preprocessing was essential to prevent **data leakage**, ensuring that no information from the validation or test sets influenced the training process.

Once the datasets were separated, the **data cleaning phase** was performed independently on each split. This stage involved identifying and handling missing values, removing duplicate records, correcting inconsistencies, ensuring appropriate data types, and applying the same preprocessing pipeline to the validation and test sets using the transformations learned from the training data. This approach preserves the integrity of the experimental setup while guaranteeing that the model is evaluated on previously unseen data.

Overall, careful data preparation and preprocessing provide a reliable foundation for machine learning by improving data quality, enhancing model robustness, and ensuring that the obtained results are accurate, reproducible, and representative of real-world performance.

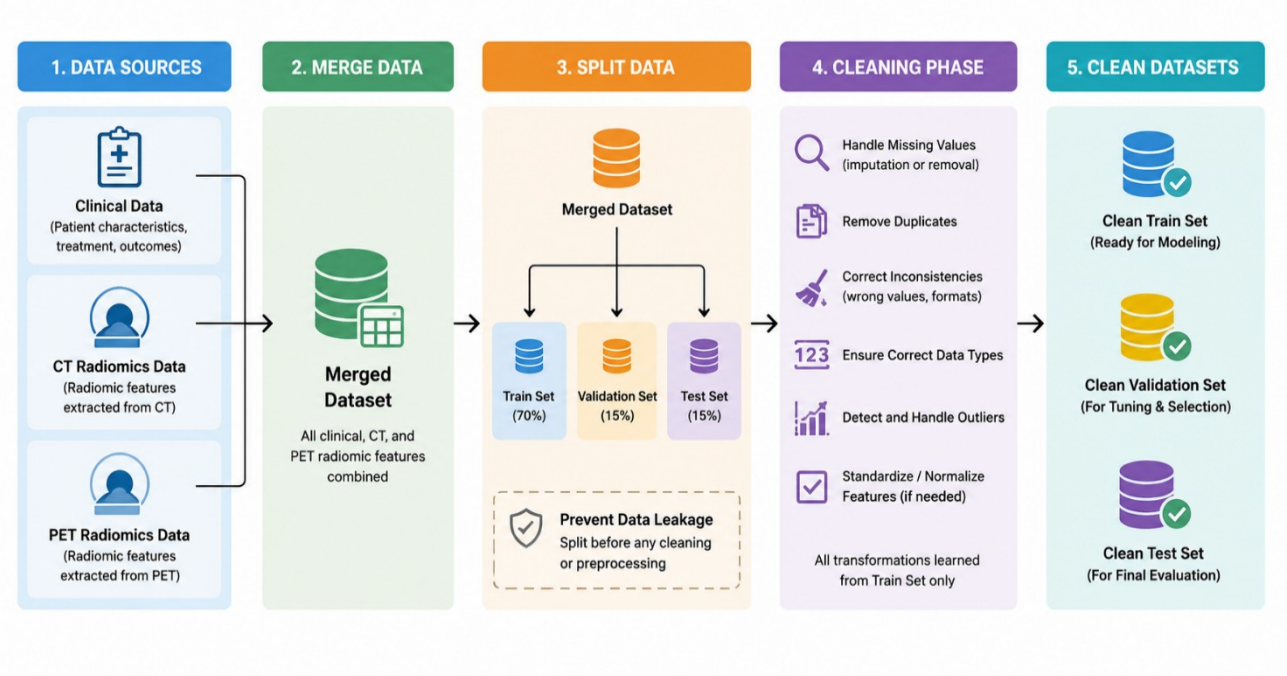


Figure 17 splitting data workflow

WORK APPLIED :

DATA MERGING

STEP 1:

Data unification combines multiple datasets into a single consistent structure for analysis. In this project, four datasets—clinical, PT, CT, and PET/CT—are merged based on the clinical patient ID. Only patients with available radiomic features are retained, while unmatched or incomplete records are removed to ensure a clean and coherent dataset

After unifying the datasets, three CSV files were generated:
-Final unified dataset: containing the selected patients with both clinical data and radiomic features
-74 clinical patient IDs with no corresponding radiomic features (excluded)
-138 patients with complete radiomic features but no clinical data (also excluded)

```
Combined HECKTOR-priority CSV head:
  PatientID  ...  petct_features__PET_original_shape_VoxelVolume
0  CHUM-012  ...                                         7806.0
1  CHUM-014  ...                                         7692.0
2  CHUM-016  ...                                         3169.0
3  CHUM-021  ...                                        17007.0
4  CHUM-022  ...                                         7415.0

[5 rows x 438 columns]

HECKTOR patients with no radiomics features:
74

Radiomics patients not found in HECKTOR:
ct_features.csv       138
pet_features.csv      138
petct_features.csv    138
dtype: int64
```

Figure 18 unified data

STEP 2:

Data splitting is an essential step in preparing data for machine learning, as it ensures proper training and evaluation of the model. It involves dividing the dataset into three main parts:

- Training set: used to learn patterns from the data
- Validation set: used to tune and optimize the model
- Test set: used to evaluate performance on unseen data

To properly split the dataset, we follow a structured and balanced approach:

Split the dataset into three groups:

- Training set → 70%
- Validation set → 15%
- Test set → 15%

➢ Ensure the split is balanced across classes / HPV Stratification (e.g., HPV = 0 and HPV = 1)

➢ Maintain a similar proportion of each class in all three sets

➢ Avoid putting most of one class in a single group

After performing the data split, three CSV files were generated:

- Training set → hecktor_union_train
- Validation set → hecktor_union_validation
- Test set → hecktor_union_test

These files correspond to the three subsets used for model training, validation, and final evaluation.

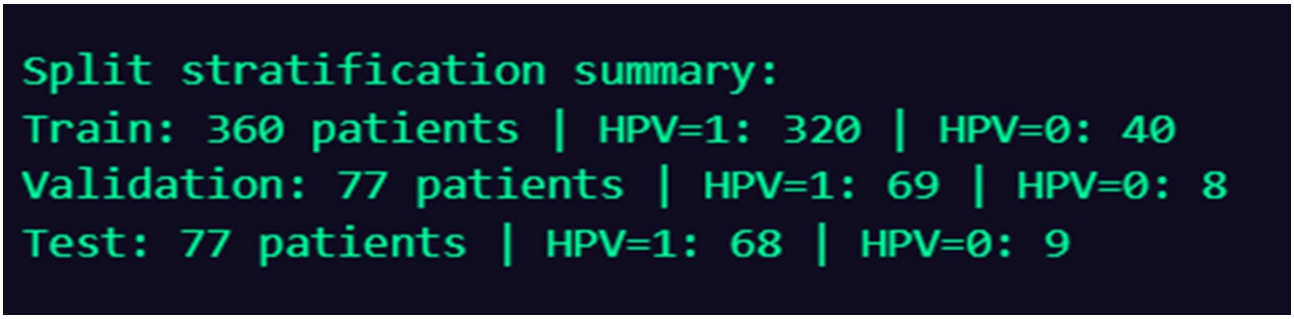


Figure 19 split stratification summary

- Training set: 360 patients HPV = 1 → 320HPV = 0 → 40
- Validation set: 77 patients HPV = 1 → 69HPV = 0 → 8
- Test set: 77 patients HPV = 1 → 68HPV = 0 → 9

The distribution shows that the class balance (HPV = 1 and HPV = 0) has been maintained consistently across all three datasets.

DATA PREPROCESSING :

Missing-data analysis was performed on the training, validation, and test sets before preprocessing to identify missing values, quantify their extent, and determine which variables required handling. This analysis was conducted separately for each dataset to ensure that missing-value patterns were understood before applying any preprocessing techniques. The results provided the basis for selecting appropriate imputation methods while maintaining a consistent preprocessing pipeline across all three data splits.

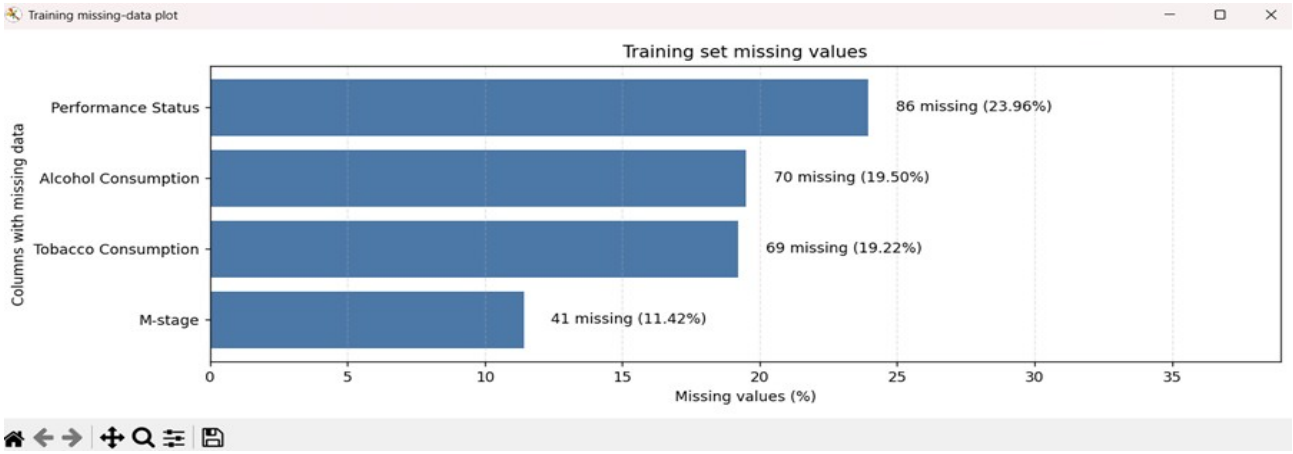


Figure 20 train set missing values

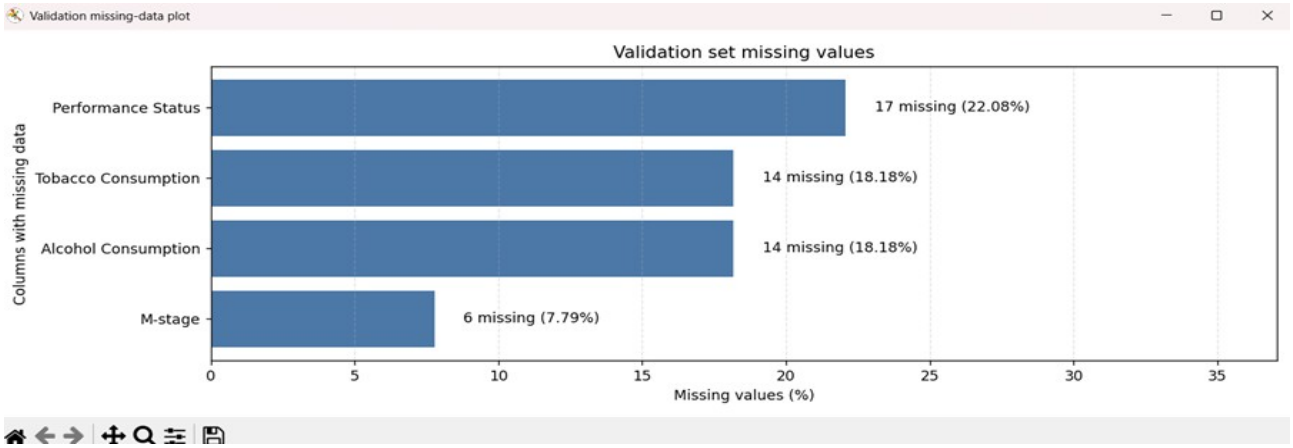


Figure 21 validation set missing values

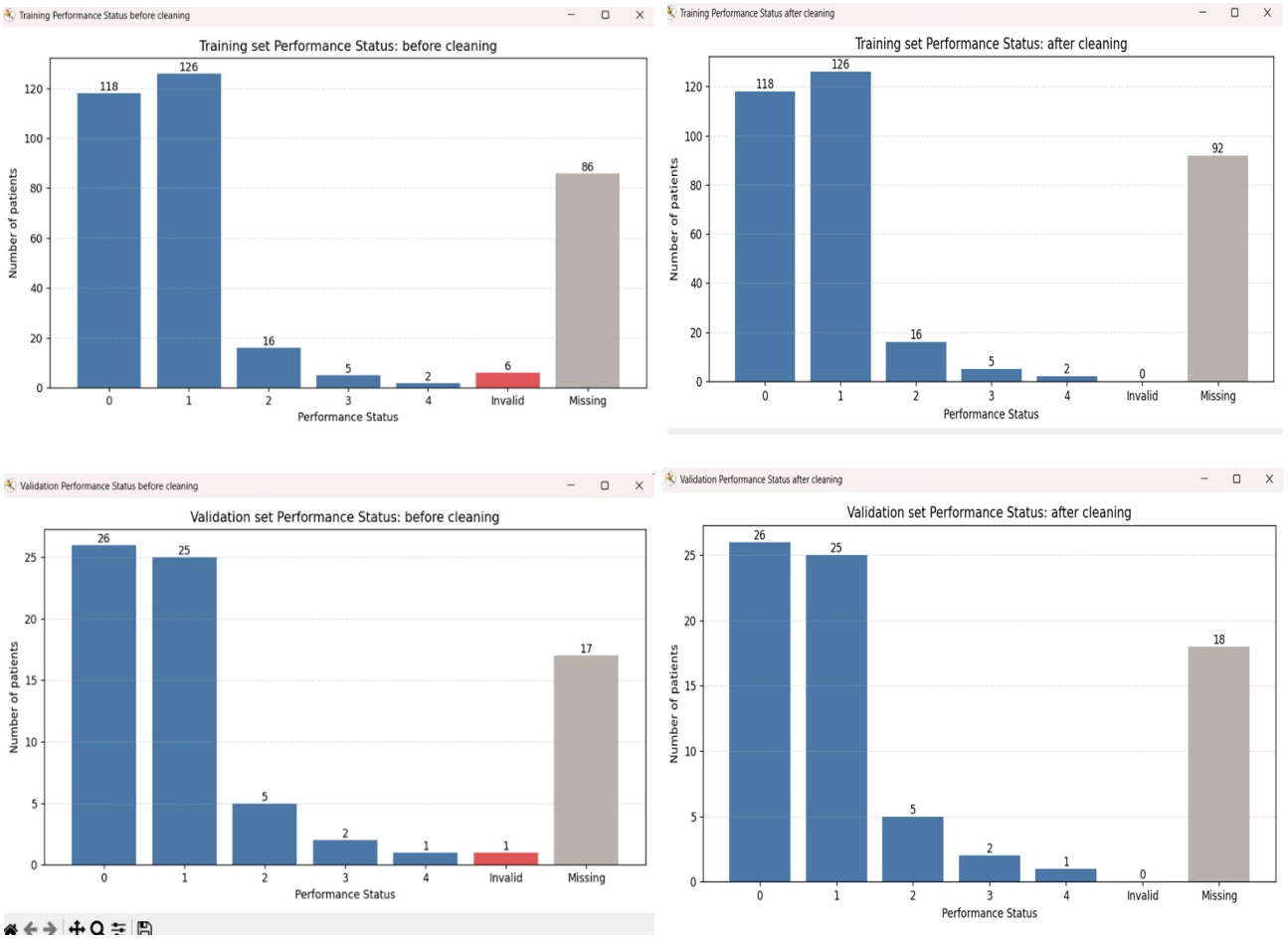


Figure 22 test set missing values

PREPROCESSING DATA:

1) Rule-Based Data Cleaning:

Rule-Based Data Cleaning: Invalid values in the Performance Status column were identified and converted to missing values (NaN) across the training, validation, and test sets. Only the values 0, 1, 2, 3, and 4 were considered valid, while any other values (e.g., 80, 90, 100) were treated as invalid and replaced with NaN.



2) Missing Indicator Method:

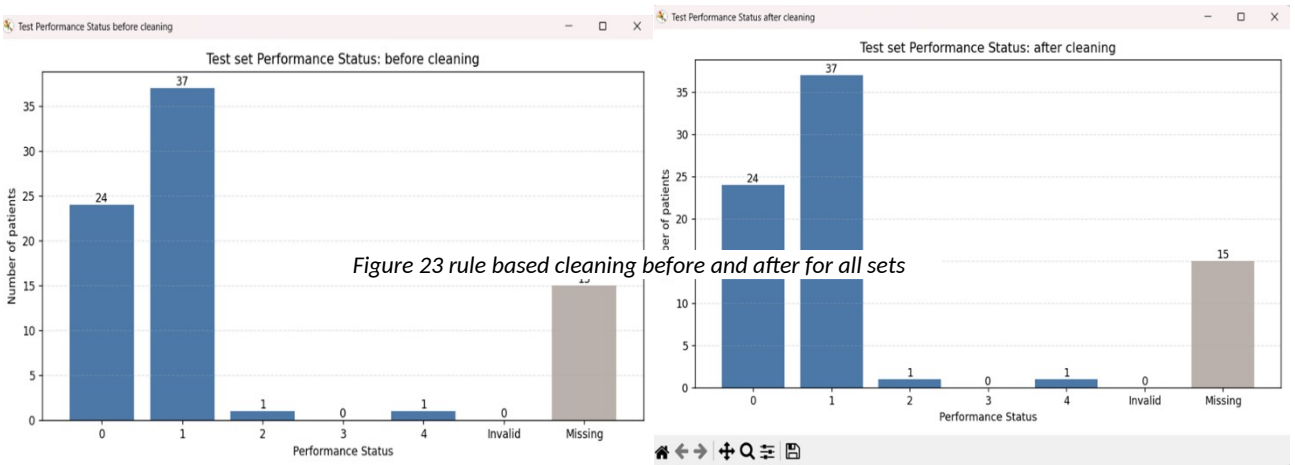


Figure 23 rule based cleaning before and after for all sets

Missing indicator columns were created for Tobacco Consumption, Alcohol Consumption, and Performance Status to preserve information about the original missing values. A value of 1 indicates that the original data was missing, while 0 indicates that the original value was present

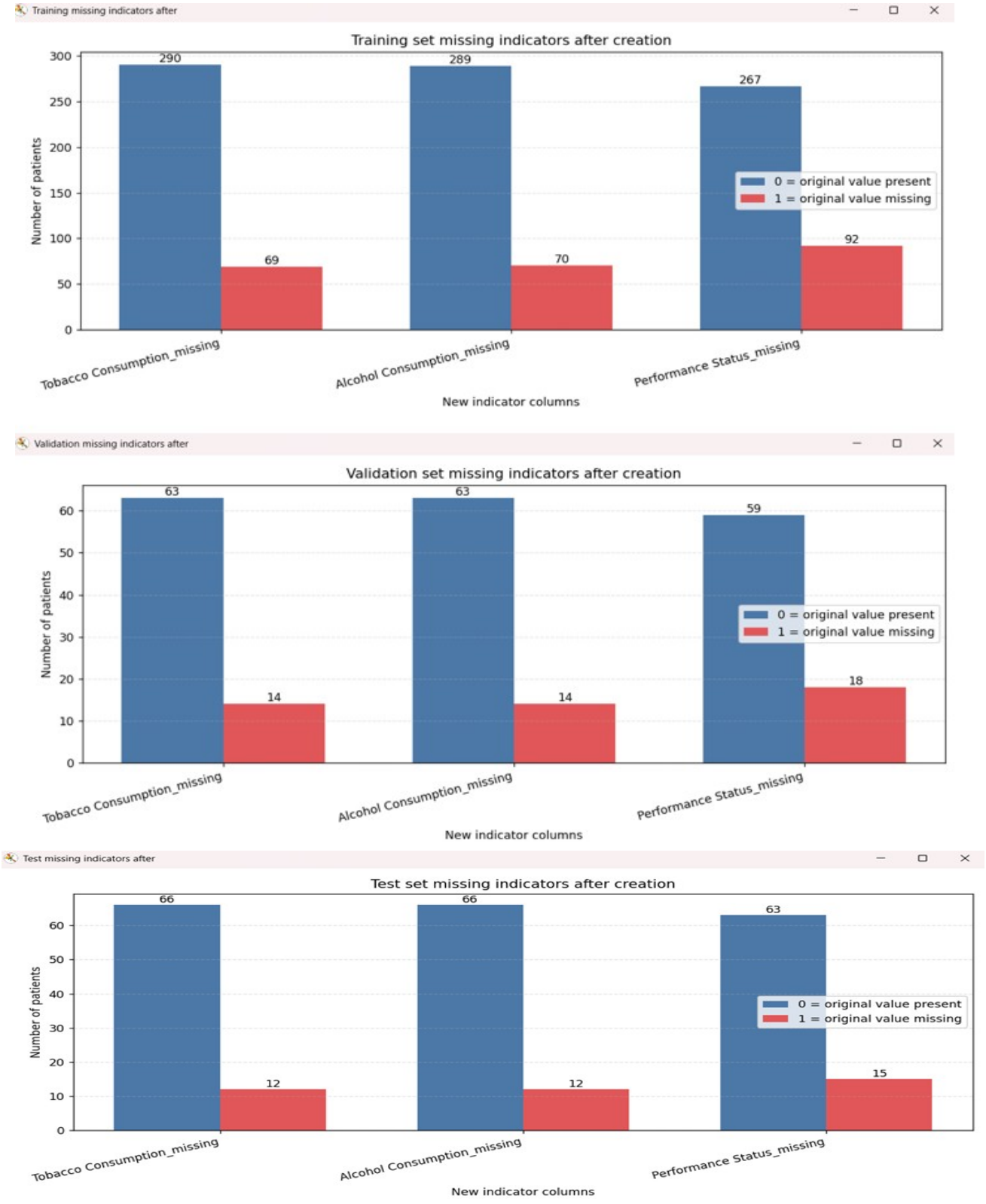


Figure 24 missing indicator before/after for all sets

3) Most-Frequent Imputation

Missing values in Gender, Tobacco Consumption, Alcohol Consumption, Treatment, and Performance Status were replaced using the most frequent value (mode) calculated from the training set only. The same training-based values were then applied to the validation and test sets. This approach was used because these variables are binary, categorical, or ordinal/discrete clinical features rather than continuous variables.

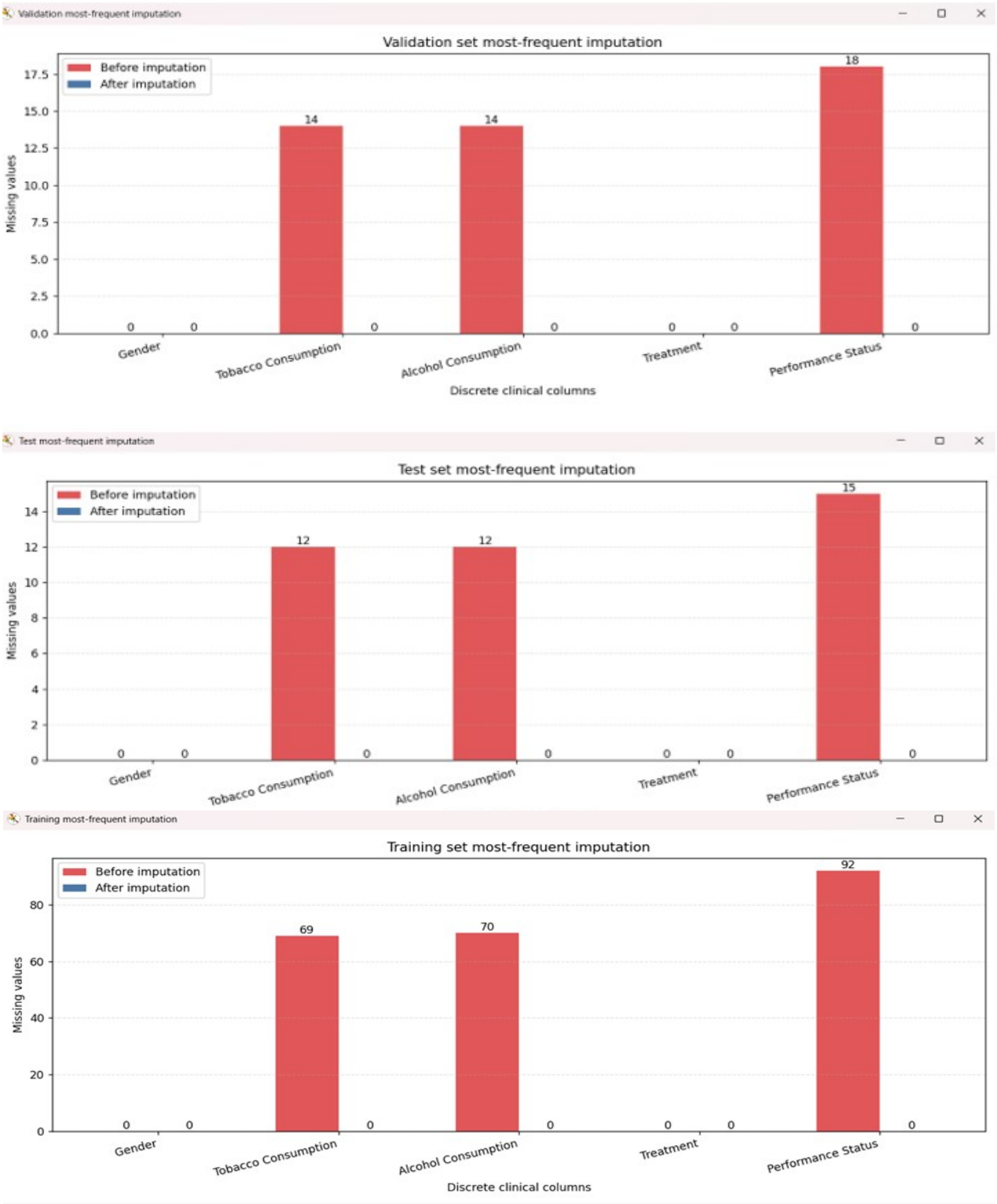


Figure 25 Most frequent imputation before/after for all sets

4) Constant Imputation

Missing values in the M-stage column were replaced with a fixed category, Unknown, across the training, validation, and test sets. This approach was selected because M-stage is a categorical variable, and assigning a dedicated category preserves missing information without affecting the existing categories.

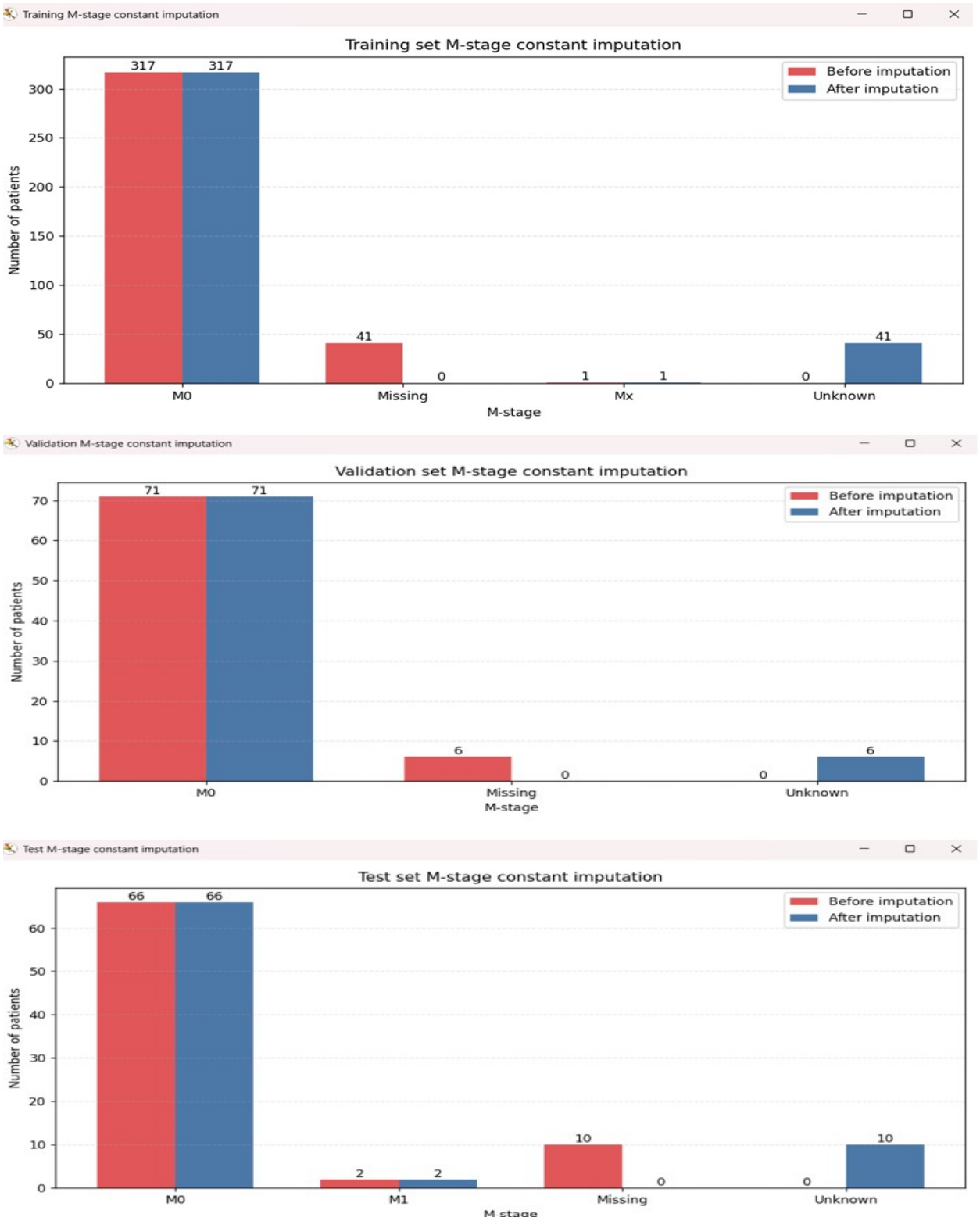


Figure 26 constant imputation before/after for all sets

5) MIN-MAX AGE SCALING

After median imputation, the Age column was standardized using Min-Max scaling to transform values into the range [-1,1]. The minimum and maximum values were calculated from the training set only, and the same scaler was then applied to the validation and test sets. This method was applied only to Age because it is a continuous numerical variable, while binary variables and missing indicators were left unchanged.

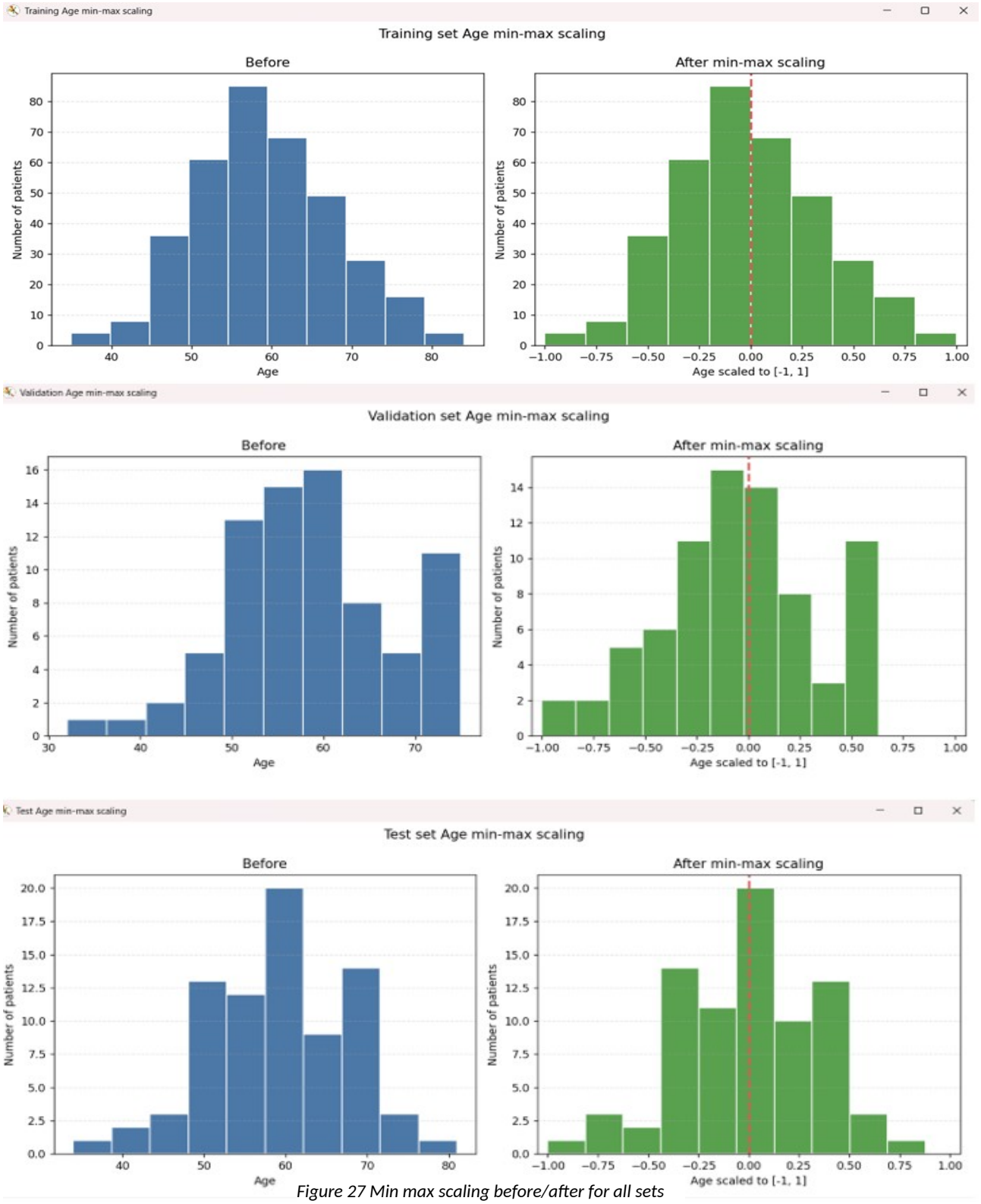


Figure 27 Min max scaling before/after for all sets

6) One-Hot Encoding

After replacing missing values in M-stage with Unknown, one-hot encoding was applied to convert the categorical variable into separate binary columns such as M-stage_M0, M-stage_M1, M-stage_Mx, and M-stage_Unknown. The encoder was fitted using the training set only and then applied to the validation and test sets. The parameter handle_unknown="ignore"

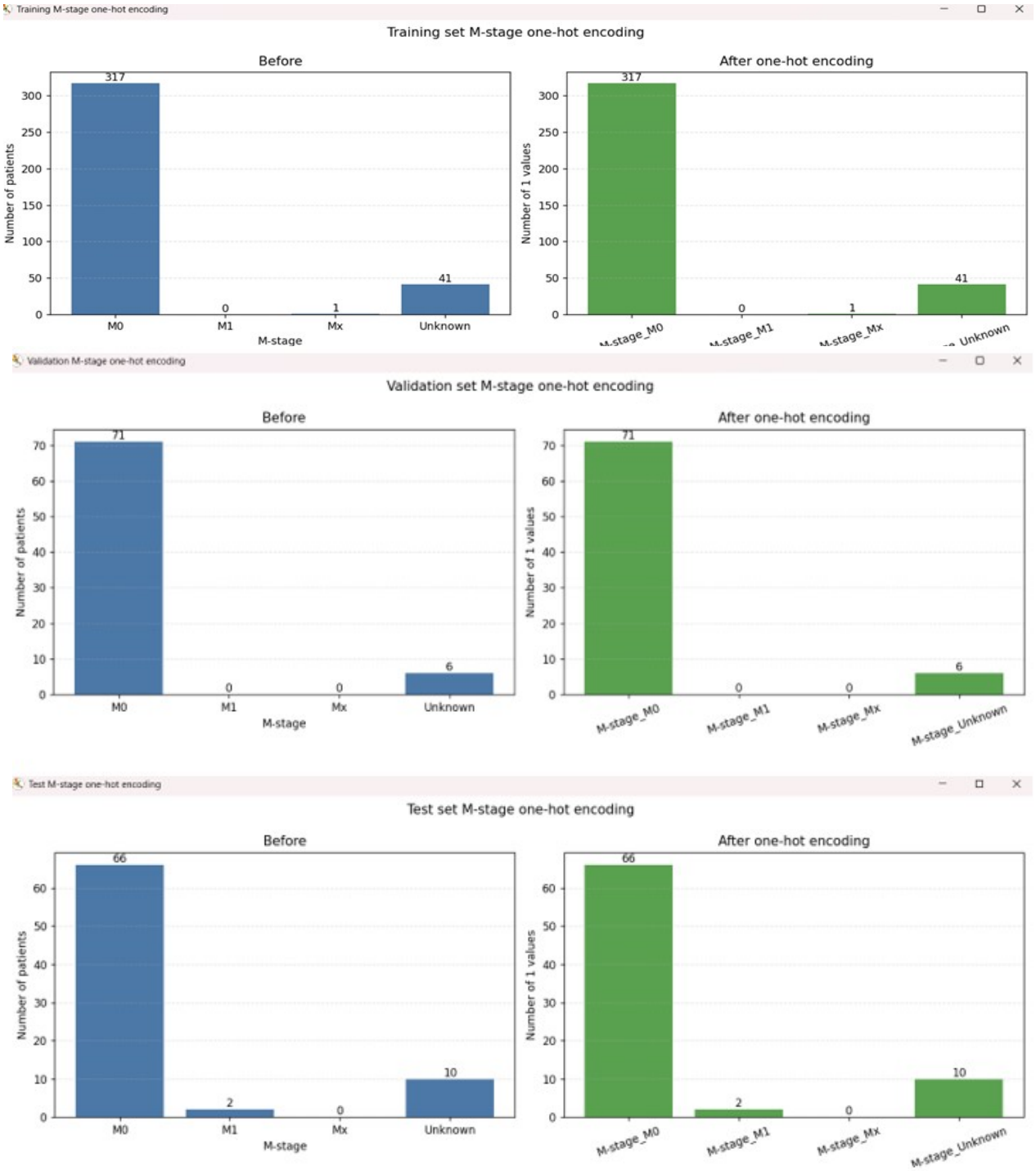


Figure 28 one hot encoding before/after for all sets

After verifying that all preprocessing steps had been successfully completed and ensuring that the datasets were clean, consistent, and free of missing values, the data were considered ready for the machine learning phase. At this stage, the prepared training, validation, and test sets provided a reliable foundation for developing and evaluating the required classification models for HPV status prediction. With the preprocessing pipeline finalized, the project proceeded to the implementation and assessment of the classification algorithms.

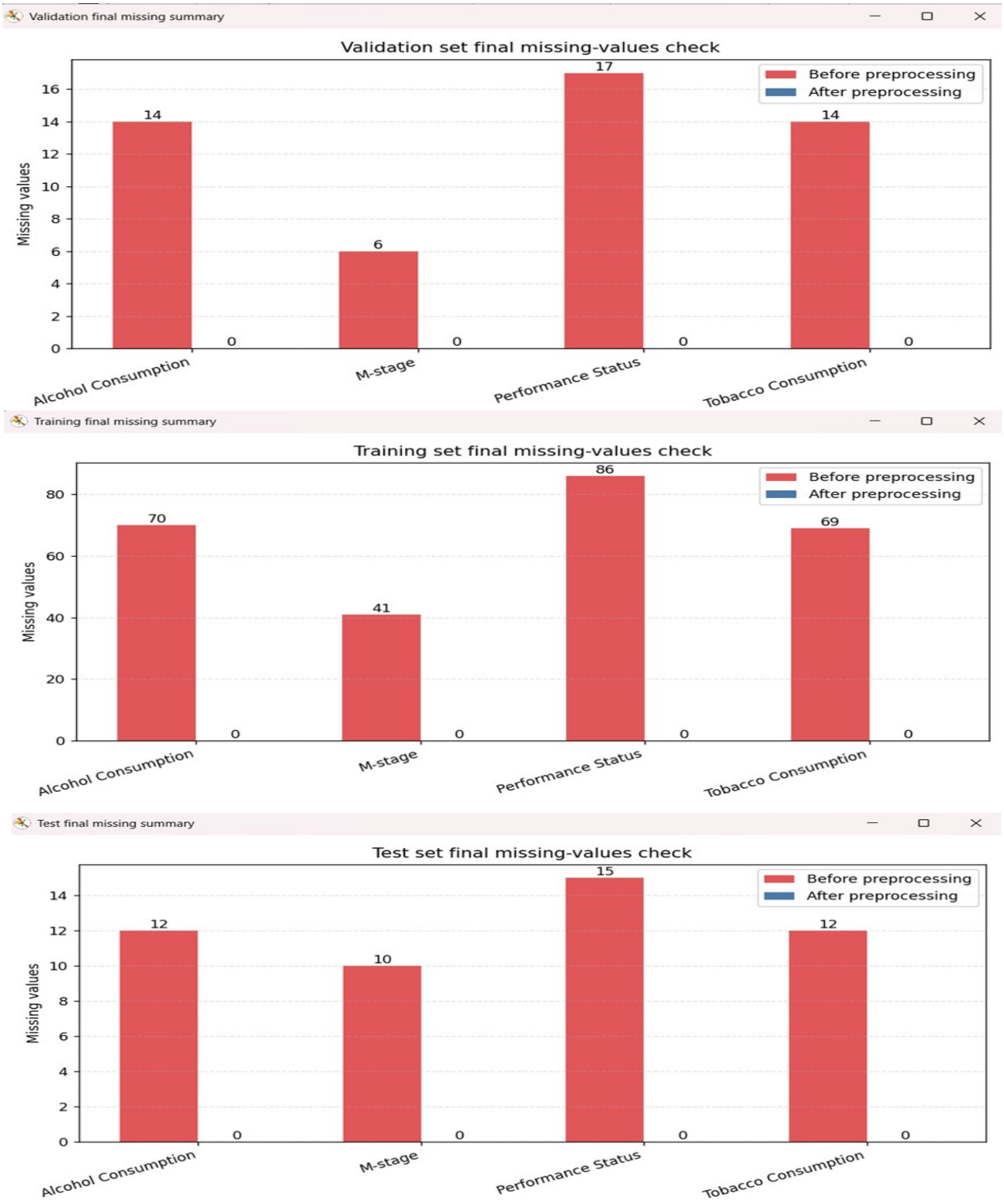


Figure 29 final missing values fro all sets

LIGHTGBM CLASSIFICATION:

LightGBM Classification

LightGBM (Light Gradient Boosting Machine) is a supervised machine learning algorithm based on the gradient boosting framework. It is designed to build highly accurate predictive models by combining the outputs of multiple decision trees. Instead of creating a single complex tree, LightGBM constructs a sequence of decision trees, where each new tree is trained to correct the errors made by the previous ones. This iterative learning process gradually improves the model's predictive performance.

Unlike traditional gradient boosting methods, LightGBM grows trees **leaf-wise** rather than level-wise. At each iteration, it expands the leaf that provides the greatest reduction in prediction error, allowing the model to achieve higher accuracy while using fewer trees. Additionally, LightGBM incorporates techniques such as **Gradient-based One-Side Sampling (GOSS)** and **Exclusive Feature Bundling (EFB)** to reduce computational cost and memory usage, making it particularly efficient for datasets with a large number of features, such as radiomics datasets.

In this project, LightGBM was selected to classify the **HPV status** of patients using clinical and radiomic features extracted from CT and PET images. Its ability to handle high-dimensional data, capture complex nonlinear relationships, and provide feature importance scores makes it well suited for medical classification tasks. Furthermore, LightGBM offers fast training, efficient memory usage, and strong predictive performance, making it an appropriate choice for developing a robust HPV status classification model.

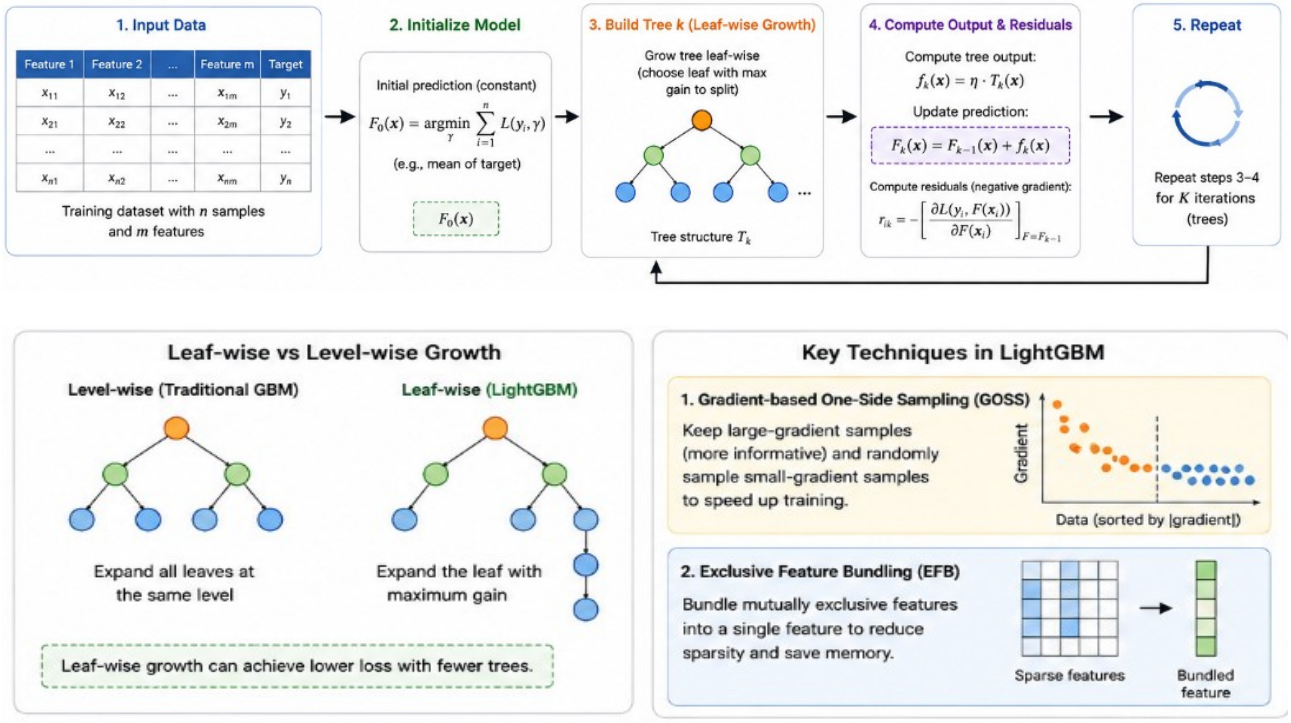


Figure 30 lightgbm workflow

WORK APPLIED :

Classification Experiments

After completing the data preprocessing stage, the classification phase was performed using the LightGBM algorithm. The model was trained using the training set, optimized and evaluated on the validation set, and finally assessed on the independent test set to ensure an unbiased evaluation of its predictive performance.

Three classification experiments were designed to evaluate the contribution of different feature sets to HPV status prediction:

Experiment 1: Clinical features only.

Experiment 2: PET radiomic features only.

Experiment 3: Combined clinical and PET radiomic features.

Following the implementation of these experiments, their performance will be analyzed and compared using several evaluation metrics. This comparison will help determine which feature set provides the highest predictive performance and whether combining clinical and PET radiomic features offers an improvement over using each data source individually.

RESULTS:

OUTCOME	CLINICAL ONLY		PET ONLY		COMBINED	
	VALIDATION	TEST	VALIDATION	TEST	VALIDATION	TEST
ROC-AUC	0.9420	0.9404	0.9004	0.5491	0.974638	0.856683
ACCURACY	0.9351	0.8974	0.8961	0.8718	0.922078	0.935897
BALANCED ACCURACY	0.7428	0.7488	0.5553	0.5411	0.680254	0.770531
SENSITIVITY	0.9855	0.9420	0.9855	0.9710	0.985507	0.985507
SPECIFICTY	0.5000	0.5556	0.1250	0.1111	0.375000	0.555556
PRECISION	0.9444	0.9420	0.9067	0.8933	0.931507	0.944444
F1 – SCORE	0.9645	0.9420	0.9444	0.9306	0.957746	0.964539
PR- AUC	0.9933	0.992	0.9880	0.9061	0.997189	0.975493
BRIER SCORE	0.0626	0.0784	0.0850	0.1202	0.057970	0.065225

Table 2 lightgbm metrics table

Clinical-only interpretation

The clinical-only model performed strongly in both validation and test.

The ROC-AUC stayed almost the same:

- Validation ROC-AUC = 0.942
- Test ROC-AUC = 0.940

This is a good sign because it means the model's ability to rank patients stayed stable when moving from validation to test.

Accuracy decreased from 0.935 on validation to 0.897 on test. This means the test set was slightly harder for the model, or the model made more threshold-based classification mistakes on the test patients.

Sensitivity was very high in both sets:

- Validation sensitivity = 0.986
- Test sensitivity = 0.942

This means the model was very good at detecting HPV positive patients. On validation, it missed only 1 HPV positive patient. On test, it missed 4 HPV positive patients.

Specificity was lower:

- Validation specificity = 0.500
- Test specificity = 0.556

This means the model had more difficulty detecting HPV negative patients. On validation, it correctly identified 4 out of 8 HPV negative patients. On test, it correctly identified 5 out of 9 HPV negative patients.

Overall:

The clinical-only model was strong and stable. It was very good for HPV positive detection and had excellent ROC-AUC and PR-AUC. Its main weakness was moderate specificity, meaning it sometimes predicted HPV negative patients as HPV positive.

PET-only interpretation

The PET-only model had a clear difference between validation and test.

On validation, ROC-AUC was strong:

- Validation ROC-AUC = 0.900

But on test, ROC-AUC dropped strongly:

- Test ROC-AUC = 0.549

This is close to random ranking. It means that on the test set, the PET-only model was not good at ranking HPV positive patients above HPV negative patients.

Accuracy looked high:

- Validation accuracy = 0.896
- Test accuracy = 0.872

But this is misleading because the dataset has many HPV positive patients. The model predicted most patients as HPV positive, so it got many positive patients correct but failed on most negative patients.

This is shown by the specificity:

- Validation specificity = 0.125
- Test specificity = 0.111

These are very low compared with the ideal value of 1.0. On validation, the model correctly detected only 1 out of 8 HPV negative patients. On test, it correctly detected only 1 out of 9 HPV negative patients.

Sensitivity was very high:

- Validation sensitivity = 0.986
- Test sensitivity = 0.971

This means the model detected most HPV positive patients. But because specificity was very low, the model was biased toward predicting HPV positive.

Brier Score and Expected Calibration Error were also worse on test than on validation:

- Brier Score increased from 0.085 to 0.120
- Expected Calibration Error increased from 0.087 to 0.130

Because lower is better for both metrics, this means the PET-only probabilities were less reliable on the test set.

Overall:

The PET-only model did not generalize well to the test set. It was good at finding HPV positive patients, but it was weak at identifying HPV negative patients. The large ROC-AUC drop from validation to test suggests that PET radiomics features alone were not stable enough for this classification task in this split.

Clinical + PET interpretation

The combined model performed very well overall, especially on the test set.

Validation ROC-AUC was excellent:

- Validation ROC-AUC = 0.975

Test ROC-AUC was lower:

- Test ROC-AUC = 0.857

This means the model ranked patients extremely well on validation, but less perfectly on test. Still, a test ROC-AUC of 0.857 is good and clearly better than random.

Accuracy improved from validation to test:

- Validation accuracy = 0.922
- Test accuracy = 0.936

This means the combined model classified the test patients slightly better than the validation patients using the 0.5 threshold.

Balanced accuracy also improved:

- Validation balanced accuracy = 0.680
- Test balanced accuracy = 0.771

This is important because balanced accuracy considers both HPV positive and HPV negative patients. The improvement happened because the model handled HPV negative patients better on test than on validation.

Sensitivity was excellent and stable:

- Validation sensitivity = 0.986

- Test sensitivity = 0.986

In both validation and test, the model missed only 1 HPV positive patient.

Specificity improved:

- Validation specificity = 0.375

- Test specificity = 0.556

On validation, the model correctly detected 3 out of 8 HPV negative patients. On test, it correctly detected 5 out of 9 HPV negative patients.

The Brier Score stayed low:

- Validation Brier Score = 0.058

- Test Brier Score = 0.065

This means the predicted probabilities were fairly good, although not perfect.

Expected Calibration Error stayed similar and slightly improved:

- Validation ECE = 0.072

- Test ECE = 0.069

Because lower is better, this suggests the test probabilities were slightly more calibrated than validation probabilities.

Overall:

The combined model was strong. It had the best test accuracy, best test balanced accuracy, best test sensitivity, best test F1-score, and good calibration. Its main weakness is that test ROC-AUC was lower than validation ROC-AUC and lower than the clinical-only test ROC-AUC.

FINAL INTERPRETATION :

The ideal model would have values close to 1.0 for ROC-AUC, accuracy, balanced accuracy, sensitivity, specificity, precision, F1-score, and PR-AUC, while having values close to 0.0 for Brier Score and Expected Calibration Error.

In our results, no model was perfect. All models were better at detecting HPV positive patients than HPV negative patients because the dataset had many more HPV positive cases.

The clinical-only model gave the best probability ranking on the test set.

The PET-only model struggled the most, especially with HPV negative patients.

The clinical + PET model gave the strongest overall classification result on the test set, but adding PET features did not improve every metric compared with clinical-only.

VISUALIZATION OUTPUTS:

CLINICAL:

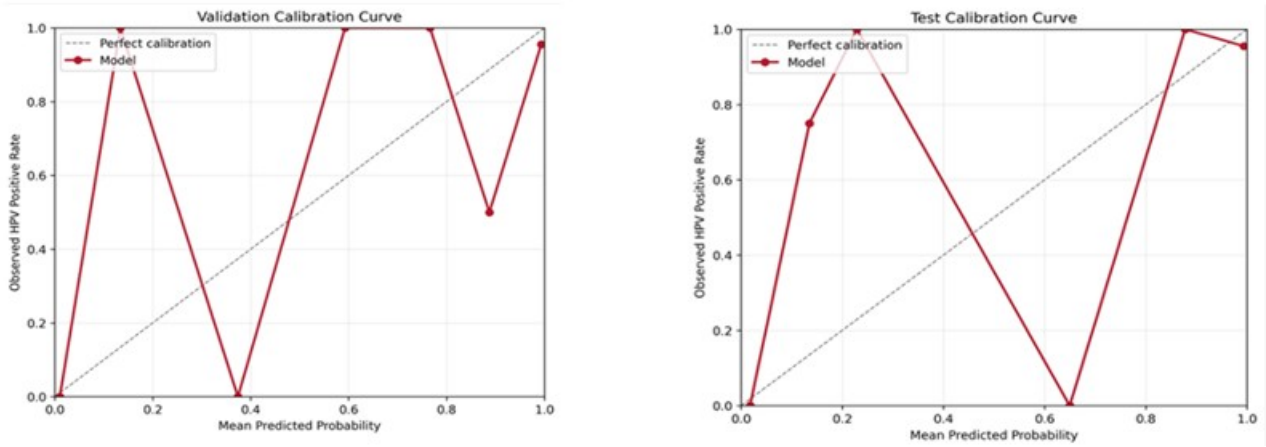


Figure 31 lightgbm clinical validation curve val/test

Validation Calibration Curve

The model's predicted probabilities do not consistently follow the perfect calibration line.

Some probability ranges are overestimated, while others are underestimated.

This indicates that the model's confidence is not perfectly calibrated on the validation set.

Test Calibration Curve

The calibration on the test set shows a similar pattern to the validation set.

The model is well calibrated for some probability ranges, particularly at high predicted probabilities, but deviates from the ideal line in the middle range.

Overall, the model demonstrates moderate calibration, although some prediction probabilities could be better aligned with the observed outcomes.

PET:

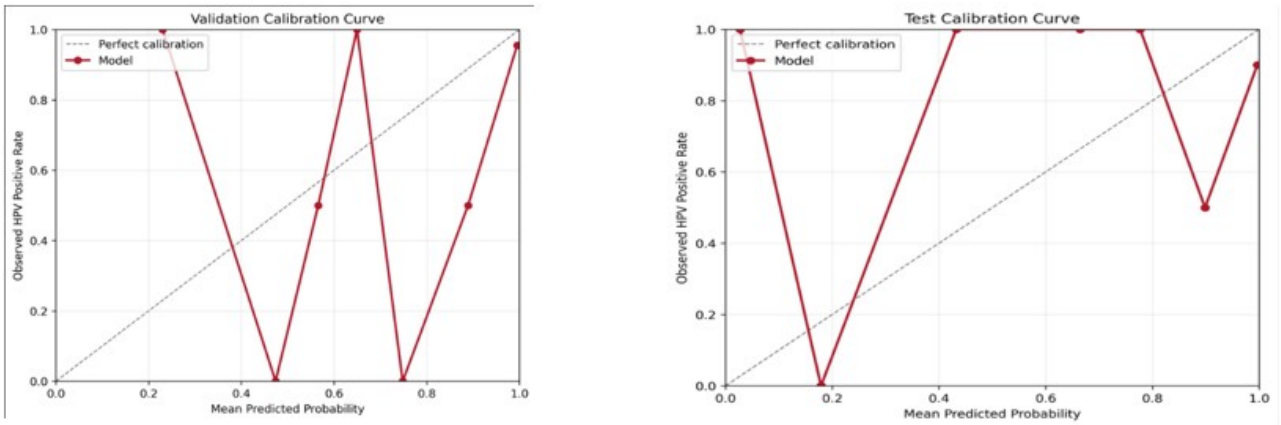


Figure 32 lightgm pet calibration curve val/test

Validation Calibration Curve (PET Features)

- The model's predicted probabilities show noticeable deviations from the perfect calibration line.
- Some probability ranges are overestimated, while others are underestimated.
- This indicates that the model's confidence is not consistently calibrated on the validation set.

Test Calibration Curve (PET Features)

- The test calibration curve follows a similar trend, with the model performing well at high predicted probabilities.
- However, deviations from the ideal calibration line are observed at lower and intermediate probability ranges.
- Overall, the model demonstrates acceptable but imperfect calibration on the test set.

MERGED:

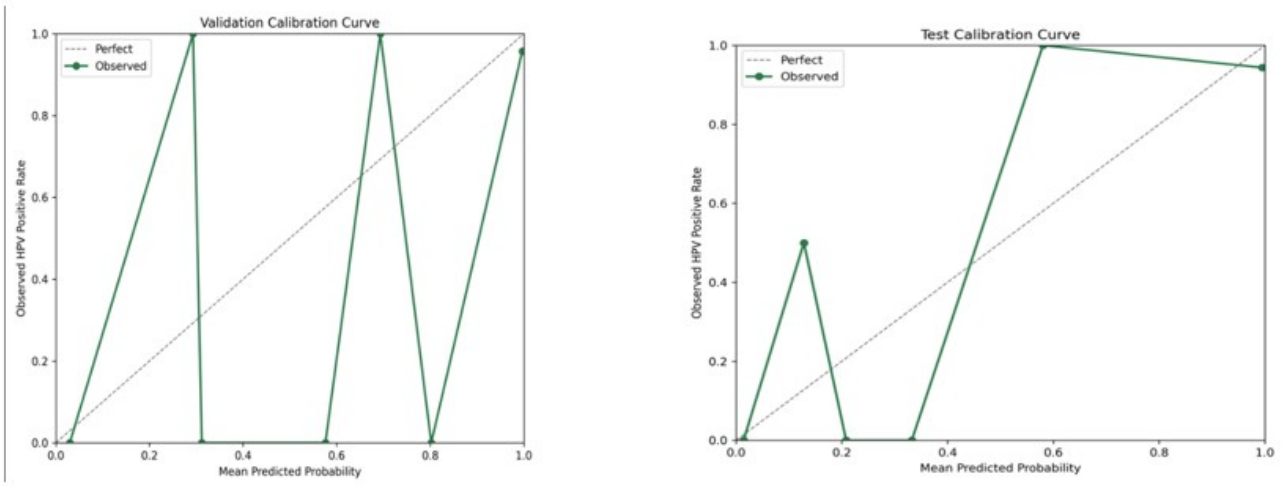


Figure 33 merged lightgbm calibration curve val/test

The calibration curves indicate that the combined Clinical + PET model provides more reliable probability estimates, particularly for high-confidence predictions. Although some fluctuations remain at lower and intermediate probability ranges, the model generally follows the ideal calibration trend better than the individual feature-set models, suggesting that combining clinical and PET radiomic features improves the reliability of the predicted probabilities.

CLINICAL:

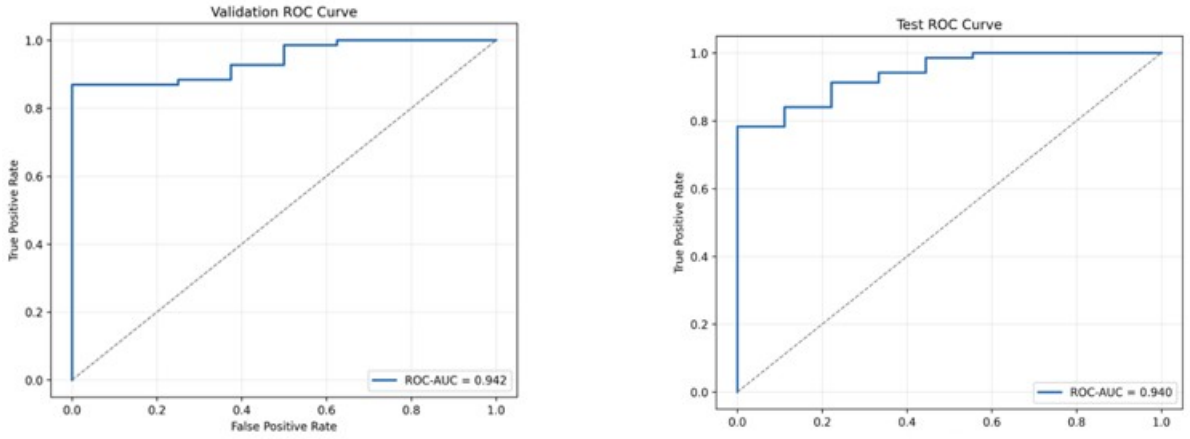


Figure 34 lightgbm clinical roc curve val/test

Validation ROC Curve

- The model achieved an **AUC of 0.942**, indicating **excellent classification performance**.
- The ROC curve remains close to the **top-left corner**, demonstrating a high true positive rate with a low false positive rate.
- This suggests that the model effectively distinguishes between HPV-positive and HPV-negative patients on the validation set.

Test ROC Curve

- The model achieved an **AUC of 0.940**, which is very similar to the validation performance.
- The ROC curve also lies close to the **top-left corner**, indicating strong discrimination between the two classes.
- The comparable AUC values on the validation and test sets suggest that the model generalizes well to unseen data.

PET:

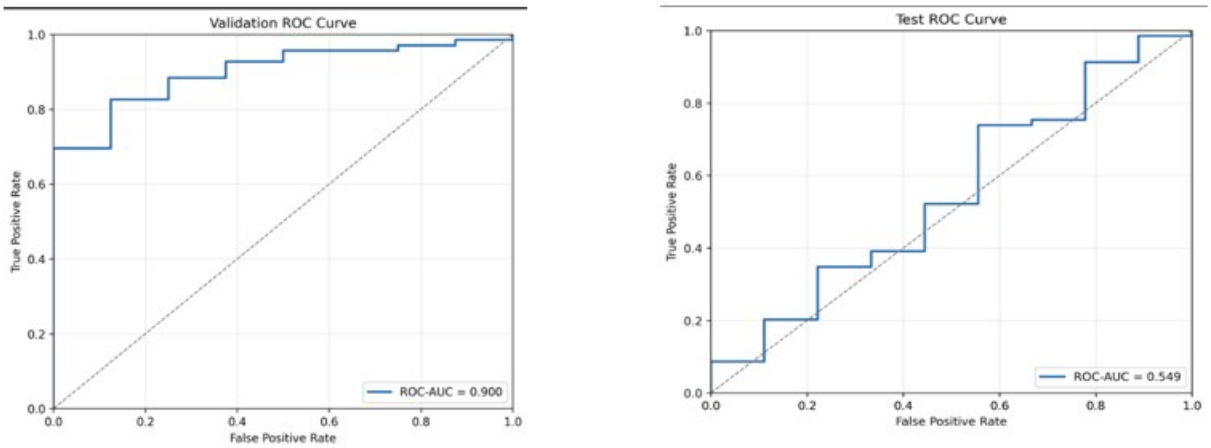


Figure 35 lightgbm pet roc curve val/test

Validation ROC Curve (PET Features)

- The model achieved an **AUC of 0.900**, indicating **good classification performance** on the validation set.
- The ROC curve is positioned well above the diagonal reference line, showing that the model can effectively distinguish between HPV-positive and HPV-negative patients during validation.
- This suggests that the model learned meaningful patterns from the PET radiomic features.

Test ROC Curve (PET Features)

- The model achieved an **AUC of 0.549**, indicating **poor classification performance** on the independent test set.
- The ROC curve lies close to the diagonal line, meaning the model performs only slightly better than random guessing.
- This suggests that the predictive ability observed during validation did not generalize well to unseen data.

MERGED:

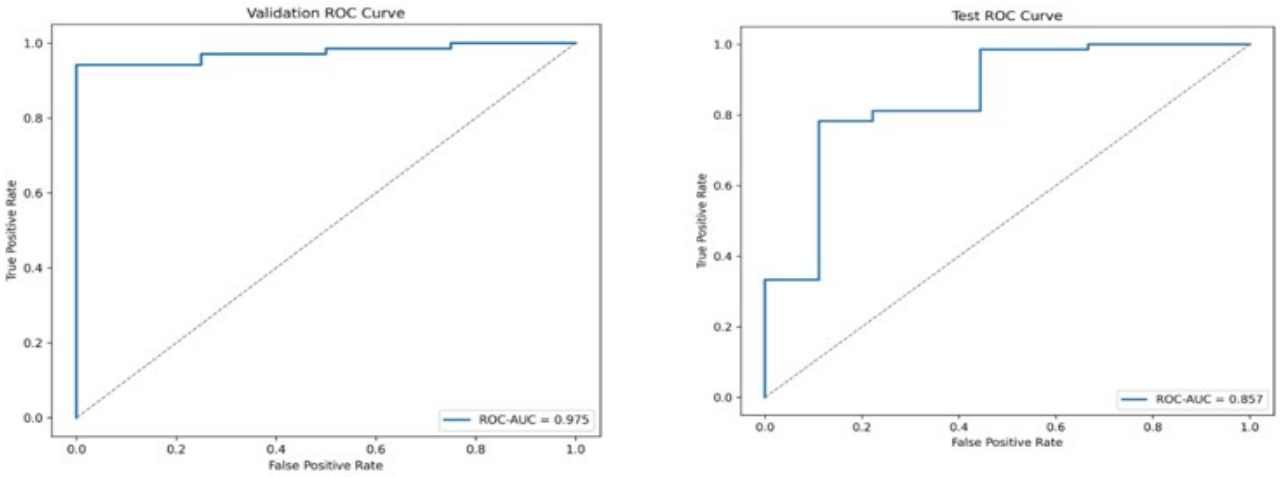


Figure 36 lightgbm merged roc curve val/test

Validation ROC Curve (Combined Clinical + PET Features)

The model achieved an AUC of 0.975, indicating outstanding classification performance on the validation set.

The ROC curve is very close to the top-left corner, demonstrating excellent discrimination between HPV-positive and HPV-negative patients.

This indicates that the model accurately identifies both classes with a high true positive rate and a low false positive rate.

Test ROC Curve (Combined Clinical + PET Features)

The model achieved an AUC of 0.857, indicating good classification performance on the independent test set.

The ROC curve remains well above the diagonal reference line, showing that the model generalizes well to unseen data.

Although the test performance is slightly lower than the validation performance, it still demonstrates strong discriminative ability.

CLINICAL:

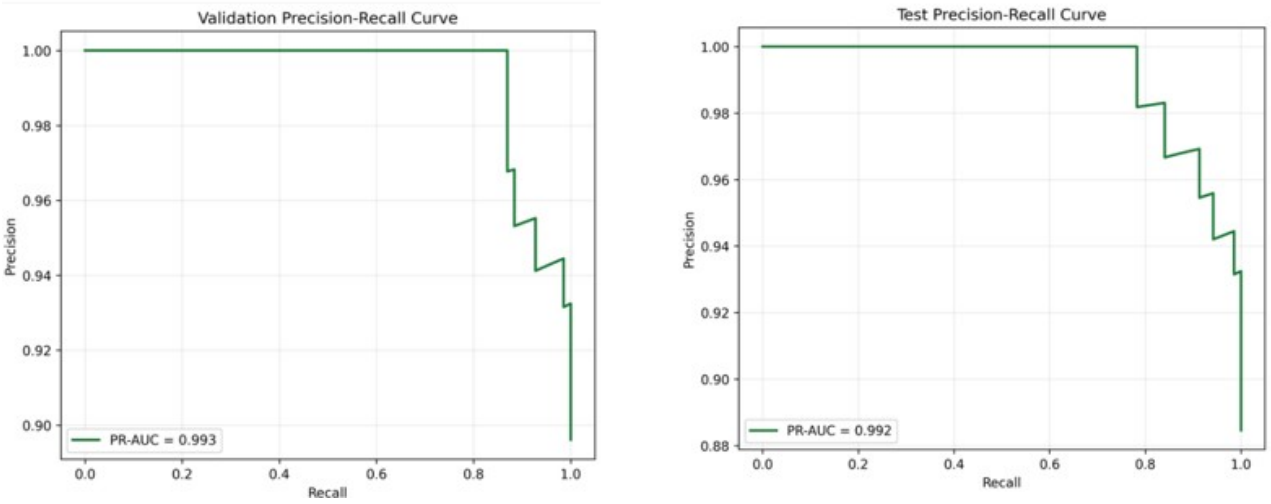


Figure 37 lightgbm clinical pr-auc val/test

Validation Precision–Recall Curve

The model achieved a PR-AUC of 0.993, indicating excellent precision–recall performance on the validation set.

Precision remains close to 1.0 across most recall values, showing that the model correctly identifies HPV-positive patients with very few false positives.

A slight decrease in precision is observed only at very high recall values.

Test Precision–Recall Curve

The model achieved a PR-AUC of 0.992, demonstrating excellent performance on the independent test set.

Precision remains consistently high throughout most of the recall range, with only a minor decline near maximum recall.

This indicates that the model maintains a high level of accuracy while successfully identifying the majority of positive cases.

PET:

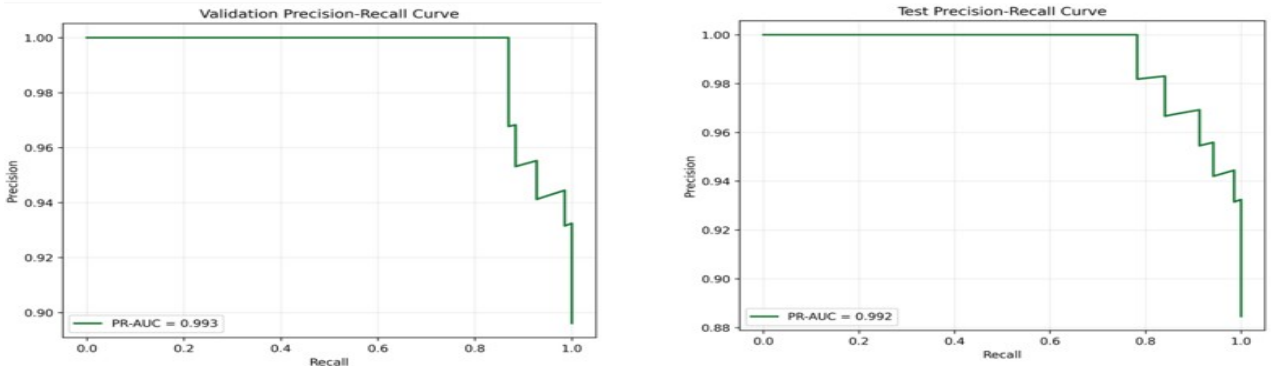


Figure 38 lightgbm pr-auc pet val/test

Validation Precision–Recall Curve (Clinical Features)

- The model achieved a **PR-AUC of 0.993**, indicating **excellent precision–recall performance** on the validation set.
- Precision remains close to **1.0** across most recall values, showing that the model correctly identifies HPV-positive patients with very few false positives.
- A slight decrease in precision is observed only at very high recall values.

Test Precision–Recall Curve (Clinical Features)

- The model achieved a **PR-AUC of 0.992**, demonstrating **excellent performance** on the independent test set.
- Precision remains consistently high throughout most of the recall range, with only a minor decline near maximum recall.
- This indicates that the model maintains a high level of accuracy while successfully identifying the majority of positive cases.

MERGED:

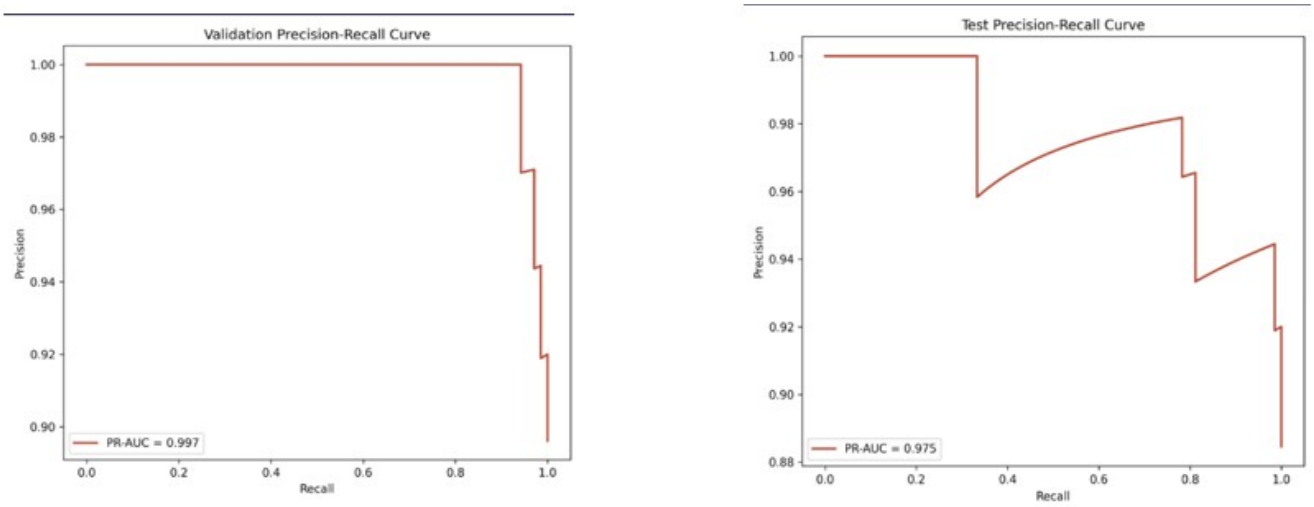


Figure 39 lightgbm merged pr-auc val/test

The Precision–Recall curves show that the combined Clinical + PET LightGBM model provides excellent predictive performance, achieving PR-AUC values of 0.997 on the validation set and 0.975 on the test set. The high and consistent precision across both datasets indicates that combining clinical and PET radiomic features enables the model to accurately identify HPV-positive patients while minimizing false-positive predictions. These results demonstrate strong

generalization and confirm the effectiveness of integrating clinical and imaging information for HPV status classification.

CLINICAL:

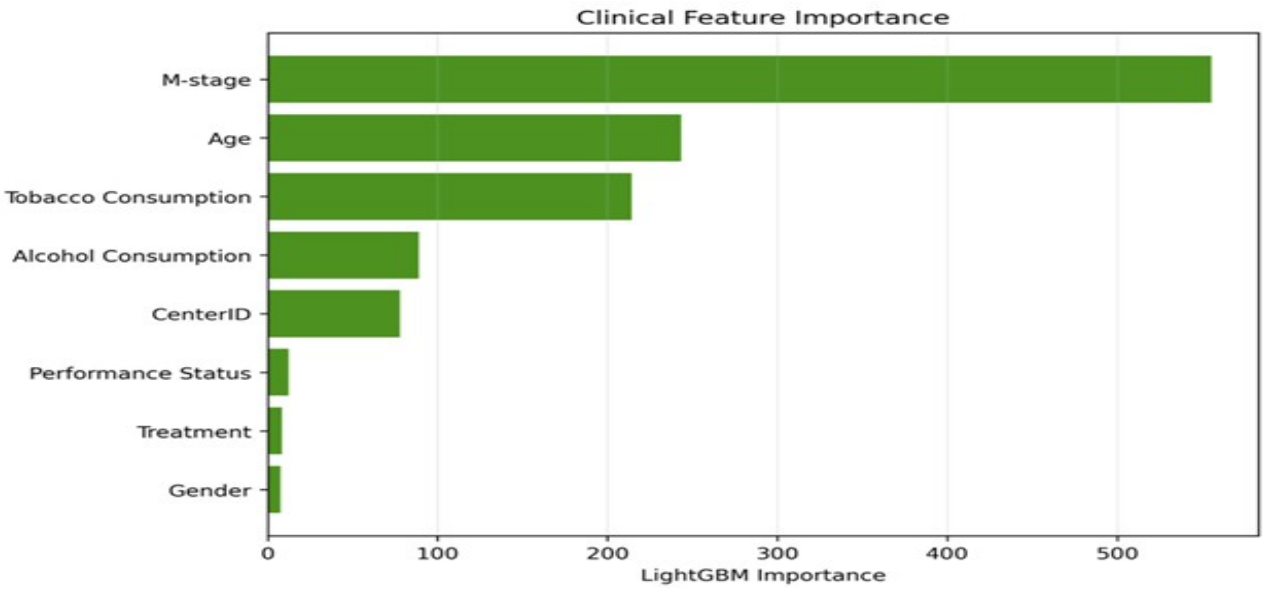


Figure 40 lightgbm clinical features importance

M-stage was the most important feature by far. It represented about 46 percent of the total gain importance in the clinical-only model.

Age and Tobacco Consumption were also important. Together, M-stage, Age, and Tobacco Consumption carried most of the model's useful clinical information. Treatment, Gender, and Performance Status had much smaller importance. This does not mean they are useless medically. It only means this LightGBM model used them less for prediction in this dataset.

PET:

The PET-only model used a mix of shape features and texture features

The top feature was Maximum2DDiameterSlice, a shape-related measurement. Several

other top features were also shape-related, such as Sphericity and Elongation.

Texture features from GLCM and GLRLM groups were also important. This means the

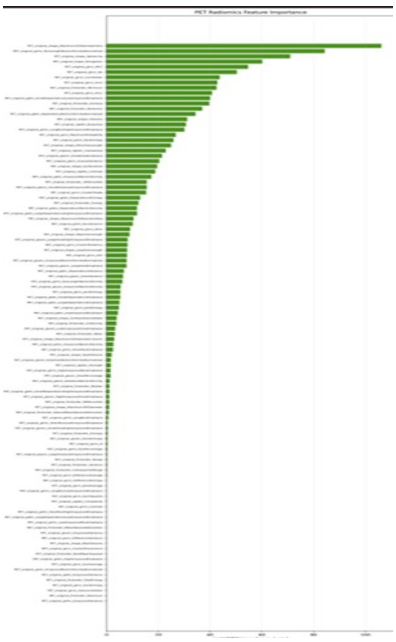


Figure 41 lightgbm pet features importance

model tried to use both tumor shape and image texture patterns.

However, even though the model found important PET features, the test results showed poor separation between HPV negative and HPV positive patients. So these PET features were not enough by themselves to generalize well.

MERGED:

Interpretation:

The combined model still depended strongly on clinical information. M-stage was the most important feature, with about 35.5 percent of total gain importance.

The top 4 features were clinical:

- M-stage
- CenterID
- Tobacco Consumption
- Alcohol Consumption

PET features also appeared in the top 10, especially shape features such as Elongation, Sphericity, Maximum2DDiameterSlice, and LeastAxisLength.

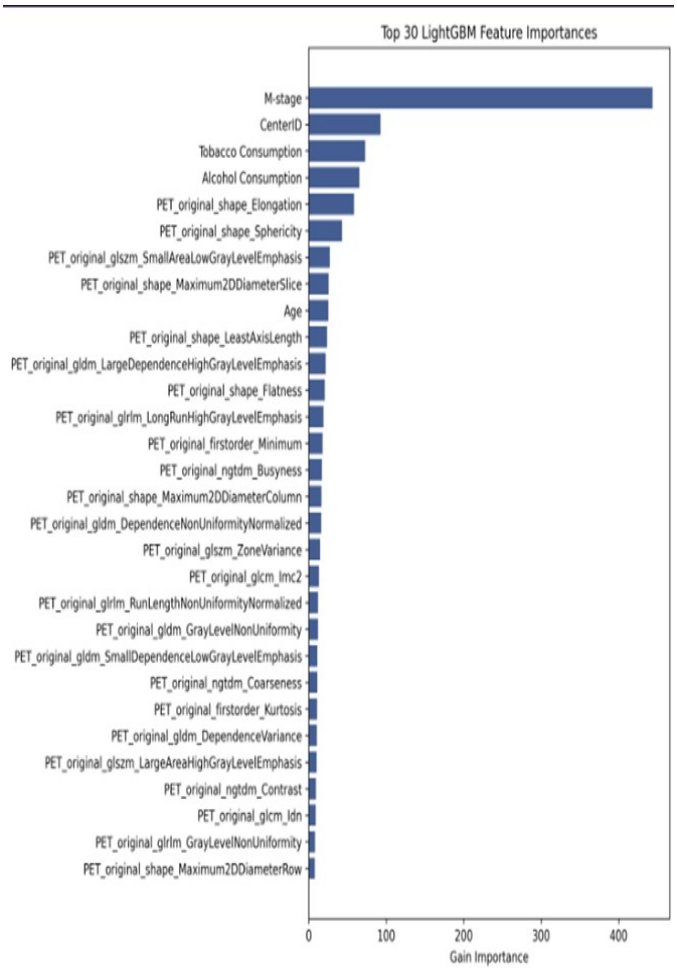


Figure 42 lightgbm merged features importance

This means the combined model used PET information, but the strongest signal was still clinical.

CLINICAL:

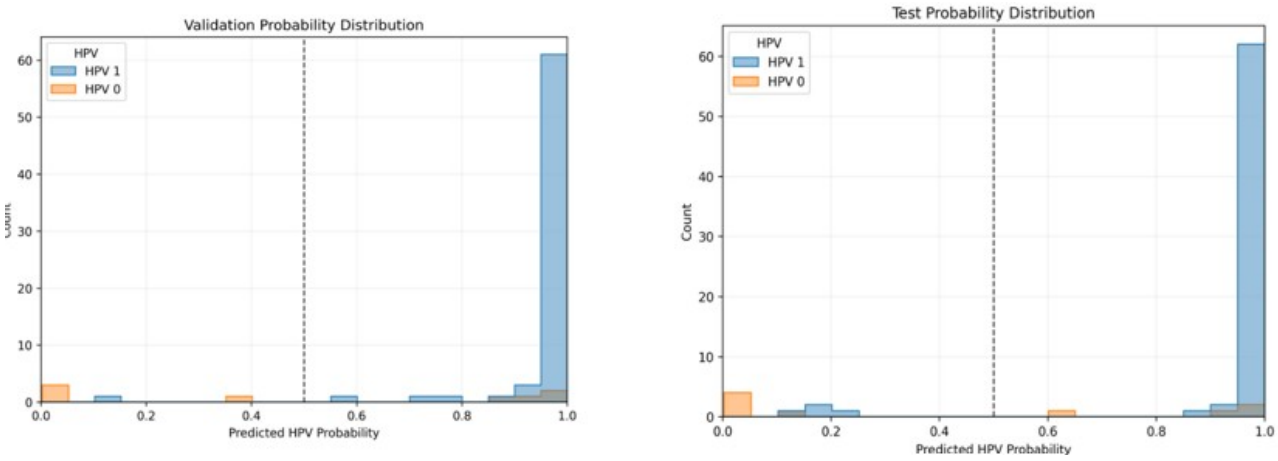


Figure 43 lightgbm clinical probability distribution val/test

The clinical-only probability distribution was strong for HPV positive patients.

Most true HPV positive patients were near probability 1.0.

The weakness was the HPV negative group. In validation, half of the HPV negative patients were still predicted as HPV positive. In test, 4 out of 9 HPV negative patients were predicted as HPV positive.

So the plot probably shows a clear cluster of HPV positive patients near 1.0, but also some HPV negative patients overlapping with that same high-probability region.

PET:

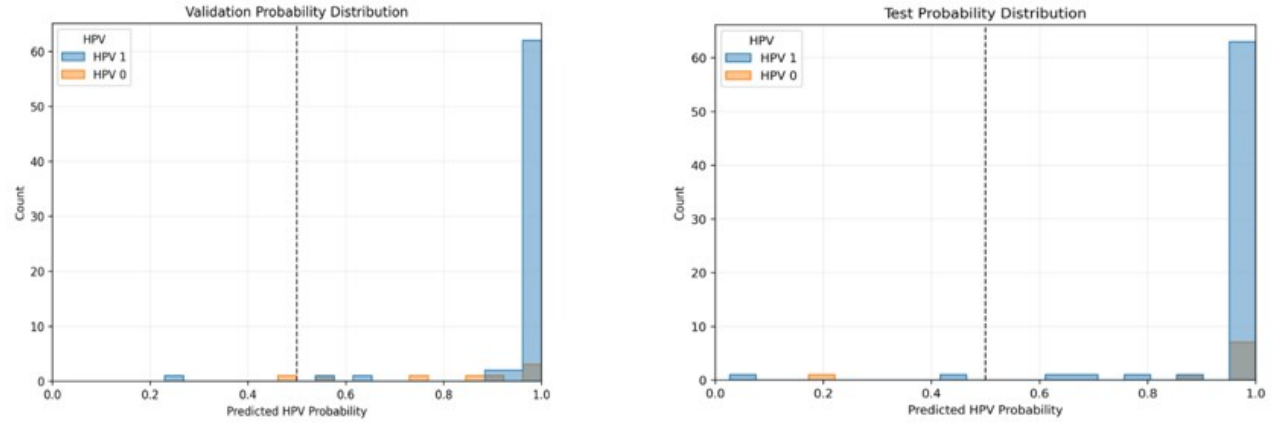


Figure 44 lightgbm pet probability distribution val/test

The PET-only probability distribution shows the main weakness of the PET-only model. It gave high HPV-positive probabilities to almost everyone.

For the test set, 7 out of 9 real HPV negative patients had probability at least 0.9. This means the model was very confident but wrong for many negative Patients.

So the probability distribution likely shows heavy overlap near 1.0 between HPV negative and HPV positive patients. This explains the very low specificity.

MERGED:

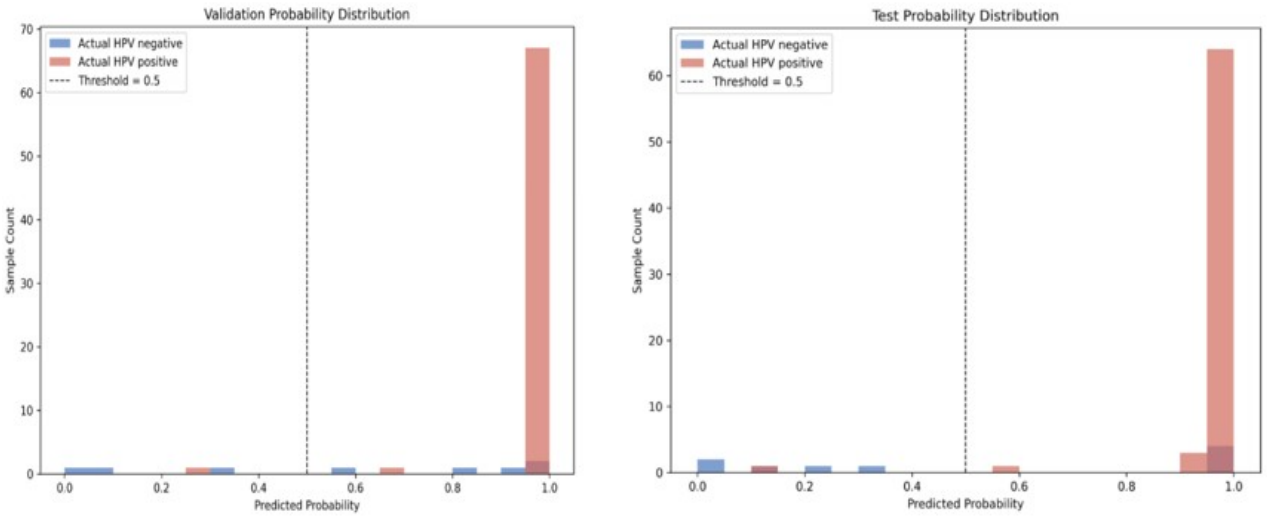


Figure 45 lightgbm merged probability distribution val/test

The combined probability distribution was strong for HPV positive patients. Most true HPV positive patients were placed very close to probability 1.0.

For HPV negative patients, the model improved on test compared with validation. On test, 5 out of 9 negatives were below the 0.5 threshold. Still, 4 out of 9 negatives were given high probabilities and predicted as HPV positive. This means the combined model was excellent for detecting HPV positive patients, but still not perfect at separating HPV negative patients.

CONFUSION MATRIX :

	Validation	Test
Clinical Features	**TN = 4, FP = 4, FN = 1, TP = 68**	**TN = 5, FP = 4, FN = 4, TP = 65**
PET FEATURES	**TN = 1, FP = 7, FN = 1, TP = 68**	**TN = 1, FP = 8, FN = 2, TP = 67**
MERGED	**TN = 3, FP = 5, FN = 1, TP = 68**	**TN = 5, FP = 4, FN = 1, TP = 68**

- **The clinical model correctly classified most HPV-positive patients, with a high number of true positives and few false negatives. However, it showed moderate difficulty in correctly identifying HPV-negative patients due to several false-positive predictions.**
- **The PET-only model achieved excellent detection of HPV-positive patients but performed poorly in identifying HPV-negative patients, resulting in a large number of false positives and very few true negatives.**
- **The combined model provided the best overall balance between detecting HPV-positive and HPV-negative patients. It maintained a high number of true positives while reducing false negatives and correctly classifying more HPV-negative patients than the PET-only model, demonstrating improved overall classification performance.**

clinical:

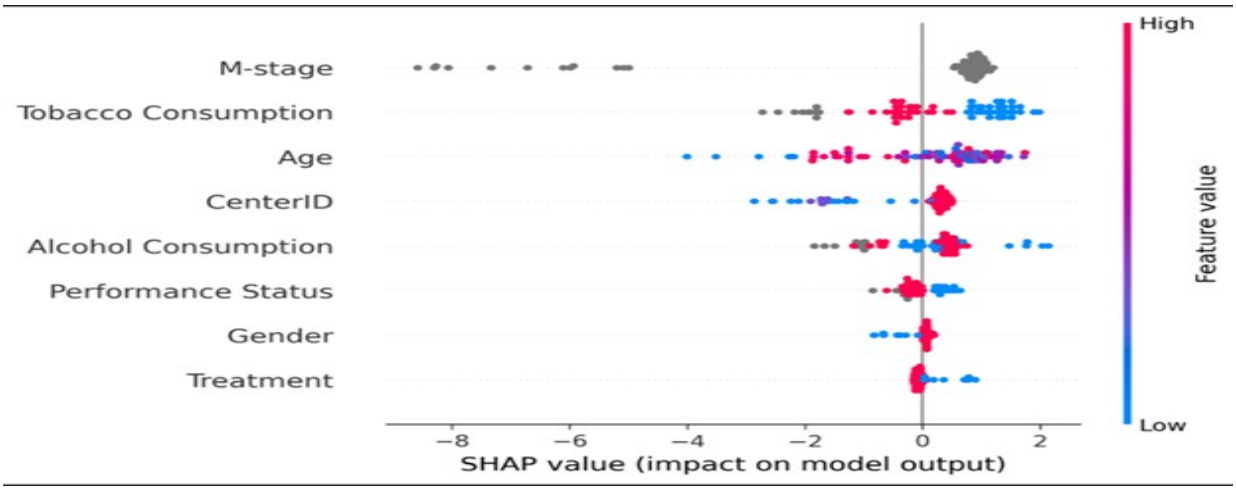


Figure 46 lightgbm clinical shape value val/test

Clinical Features Only – SHAP Summary

M-stage was the most influential feature.

Tobacco Consumption and Age also had a strong impact.

CenterID and Alcohol Consumption contributed moderately.

Performance Status, Gender, and Treatment had the least influence.

Interpretation: The model mainly relied on a few important clinical features, with M-stage being the strongest predictor of HPV status.

PET:



Figure 48 lightgbm pet shape value val/test

COMBINED:

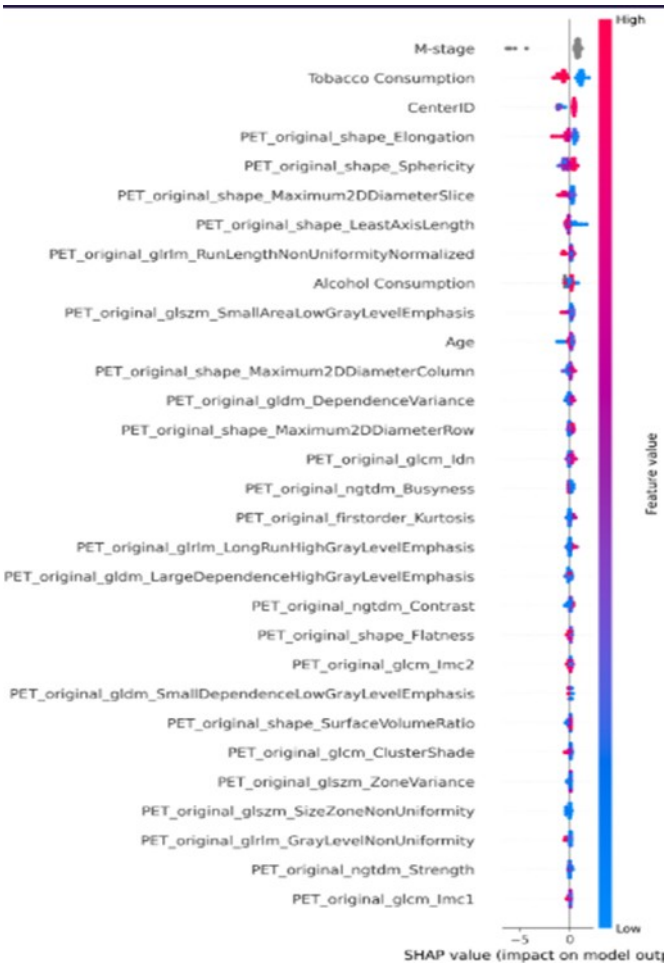


Figure 47 lightgbm mergedshape value val/test

PET Features Only – SHAP Summary

- **Maximum2DDiameterSlice** was the most influential PET feature.
- **Sphericity** was the second most important feature.
- Several **texture features (GLCM)** also contributed to the predictions.
- **Shape** and **texture** features were the main sources of information.

- **Interpretation:** The model mainly relied on PET radiomic shape and texture features, but these features alone were not sufficient to reliably distinguish HPV-positive from HPV-negative patients

Combined Clinical + PET Features – SHAP Summary

- **M-stage** remained the most influential feature.
- **Tobacco Consumption** and **CenterID** were also important.
- PET **shape features** such as **Elongation**, **Sphericity**, and **Maximum2DDiameterSlice** contributed to the predictions.
- Both **clinical** and **PET radiomic** features were used by the model.
- **Interpretation:** The combined model was mainly driven by clinical features, while PET radiomic features provided additional information that improved the final predictions.

CLINICAL:

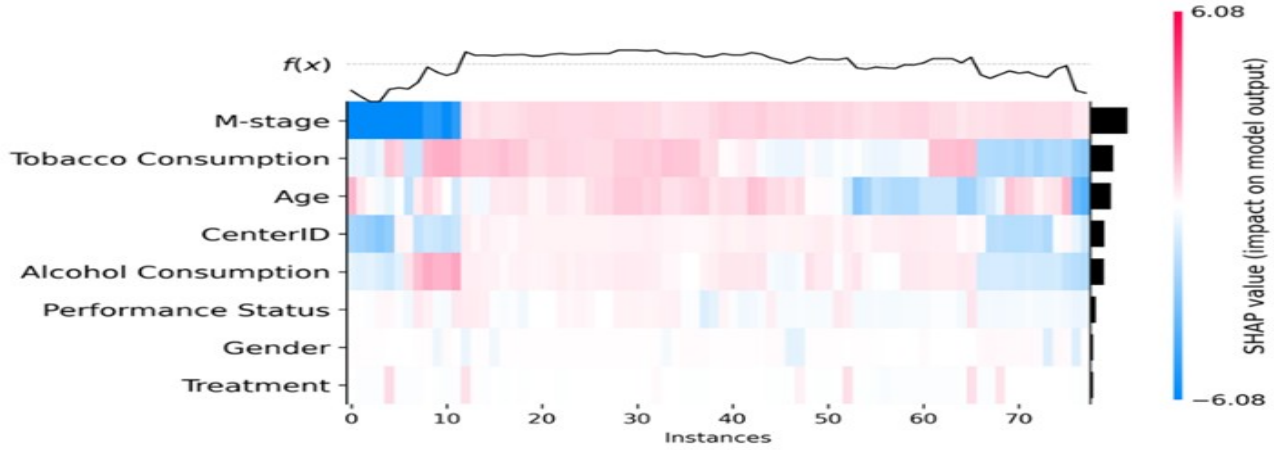


Figure 49 lightgbm clinical shap heatmap

Clinical Features Only – SHAP Heatmap

- M-stage had the strongest influence across most patients.
- Tobacco Consumption, Age, CenterID, and Alcohol Consumption also contributed to the predictions.
- The remaining clinical features had a smaller impact.
- Interpretation: The heatmap shows that the model relied mainly on a few key clinical features, especially M-stage, to classify HPV status.

PET:

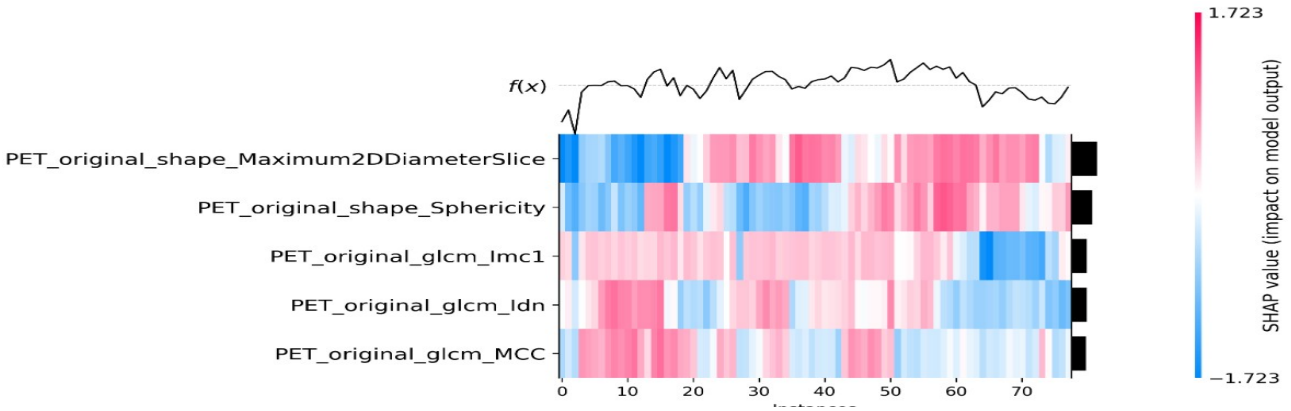


Figure 50 V lightgbm pet shap heatmap

PET Features Only – SHAP Heatmap

- Several **PET radiomic features** contributed to the predictions.
- Both **shape** and **texture** features influenced the model.
- Many patients received strong positive contributions from PET features.
- **Interpretation:** The heatmap shows that the model used multiple PET radiomic features, but these features did not clearly separate HPV-positive from HPV-negative patients.

MERGED:

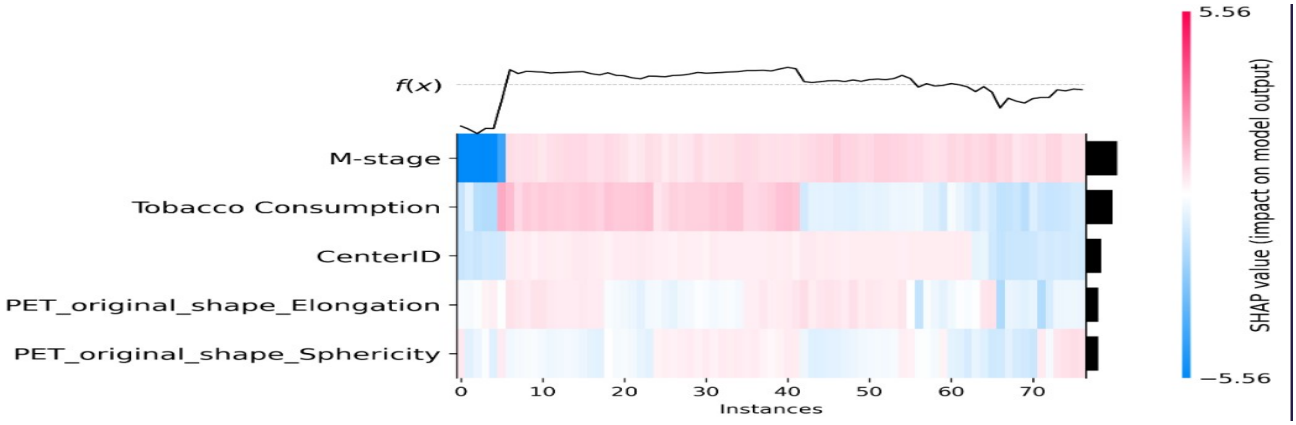
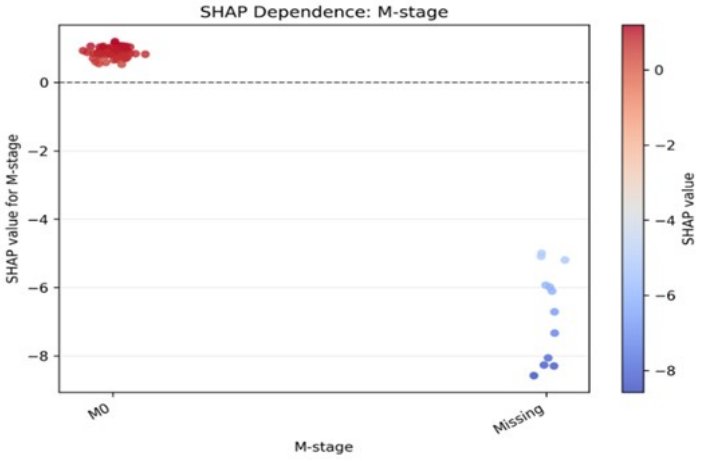


Figure 51 lightgbm merged shap heatmap

M-stage and Tobacco Consumption remained the most influential features.

- PET shape features provided additional positive or negative contributions depending on the patient.
- Both clinical and PET radiomic features contributed to the predictions.
- Interpretation: The heatmap shows that the combined model primarily relied on clinical information, while PET radiomic features refined the predictions by adding patient-specific information.



- **CLINICAL:**

Clinical Features Only – SHAP Dependence

- The dependence plot focused on **M-stage**.
- **M0** values increased the probability of predicting **HPV-positive**.
- **Missing M-stage** values strongly decreased the HPV-positive prediction score.
- **Interpretation:** M-stage had the strongest influence on the model, making it the main clinical feature driving HPV status prediction.

PET :

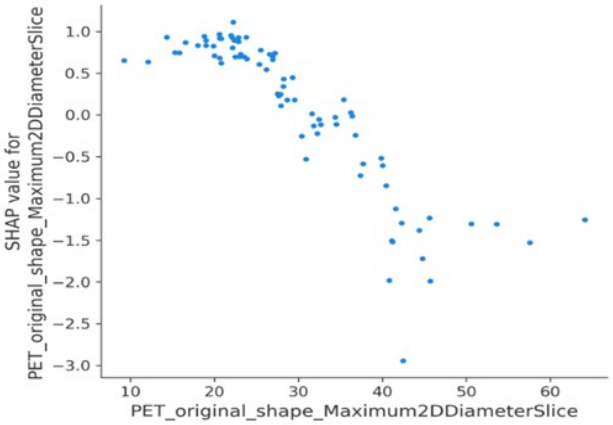


Figure 53 lightgbm pet shap dependence

PET Features Only – SHAP Dependence

- The dependence plot focused on PET_original_shape_Maximum2DDiameterSlice.
- As this feature increased, its SHAP value generally became more negative.
- This means larger values tended to decrease the probability of predicting HPV-positive.
- Interpretation: Although this was the most influential PET feature, it alone was not sufficient to achieve reliable HPV status classification

MERGED:

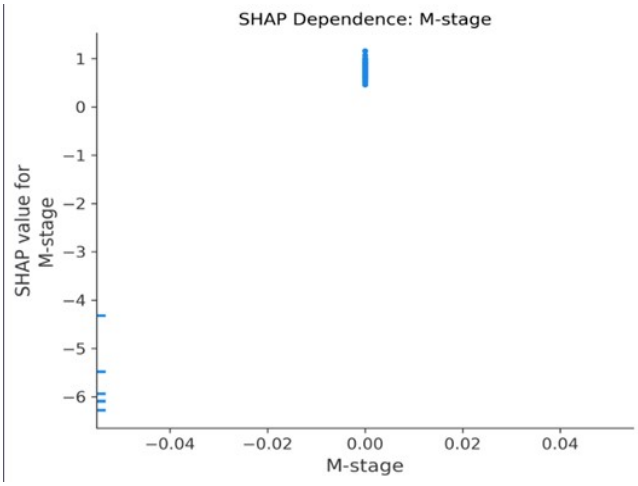


Figure 54 lightgbm merged shap dependence

Combined Clinical + PET Features – SHAP Dependence

- The dependence plot focused on M-stage.
- M0 values increased the probability of predicting HPV-positive.
- Missing M-stage values strongly decreased the HPV-positive prediction score.
- Interpretation: Even after combining PET radiomic and clinical features, M-stage remained the strongest factor influencing the model's predictions, confirming the dominant role of clinical information in HPV status classification.

FINAL INTERPRETATION:

Three LightGBM classification experiments were conducted to evaluate the ability of different feature sets to predict HPV status in head and neck cancer patients. The first experiment used clinical features only, the second used PET radiomic features only, and the third combined clinical and PET radiomic features. Their performances were assessed using multiple evaluation metrics, including ROC-AUC, Accuracy, Balanced Accuracy, Sensitivity, Specificity, Precision, F1-score, PR-AUC, Brier Score, Expected Calibration Error (ECE), confusion matrices, calibration curves, feature importance, and SHAP analysis.

The clinical-only model demonstrated stable and reliable performance, achieving the highest test ROC-AUC while relying mainly on important clinical variables such as M-stage, Tobacco Consumption, and Age. However, its specificity remained moderate, indicating some difficulty in identifying HPV-negative patients.

The PET-only model successfully identified most HPV-positive patients but showed poor generalization on the independent test set. It produced a large number of false-positive predictions and exhibited low specificity, demonstrating that PET radiomic features alone were insufficient for robust HPV status classification. The SHAP analysis revealed that the model mainly relied on tumor shape and texture features, particularly Maximum2DDiameterSlice and Sphericity.

The combined Clinical + PET model achieved the best overall classification performance. It obtained the highest test Accuracy, Balanced Accuracy, Sensitivity, Precision, F1-score, and the lowest Brier Score and Expected Calibration Error, indicating improved predictive performance and more reliable probability estimates. Although the model incorporated useful PET radiomic features, the SHAP analysis showed that clinical variables, particularly M-stage, remained the primary drivers of the predictions, while PET features provided complementary patient-specific information.

Overall, these experiments demonstrate that clinical information provides the strongest predictive signal for HPV status, while PET radiomic features alone are not sufficient to produce robust classification. However, integrating PET radiomic features with clinical data improves the overall classification performance and calibration of the model, resulting in a more reliable predictive framework. Based on these findings, the next stage of the project investigates whether a Multi-Layer Perceptron (MLP) neural network can further improve HPV status classification by learning more complex nonlinear relationships within the same datasets.

MLP DEEP LEARNING CLASSIFICATION:

Phase 3: Deep Learning Classification Using a Multi-Layer Perceptron (MLP)

Following the completion of the LightGBM experiments, the internship entered its third phase, which focused on **deep learning-based classification** using a **Multi-Layer Perceptron (MLP)** model. The objective was to investigate whether a neural network could learn more complex nonlinear relationships between the input features and improve the prediction of HPV status compared with the gradient boosting approach.

To ensure a fair comparison with the previous experiments, the same experimental design was maintained. The same **training, validation, and test datasets**, preprocessing pipeline, and evaluation metrics were used throughout the study. Three classification experiments were performed using the MLP model:

- **Clinical features only**
- **PET radiomic features only**
- **Combined Clinical + PET radiomic features**

Each experiment was trained and evaluated independently using the same input features as those employed in the LightGBM analysis. Model performance was assessed using the same evaluation metrics, including **ROC-AUC, Accuracy, Balanced Accuracy, Sensitivity, Specificity, Precision, F1-score, PR-AUC, Brier Score, Expected Calibration Error (ECE), confusion matrices, calibration curves, feature importance, and SHAP analysis**.

This phase aimed to determine whether the MLP architecture could better exploit the available clinical and radiomic information, improve the classification of HPV status, and provide additional insights into the contribution of each feature set to the prediction task.

How MLP works?

A Multi-Layer Perceptron (MLP) is one of the most basic and widely used types of artificial neural networks. It consists of multiple layers of interconnected neurons that learn patterns from data and can be used for classification or regression tasks.

An MLP typically contains:

1. Input Layer – Receives the input features (e.g., clinical variables or PET features).
2. Hidden Layer(s) – Perform computations and learn complex relationships within the data using weights, biases, and activation functions.
3. Output Layer – Produces the final prediction.

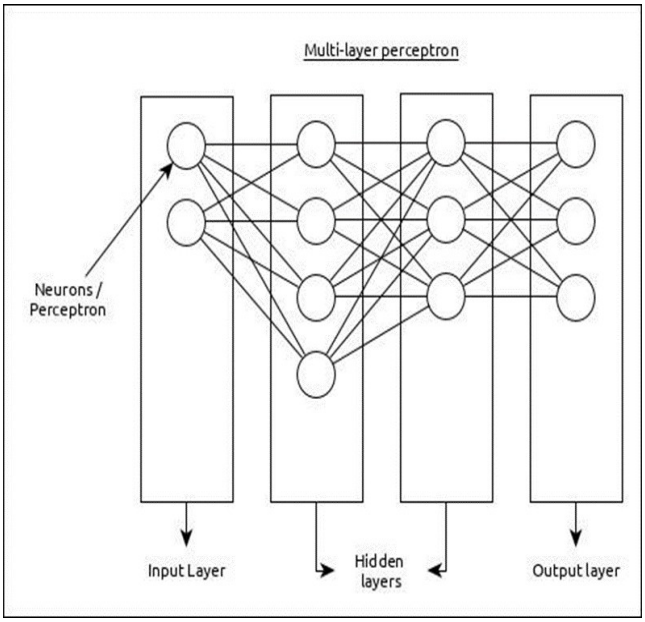


Figure 55 multi-layer perceptron

During training, the MLP:

- Receives input data.
- Computes predictions through forward propagation.
- Compares predictions with the true labels using a loss function.
- Updates weights and biases using backpropagation and an optimizer such as Adam.
- Repeats this process over multiple epochs to improve performance.

MLPs are capable of learning both simple and complex patterns in structured data and are commonly used for medical, financial, and classification applications.

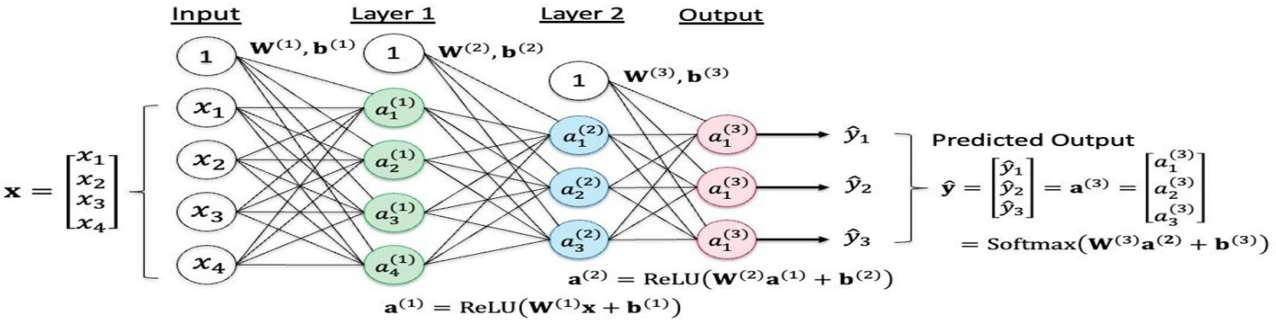


Figure 56 MLP workflow

DEEP LEARNING FUNDEMANTELS :

1. What Is Deep Learning?
Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple layers to learn complex patterns from data.

- Traditional Machine Learning
  - Logistic Regression
  - Decision Trees
  - Random Forest
  - LightGBM
- Deep Learning
  - Neural Networks
  - MLPs (Multi-Layer Perceptrons)
  - CNNs (Convolutional Neural Networks)
  - RNNs (Recurrent Neural Networks)
  - Transformers

The key idea is that instead of manually designing features, neural networks learn useful representations automatically.

- 2. The Linear Unit — Weights and Bias

The basic building block of a neural network is the linear unit, defined by four elements:

- x — the input
- w — the weight
- b — the bias
- y — the output

- What Is the Weight (w)?

The weight tells the neuron how important an input is. During training, the neural network learns the best weights.

- What Is the Bias (b)?

The bias is an extra value added at the end of the computation. It lets the neuron shift its output up or down.

Without bias, the neuron is too restricted. With bias, the neuron becomes more flexible and can learn better patterns.

Unlike weights, the number of biases is not learned during training — it is decided by the model designer when the network architecture is defined.

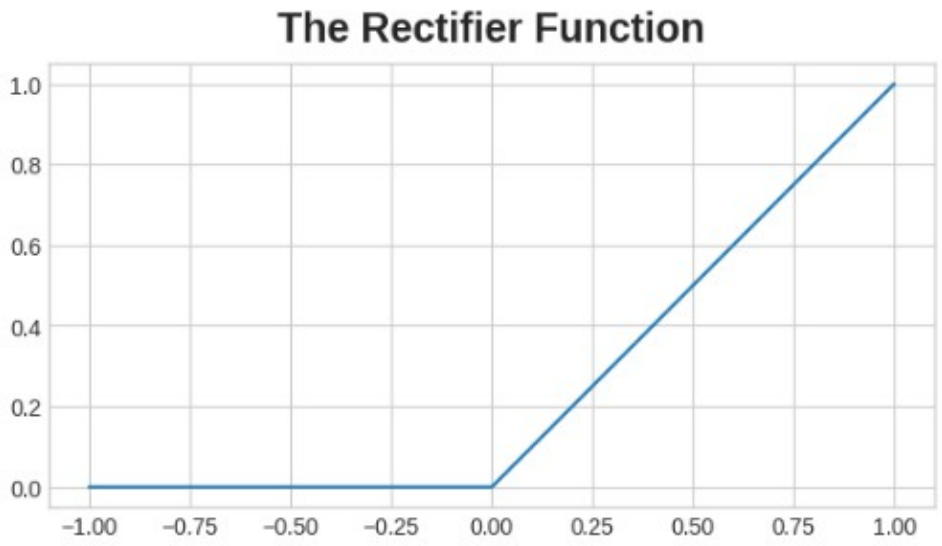


Figure 57 the rectifier function

The linear unit: input, weight, bias, and output.

- 3. From One Neuron to Multiple Inputs

A neuron with many inputs is just doing the following:

"Multiply each feature by its importance (weight), add them all up, add a bias, and produce a score."

That is the foundation of an MLP. The only difference is that an MLP contains many neurons working together, organized in multiple layers.

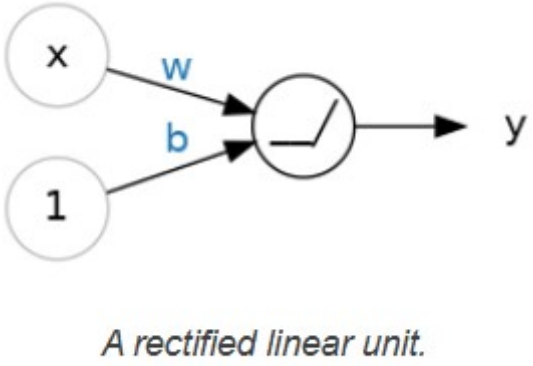


Figure 58 rectifier linear unit

A neuron receiving multiple inputs.

- 4. Building a Neuron and a Simple MLP in Keras

A single neuron can be created in Keras with a simple Dense layer. In practice this means:

3 inputs → 1 neuron → 1 output

Stacking several of these neurons and layers together produces a simple Multi-Layer Perceptron (MLP), where the only difference from a single neuron is the number of neurons and layers working together.

- 5. Layers

- Why Multiple Neurons?
  - One neuron can learn only one pattern.
  - Multiple neurons can learn multiple patterns simultaneously.

Each layer in a neural network performs a relatively simple transformation. Through a deep stack of layers, a neural network can transform its inputs in increasingly complex ways. In a well-trained neural network, each layer gets a little closer to the solution.

- 6. The Activation Function

Every neuron, on its own, computes:

$$y = wx + b$$

This is simply a straight-line equation. No matter how many layers are stacked, without an activation function the entire network remains equivalent to one single straight line.

An activation function is a function applied to each of a layer's outputs (its activations). The most common is the rectifier function (ReLU).

- The Rectifier Function (ReLU)

Before ReLU, a neuron computes:

$$y = wx + b$$

After ReLU is applied:

$$y = max(0, wx + b)$$

In words: "First calculate y. If the result is negative, replace it with 0."

- Why ReLU Is Powerful

Without ReLU, stacking linear layers collapses into one linear transformation:

- Input → Linear → Linear → Linear

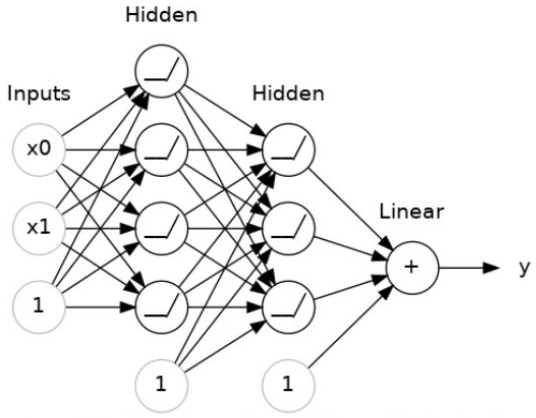
With ReLU inserted between linear layers, the network gains genuine expressive power:

- Input → Linear → ReLU → Linear → ReLU → Output

With this nonlinearity, the network can now learn:

- Curves
- Complex boundaries
- Nonlinear relationships

Now that we have some nonlinearity, let's see how we can stack layers to get complex data transformations.



A stack of dense layers makes a "fully-connected" network.

The layers before the output layer are sometimes called **hidden** since we never see their outputs directly.

Now, notice that the final (output) layer is a linear unit (meaning, no activation function). That makes this network appropriate to a regression task, where we are trying to predict some arbitrary numeric value. Other tasks (like classification) might require an activation function on the output.

Figure 59 stack of layers network

The ReLU activation function and its effect on a stack of layers.

- 7. Stacking Dense Layers

The layers before the output layer are sometimes called hidden layers, since their outputs are never seen directly.

The final (output) layer is typically a linear unit — meaning it has no activation function. This makes the network appropriate for a regression task, where the goal is to predict an arbitrary numeric value. Other tasks, such as classification, may require an activation function on the output layer.

- 8. The Loss Function
    - The loss function measures the disparity between the target's true value and the value predicted by the model.
    - A common loss function for regression problems is the Mean Absolute Error (MAE). For each prediction y_pred, MAE measures the disparity from the true target y_true using the absolute difference |y_true − y_pred|.
    - The total MAE loss on a dataset is the mean of all these absolute differences.

- 9. The Optimizer — Stochastic Gradient Descent

The optimizer is the algorithm that adjusts the weights to minimize the loss. Virtually all optimization algorithms used in deep learning belong to a family called stochastic gradient descent (SGD) — iterative algorithms that train a network in steps.

- One Training Step
    - Sample some training data and run it through the network to make predictions.
    - Measure the loss between the predictions and the true values.
    - Adjust the weights in a direction that makes the loss smaller.

This process repeats over and over until the loss is as small as desired, or until it no longer decreases.

- 10. Overfitting and Underfitting

When training a model, the loss on the training set is plotted epoch by epoch. Adding the validation loss to the same plot produces what are called learning curves — essential for interpreting how a deep learning model is training.

When a model learns signal, both curves go down. When it learns noise, a gap opens up between the curves — the size of the gap shows how much noise the model has learned.

Ideally, a model would learn all of the signal and none of the noise, but this practically never happens. Instead, there is a trade-off: the model can learn more signal at the cost of learning more noise. As long as the trade is favorable, validation loss keeps decreasing. Past a certain point the trade turns unfavorable, the cost exceeds the benefit, and validation loss begins to rise.

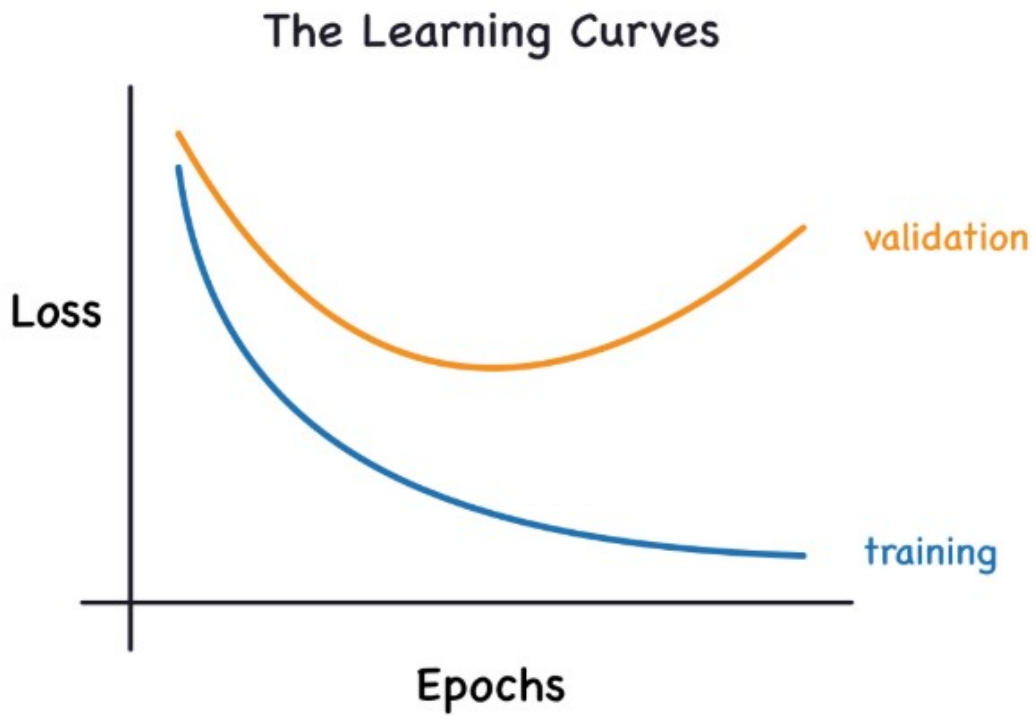


Figure 60 training loss

Learning curves: training loss versus validation loss.

- Two Failure Modes
  - Underfitting: the loss is not as low as it could be because the model has not learned enough signal.
  - Overfitting: the loss is not as low as it could be because the model has learned too much noise.

The trick to training deep learning models is finding the best balance between the two.

To prevent overfitting, training can simply be stopped once the validation loss stops decreasing — a technique called early stopping.

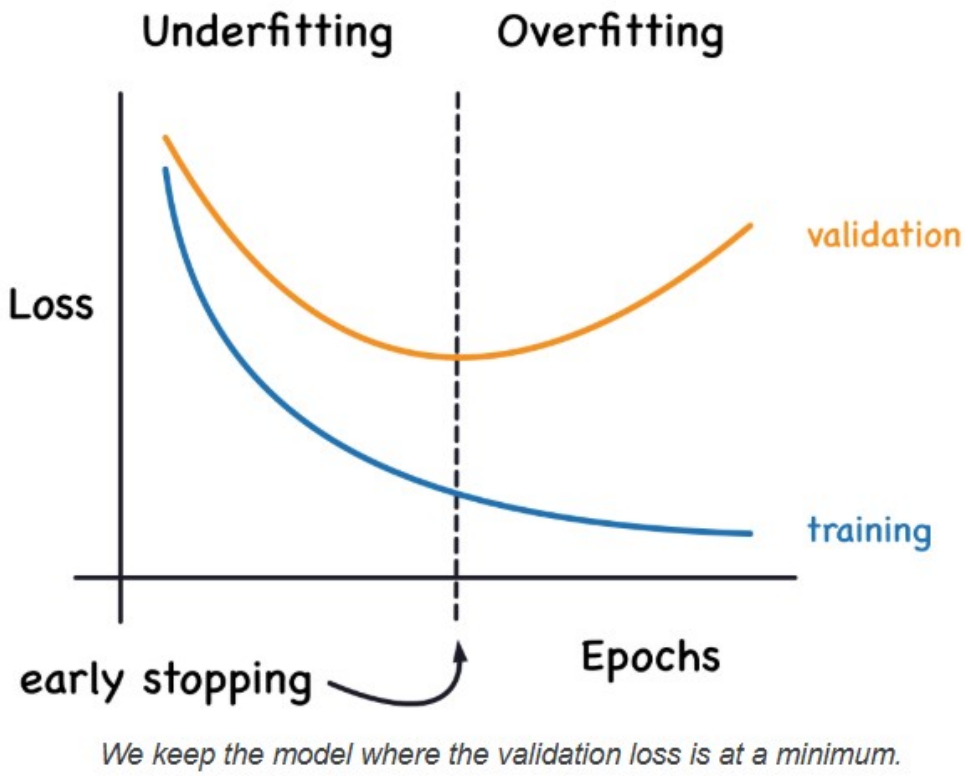


Figure 61 early stopping

Early stopping: halting training when validation loss stops improving.

- 11. Capacity

A network's capacity can be increased in two ways:

- Wider — adding more units to existing layers.
- Deeper — adding more layers.

Wider networks find it easier to learn more linear relationships, while deeper networks tend to prefer more nonlinear ones. Which approach is better depends on the dataset.

```
model = keras.Sequential([
    layers.Dense(16, activation='relu'),
    layers.Dense(1),
])

wider = keras.Sequential([
    layers.Dense(32, activation='relu'),
    layers.Dense(1),
])

deeper = keras.Sequential([
    layers.Dense(16, activation='relu'),
    layers.Dense(16, activation='relu'),
    layers.Dense(1),
])
```

Increasing capacity: wider versus deeper networks.

- 12. Dropout and Batch Normalization

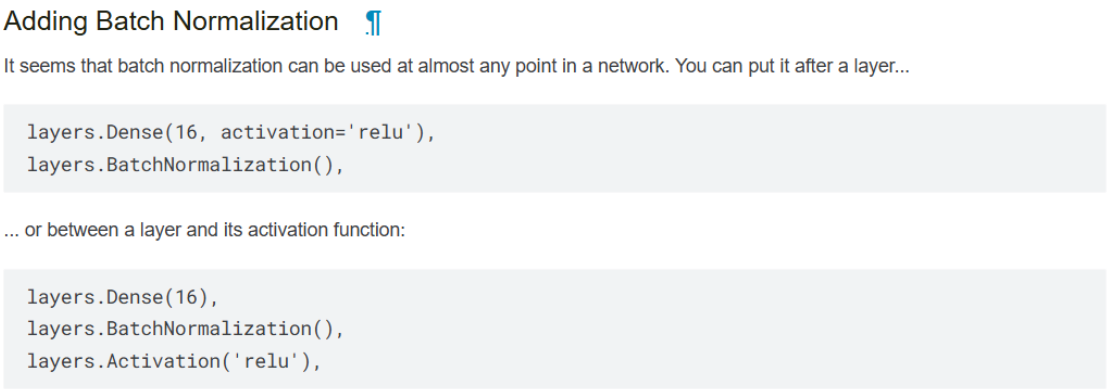
- Dropout

Dropout is a regularization technique that randomly deactivates a fraction of neurons during each training iteration. By preventing the network from relying too heavily on specific neurons, dropout encourages the model to learn more robust and generalizable patterns.

- Batch Normalization

Batch Normalization (BatchNorm) is a technique used to make training faster, more stable, and more efficient. It works by normalizing the outputs of a layer during training so that they have a consistent scale.

Before training a neural network, input features are often normalized using methods such as StandardScaler or MinMaxScaler. This matters because features with very different scales can produce activations of very different magnitudes, making optimization difficult and unstable.


Adding Batch Normalization ¶

It seems that batch normalization can be used at almost any point in a network. You can put it after a layer...

```
layers.Dense(16, activation='relu'),
layers.BatchNormalization(),
```

... or between a layer and its activation function:

```
layers.Dense(16),
layers.BatchNormalization(),
layers.Activation('relu'),
```

Batch Normalization applied within a network.

- 13. Binary Classification, Accuracy and Cross-Entropy

Classification into one of two classes is a common machine learning problem: predicting whether a customer is likely to purchase, whether a credit card transaction is fraudulent, whether deep space signals show evidence of a new planet, or whether a medical test shows evidence of a disease. These are all binary classification problems.

Before training a neural network, categorical labels must be converted into numerical values, since neural networks can only process numbers. Typically, one class is assigned 0 and the other is assigned 1.

- Accuracy

Accuracy is one of the many metrics used for measuring success on a classification problem. It is the ratio of correct predictions to total predictions:

$$accuracy = number_correct / total$$

A model that always predicts correctly would have an accuracy score of 1.0.

- Cross-Entropy

The loss function defines the objective of the network during training. In regression, the goal was to minimize the distance between the expected and predicted outcomes, measured using MAE.

For classification, the goal instead is to measure a distance between probabilities — this is what cross-entropy provides. Cross-entropy is a measure of the distance from one probability distribution to another.

WORK APPLIED:

During the third phase of the internship, a **Multi-Layer Perceptron (MLP)** model was developed to classify the **HPV status of head and neck cancer tumors**. As in the previous LightGBM study, the same experimental protocol was followed to ensure a fair comparison between the two models. The same preprocessing pipeline, training, validation, and test splits, and evaluation metrics were maintained throughout the experiments.
The dataset used in this project was **highly imbalanced**, with a much smaller number of **HPV-negative** patients compared with **HPV-positive** patients. Such class imbalance can bias the model toward predicting the majority class, leading to reduced performance in identifying HPV-negative tumors.
To address this issue, two oversampling techniques were investigated: **SMOTE (Synthetic Minority Over-sampling Technique)** and **SMOTENC (SMOTE for Nominal and Continuous features)**. SMOTE generates synthetic samples of the minority class using continuous features, whereas SMOTENC is specifically designed for datasets containing both **categorical and continuous variables**, making it well suited for the clinical dataset used in this study.
For each oversampling method, the same three classification experiments were performed using the identical data split:

- **Clinical features only**
- **PET radiomic features only**
- **Combined Clinical + PET radiomic features**

The performance of the MLP model was evaluated **before and after applying SMOTE and SMOTENC**. This comparison was carried out to determine whether balancing the training data could improve the classification of HPV-negative patients while maintaining strong overall predictive performance. The results obtained with the oversampling techniques were then compared with the original MLP results

PREPROCESSING ENTRY DATA

Methods used :

1. **Rule-Based Data Cleaning**
   Invalid values in **Performance Status** are converted to missing values (NaN), while only valid categories (0–4) are retained. This ensures data consistency before further preprocessing.
2. **Missing Indicator Method**
   Binary indicator columns are created to record whether values were originally missing before imputation. This preserves potentially useful information about missing data patterns.
3. **Median Imputation**
   Missing values in the **Age** feature are replaced with the median calculated from the training set. The median is robust to outliers and suitable for continuous numerical variables.
4. **Most-Frequent Imputation**
   Missing categorical and discrete clinical variables are filled with the most common value (mode) from the training set. This maintains valid category values while minimizing bias.
5. **Constant Imputation**
   Missing values in **M-stage** are replaced with the fixed category **"Unknown"**. This treats missingness as a separate meaningful category rather than removing information.
6. **Min–Max Normalization**
   After imputation, the **Age** feature is scaled using Min–Max normalization based on the training set minimum and maximum values. This transforms Age into a fixed range (typically 0–1) while preserving the relative differences between observations.
7. **One-Hot Encoding**
   The categorical **M-stage** variable is converted into separate binary columns for each category, including **Unknown**. This allows machine learning models to process categorical data without assuming any ordinal relationship.

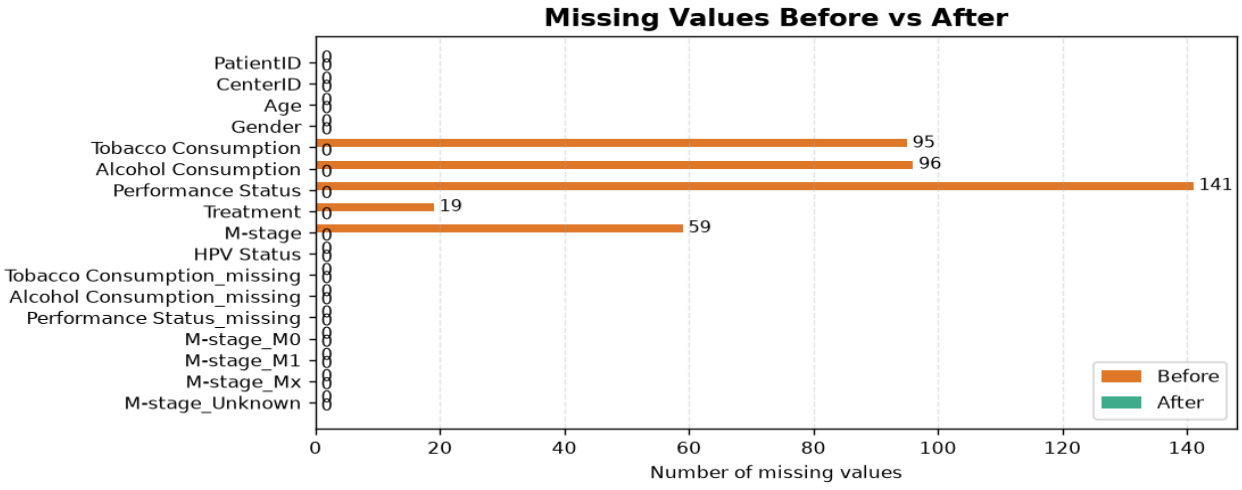


Figure 62 data missing values before and after

Merging clinical + pet :

- **Inputs used**
  Clinical: clinical_entry_preprocessed.csv
  PET: pet_features.csv
- **Merge key**
  Both files were matched by **PatientID**.
- **Join type**
  Inner join was used. This keeps only patients present in both files.
- **Dropped records**
  Clinical rows without PET features: **74**
  PET rows without clinical records: **138**
- **MLP readiness check**
  Missing cells in merged CSV: **0**
  Non-numeric feature columns: **none**
  Model features if **PatientID** is removed and **HPV Status** is the label: **121**

SMOTE and SMOTENC for MLP Classification

In the initial MLP classification experiments, the model was trained using the original imbalanced

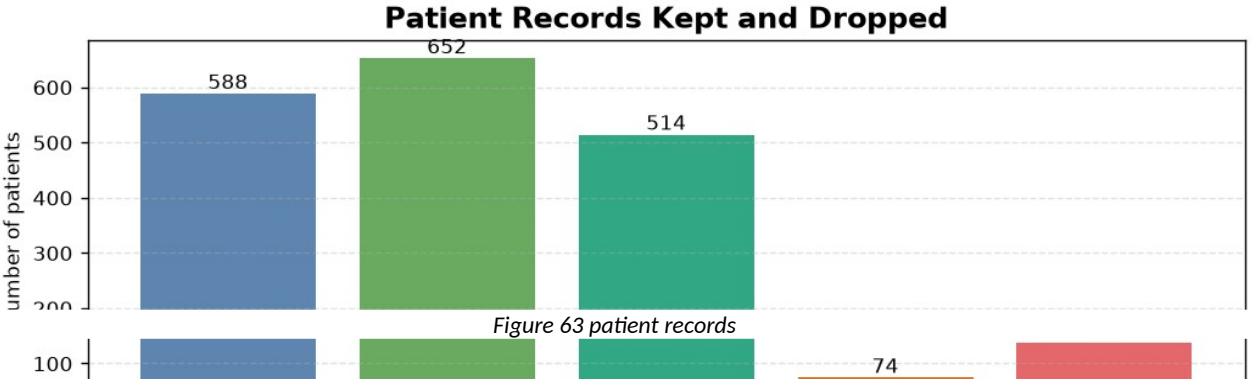


Figure 63 patient records

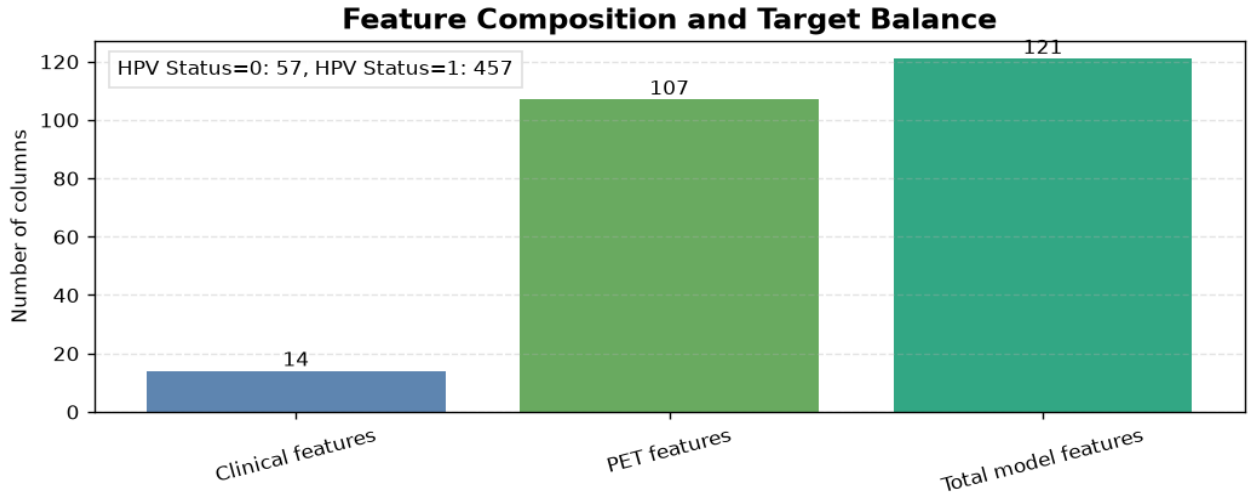


Figure 64 feature compostion and target balance

dataset without applying any oversampling techniques. The obtained results showed relatively poor performance, particularly in detecting the minority class, indicating that class imbalance negatively affected the model's learning ability.

To address this issue, two oversampling methods were introduced: **Synthetic Minority Oversampling Technique (SMOTE)** and **SMOTENC (SMOTE for Nominal and Continuous features)**.

SMOTE generates synthetic samples for the minority class by interpolating between existing minority instances, thereby balancing the class distribution. SMOTENC extends this approach by supporting datasets that contain both continuous and categorical features, making it more suitable for mixed clinical data.

The MLP classification was then repeated using these oversampling techniques on the training data. The resulting performance will be analyzed and compared with the baseline model to evaluate the impact of SMOTE and SMOTENC on classification accuracy, minority class detection, and overall model performance.

Test
0.862
0.872
0.783
0.899
0.667
0.954
0.925
[[6, 3], [7, 62]]
0.089
0.969
0.106

Metric	Validation
ROC-AUC	0.978
Accuracy	0.922
Balanced Accuracy	0.957
Sensitivity	0.913
Specificity	1.000
Precision	1.000
F1-score	0.955
Confusion Matrix	[[8, 0], [6, 63]]
Brier Score	0.048
PR-AUC	0.997
Expected Calibration Error (ECE)	0.058

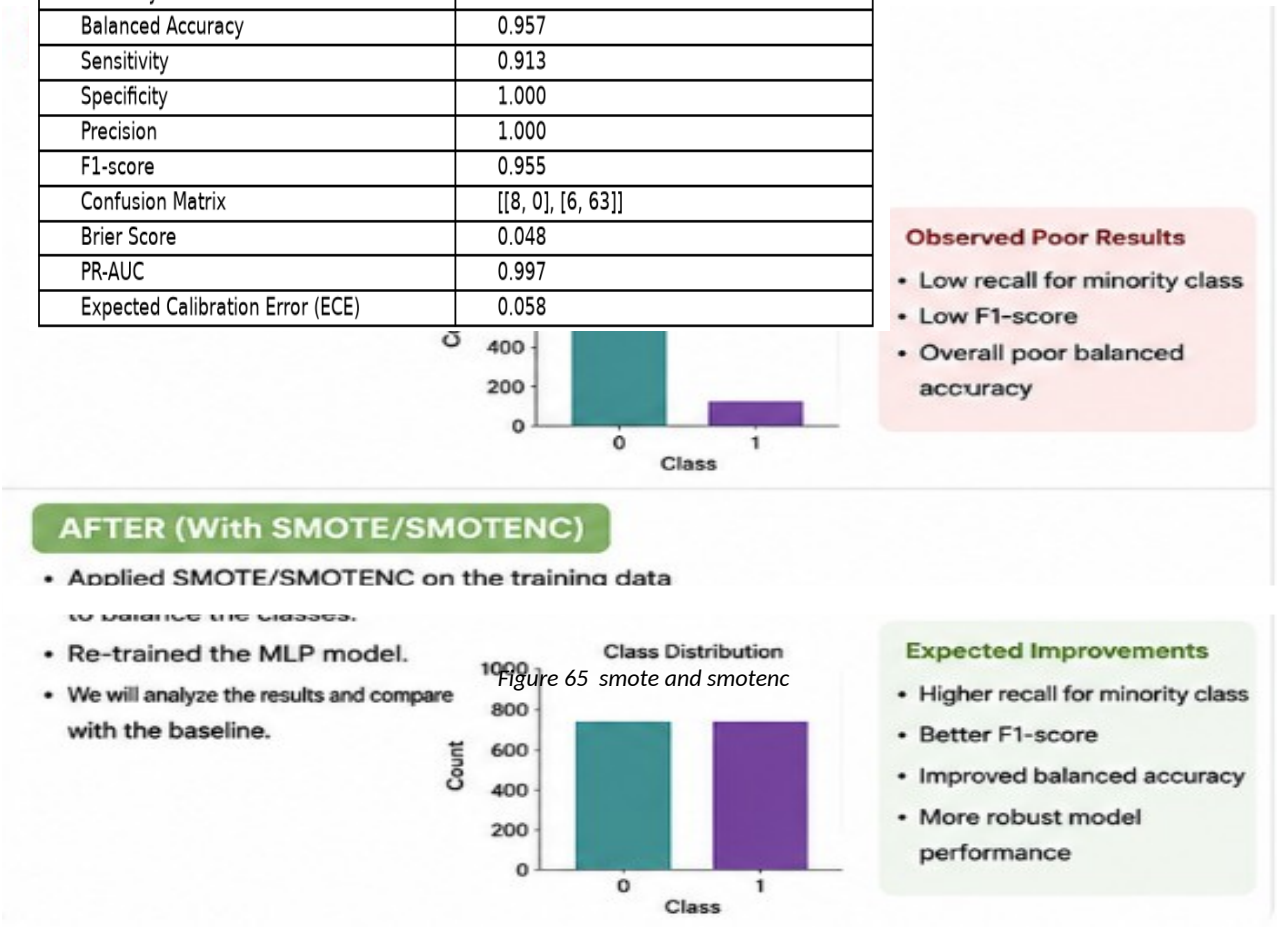


Figure 65 smote and smotenc

CLINICAL ONLY :

1. The clinical-only model is the strongest individual model. The validation ROC-AUC is 0.978 and the test ROC-AUC is 0.862, showing that the clinical features contain strong information for predicting

Validation Calibration Table (10 Bins)

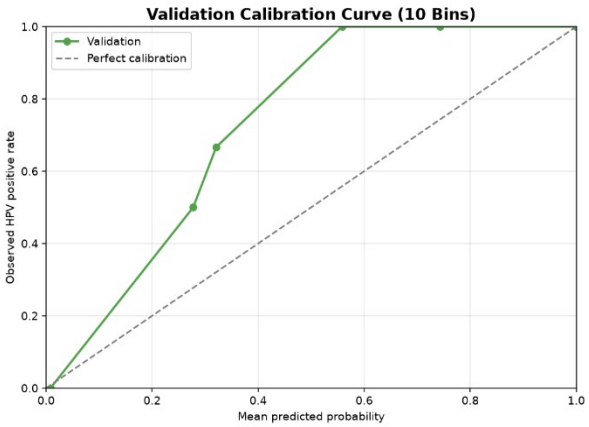
Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.008	0.000	0.008
[0.1, 0.2)	0			
[0.2, 0.3)	4	0.277	0.500	0.223
[0.3, 0.4)	6	0.321	0.667	0.346
[0.4, 0.5)	0			
[0.5, 0.6)	1	0.558	1.000	0.442
[0.6, 0.7)	0			
[0.7, 0.8)	4	0.743	1.000	0.257
[0.8, 0.9)	0			
[0.9, 1.0]	58	1.000	1.000	0.000

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.006	0.000	0.006
[0.1, 0.2)	2	0.133	0.500	0.367
[0.2, 0.3)	5	0.261	0.800	0.539
[0.3, 0.4)	2	0.313	1.000	0.687
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	3	0.659	1.000	0.341
[0.7, 0.8)	2	0.758	0.500	0.258
[0.8, 0.9)	0			
[0.9, 1.0]	60	0.998	0.967	0.032

HPV Status. The decrease from validation to test means the model generalizes well, but not perfectly.

2. Accuracy is high on both splits. Validation accuracy is 0.922, meaning 71 of 77 validation patients were classified correctly. Test accuracy is 0.872, meaning 68 of 78 test patients were classified correctly. This shows strong overall classification performance.



3. Balanced Accuracy is also strong, especially on validation. Validation Balanced Accuracy is 0.957, which means the model performed well for both HPV-negative and HPV-positive classes. Test Balanced Accuracy is 0.783, which is lower than validation but still useful. The drop is mainly caused by weaker test specificity.

4. The model keeps high sensitivity on both validation and test. Validation sensitivity is 0.913, and test sensitivity is 0.899. On the test set, this means the model correctly detected 62 of 69 HPV-positive patients and missed 7 HPV-positive patients.

Table 3 mlp clinical calibration table val/test

5. Specificity is excellent on validation but lower on test. Validation specificity is 1.000, meaning all 8 HPV-negative validation patients were correctly identified. Test specificity is 0.667, meaning 6 of 9 HPV-negative test patients were correctly identified, while 3 were false positives.

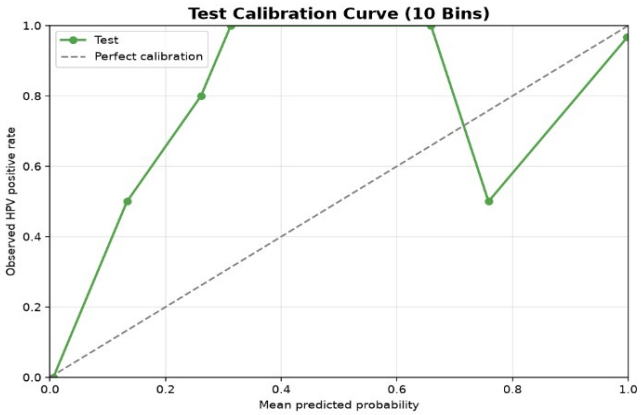


Figure 67 mlp clinical calibration curve val/test

The validation ECE is 0.058 and the test ECE is 0.106. The test calibration is weaker than validation, but still acceptable compared with PET-only before scaling. The Brier Score is 0.048 on validation and 0.089 on test, which supports the same idea: the clinical probabilities are fairly useful, but the test probabilities are less calibrated than validation

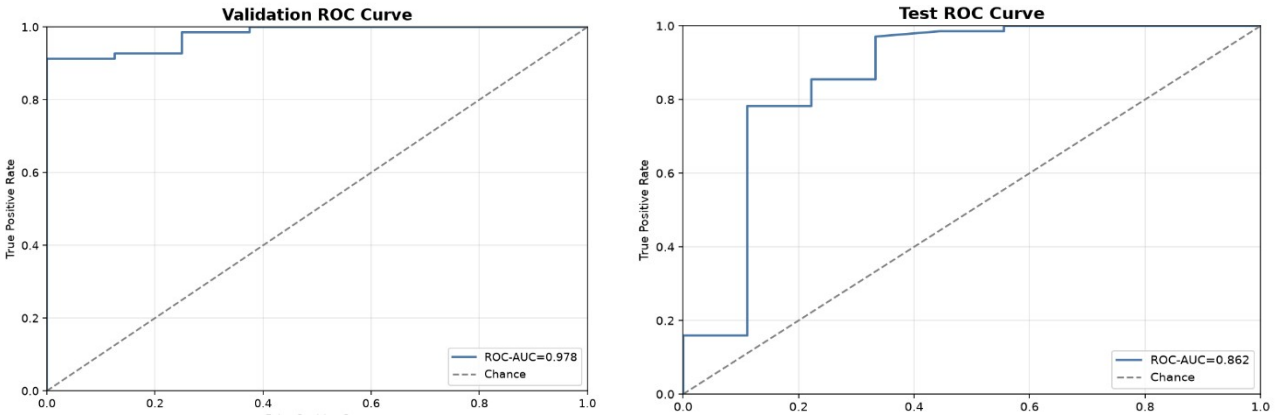


Figure 68 mlp clinical roc curve val/test

The ROC curves show strong discrimination. Validation ROC-AUC is 0.978, which is very high. Test ROC-AUC is lower at 0.862, but still good. This means the clinical model usually ranks HPV-positive patients above HPV-negative patients, although the performance drops on unseen test data.

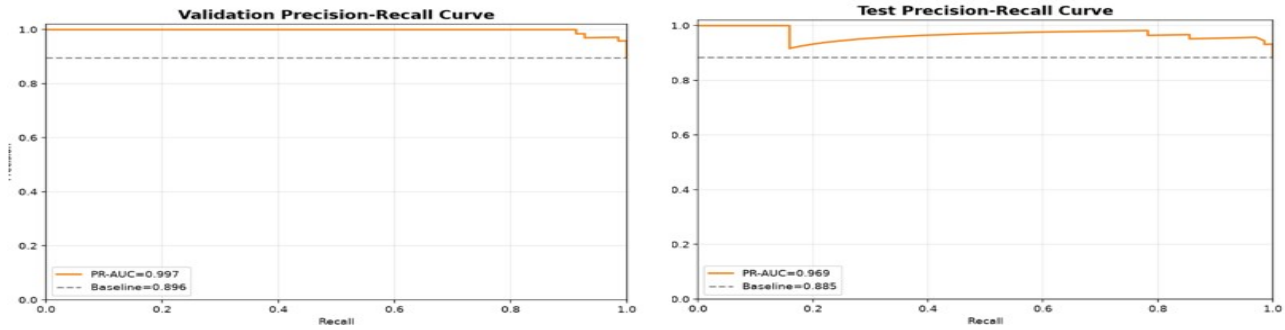


Figure 69 mlp pr-auc val/test

The PR-AUC is 0.997 on validation and 0.969 on test. Because the dataset has many more HPV-positive patients than HPV-negative patients, the precision-recall curve is important. These high PR-AUC values show that the clinical model keeps strong precision while recovering most HPV-positive patients.

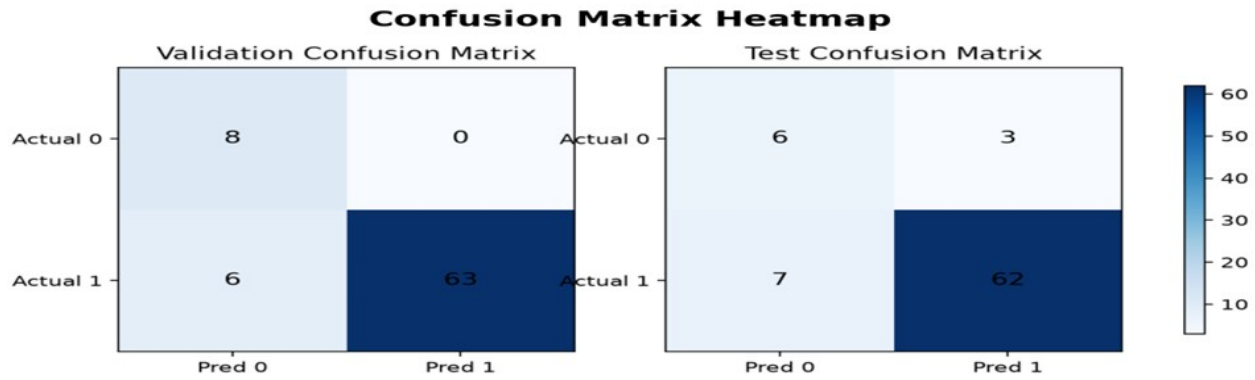


Figure 70 mlp clinical confusion matrix val/test

On validation, the model correctly identified all 8 HPV-negative patients and 63 of 69 HPV-positive patients. It made 0 false positives and 6 false negatives. On test, it correctly identified 6 of 9 HPV-negative patients and 62 of 69 HPV-positive patients. It made 3 false positives and 7 false negatives.

The probability distribution plot shows how predicted HPV-positive probabilities are separated by true class. For the clinical model, HPV-positive patients are generally pushed toward high predicted probabilities. HPV-negative patients are more often lower, but there is some overlap on test, which explains why test specificity is 0.667 rather than perfect.

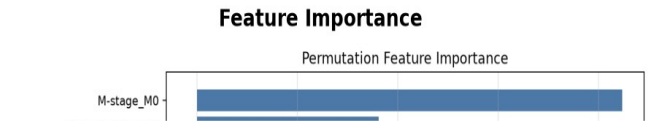


Figure 71 mlp clinical probability distribution val/test

The permutation feature importance plot shows which features caused the largest validation ROC-AUC drop when shuffled. The strongest feature was M-stage_M0 with a mean ROC-AUC drop of 0.085. Other important features included M-stage_Unknown, Alcohol Consumption_missing, Tobacco Consumption, Performance Status, and Tobacco Consumption_missing.

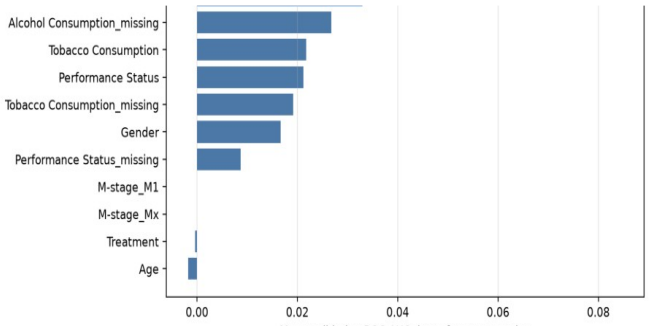


Figure 72 mlp clinical feature importance

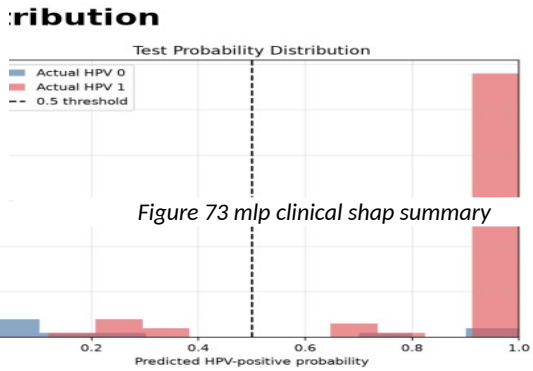


Figure 73 mlp clinical shap summary

The SHAP summary plot supports the permutation importance result. The top SHAP features were M-stage_M0, M-stage_Unknown, Alcohol Consumption_missing, Tobacco Consumption_missing, Alcohol Consumption, Performance Status_missing, Tobacco Consumption, Performance Status, Gender, and Age.

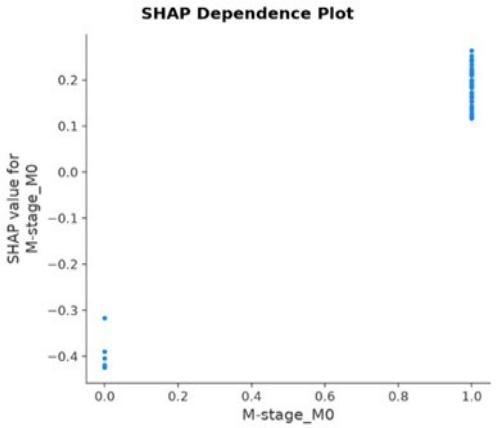
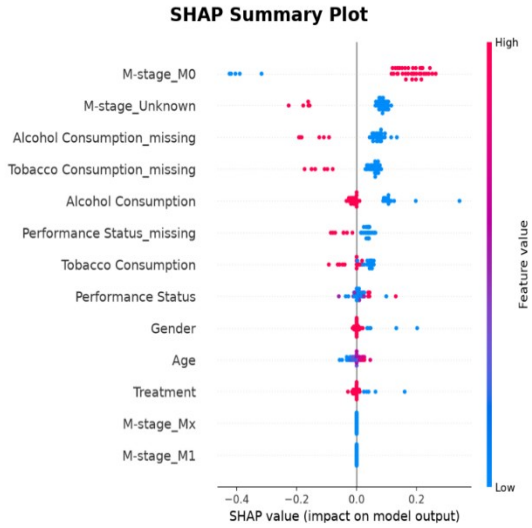


Figure 74 mlp clinical shap dependence



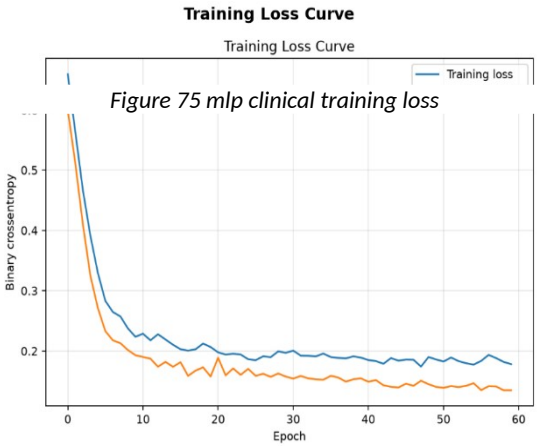


Figure 75 mlp clinical training loss

The automatically selected dependence feature was M-stage_M0. This plot shows how the SHAP contribution changes when M-stage_M0 changes. Because M-stage_M0 is one-hot encoded, the plot mainly compares patients with M-stage_M0 = 0 against M-stage_M0 = 1. The feature has a clear contribution to the predicted HPV probability.

The training loss decreased from 0.659 at the first epoch to 0.178 at the final epoch. Validation loss decreased from 0.599 to 0.135. Validation ROC-AUC finished around 0.978 in the training history. This suggests the clinical model learned a stable signal and did not show severe overfitting in the saved training curve.

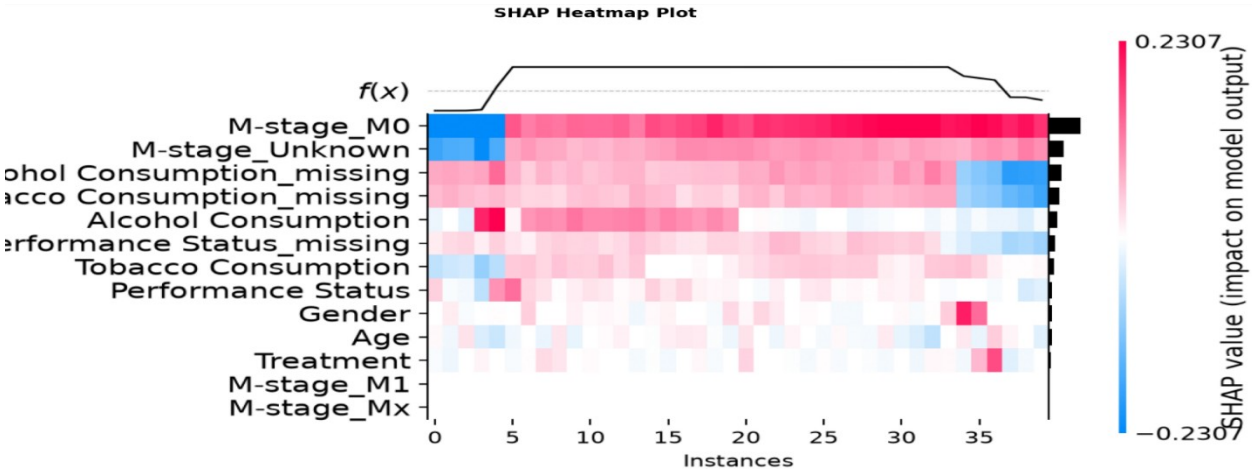


Figure 76 mlp clinicalshap heatmap

The SHAP heatmap shows patient-level explanations across the validation samples. M-stage_M0 and M-stage_Unknown are dominant rows, meaning they repeatedly influence predictions across many patients. Alcohol/tobacco missingness and consumption variables also contribute across multiple samples.

PET ONLY:

Even after applying SMOTE and SMOTENC to address class imbalance in the PET-only experiment, the model continued to exhibit poor predictive performance. Since these resampling techniques did not produce a meaningful improvement, we decided to apply standard feature scaling using the StandardScaler, fitting it only on the training set and subsequently transforming the validation and test sets using the same fitted

scaler. This additional preprocessing step was performed to evaluate whether feature standardization could improve the model's learning process and lead to better predictive performance.

StandardScaler standardizes each numerical feature by subtracting its mean and dividing by its standard deviation, resulting in features with a mean of 0 and a standard deviation of 1. The scaler is fitted only on the training set, and the same transformation is then applied to the validation and test sets to prevent data leakage. This ensures that all input features are on a comparable scale, which facilitates more stable and efficient training of the Multilayer Perceptron (MLP) model.

Before scaling, the PET-only model looked accurate only because the dataset contains many HPV-positive patients. Validation accuracy was 0.896 and test accuracy was 0.885, but Balanced Accuracy was only 0.500 on both splits. This means the model was not truly separating the two classes.

Accuracy before scaling is misleading. The model had validation accuracy of 0.896 and test accuracy of 0.885, but this happened because most patients are HPV positive and the model predicted every patient as HPV positive. So the high accuracy did not mean the model was clinically balanced.

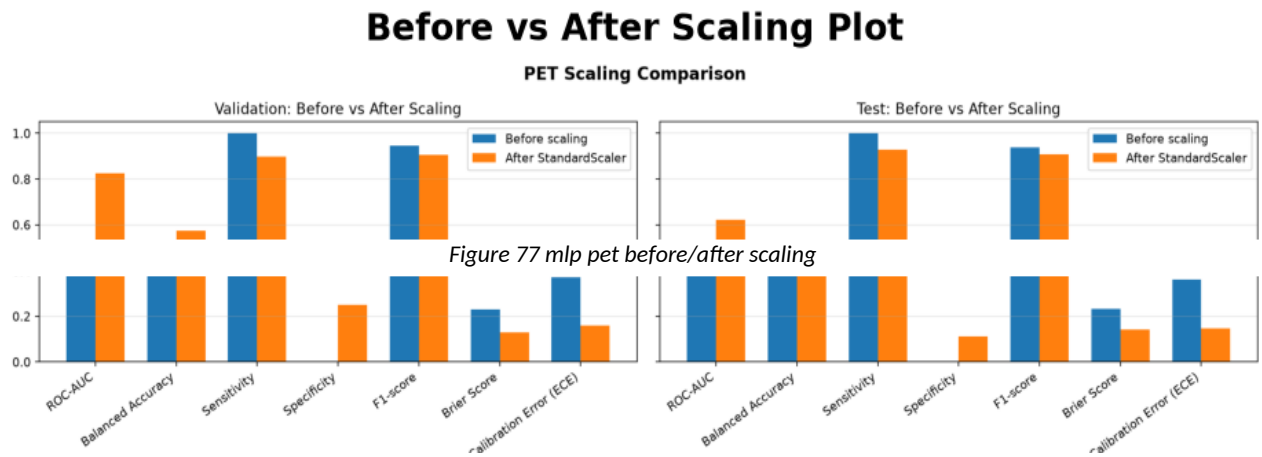


Figure 77 mlp pet before/after scaling

Balanced Accuracy before scaling showed the real problem. It was 0.500 on validation and 0.500 on test, which is chance-level performance for balanced class evaluation. This confirmed that the model ignored the

Before vs After Scaling Metrics

Experiment	Split	ROC-AUC	Bal Acc	Sens	Spec	F1	Brier	ECE
Before scaling	Validation	0.507	0.500	1.000	0.000	0.945	0.230	0.370
Before scaling	Test	0.500	0.500	1.000	0.000	0.939	0.232	0.361
After StandardSca	Validation	0.826	0.574	0.899	0.250	0.905	0.130	0.159
After StandardSca	Test	0.621	0.519	0.928	0.111	0.908	0.141	0.147

Table 4 mlp metrics results

HPV-negative class.

The main problem before scaling was specificity. Specificity was 0.000 on validation and 0.000 on test. The confusion matrices were [[0, 8], [0, 69]] on validation and [[0, 9], [0, 69]] on test, meaning the model predicted every patient as HPV positive.

After StandardScaler, the PET model improved. Validation ROC-AUC increased from 0.507 to 0.826, and test ROC-AUC increased from 0.500 to 0.621. This means the scaled PET features helped the MLP rank HPV-positive and HPV-negative patients better.

After scaling, accuracy decreased slightly because the model stopped predicting everyone as positive. Validation accuracy changed from 0.896 to 0.831, and test accuracy changed from 0.885 to 0.833. This lower accuracy is not automatically bad, because the model started identifying some HPV-negative patients.

Balanced Accuracy improved after scaling. Validation Balanced Accuracy improved from 0.500 to 0.574, and test Balanced Accuracy improved from 0.500 to 0.519. The improvement is modest, but it shows that scaling helped the model begin to handle both classes instead of only the majority HPV-positive class.

Specificity also improved after scaling, but it remained weak. Validation specificity improved from 0.000 to 0.250, and test specificity improved from 0.000 to 0.111. The test model correctly identified only 1 of 9 HPV-negative patients after scaling, so PET-only is still not strong for detecting HPV-negative cases.

Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	3	0.016	0.333	0.317
[0.1, 0.2)	2	0.171	1.000	0.829
[0.2, 0.3)	2	0.228	1.000	0.772
[0.3, 0.4)	2	0.346	1.000	0.654
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	2	0.627	1.000	0.373
[0.7, 0.8)	3	0.766	1.000	0.234
[0.8, 0.9)	2	0.835	0.000	0.835
[0.9, 1.0]	61	0.994	0.934	0.060

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	5	0.014	0.800	0.786
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	1	0.464	1.000	0.536
[0.5, 0.6)	0			
[0.6, 0.7)	2	0.671	0.000	0.671
[0.7, 0.8)	2	0.742	1.000	0.258
[0.8, 0.9)	3	0.820	0.667	0.153
[0.9, 1.0]	65	0.996	0.923	0.073

Table 5 mlp pet calibration table val/test

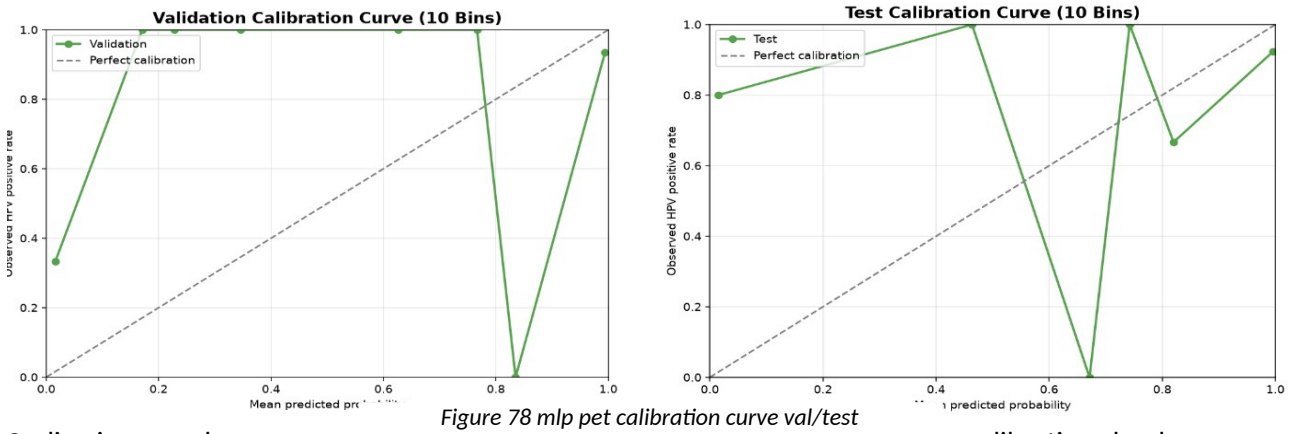
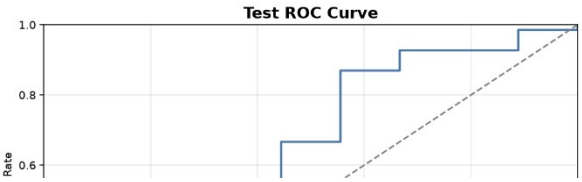


Figure 78 mlp pet calibration curve val/test

Scaling improved calibration clearly. Validation ECE improved from 0.370 to 0.159, and test ECE improved from 0.361 to 0.147. Brier Score improved from 0.230 to 0.130 on validation and from 0.232 to 0.141 on test. The probabilities are still not perfect, but the scaled PET model is much better calibrated than the unscaled PET model.

After scaling, the validation ROC-AUC is 0.826 and the test ROC-AUC is 0.621. This means scaling



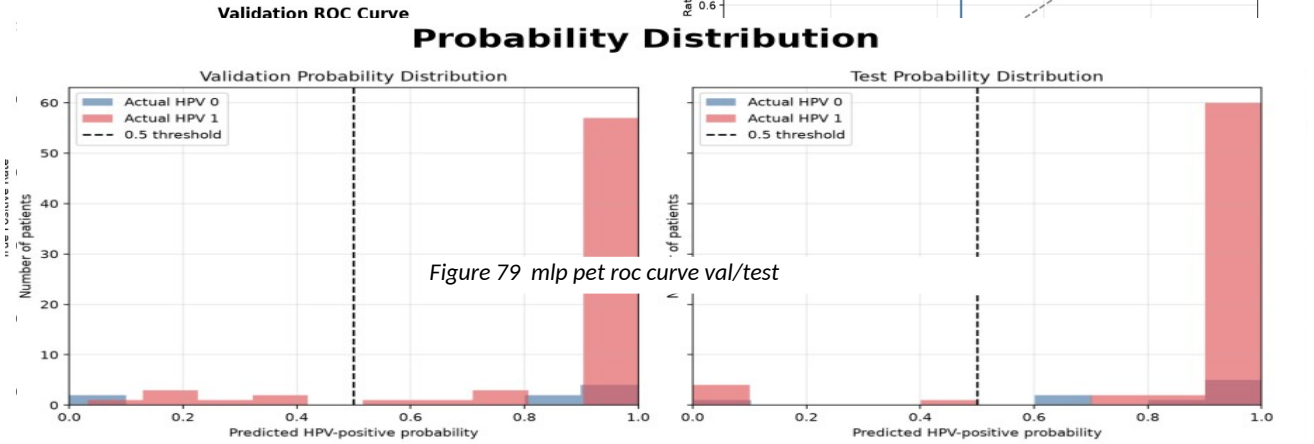
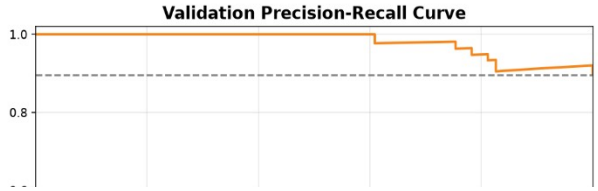


Figure 79 mlp pet roc curve val/test



helped the model rank HPV-positive and HPV-negative patients better. However, the gap between validation and test is large. The PET model has weaker generalization than the clinical model.



The PR-AUC is 0.978 on validation and 0.918 on test after scaling. These values are high because HPV-positive patients are the majority class and the model is still biased toward predicting positives. The PR curve

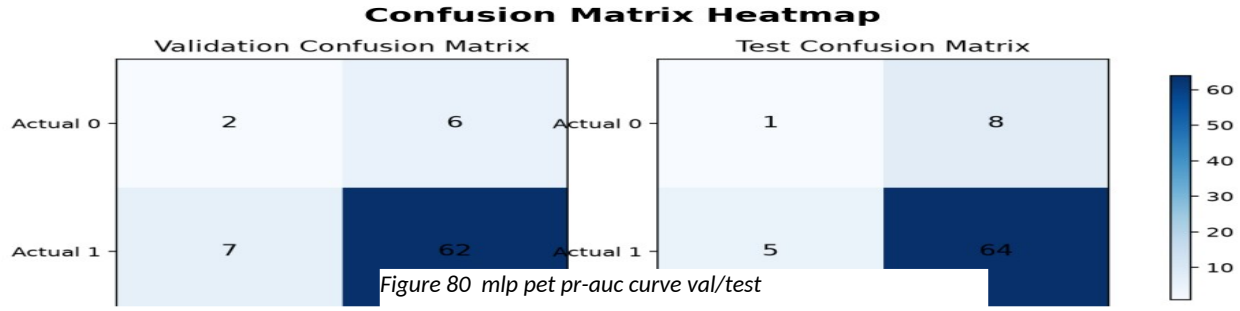


Figure 80 mlp pet pr-auc curve val/test

Figure 81 mlp pet confusion matrix val/test

is useful, but it should be interpreted together with specificity and Balanced Accuracy.

Before scaling, the PET model had confusion matrices [[0, 8], [0, 69]] on validation and [[0, 9], [0, 69]] on test. It detected every HPV-positive patient but detected no HPV-negative patient.

After scaling, validation changed to [[2, 6], [7, 62]], and test changed to [[1, 8], [5, 64]]. This is an improvement because the model now identifies some HPV-negative patients, but it still misses most HPV-negative patients.

The probability distribution plot still shows many patients receiving high HPV-positive probabilities. This explains why sensitivity remains high but specificity remains low. Scaling improves separation, but the positive and negative distributions still overlap strongly.

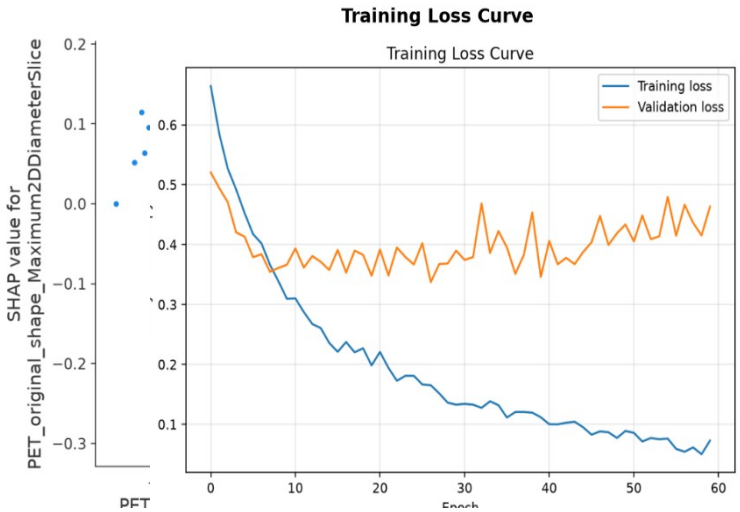


Figure 83 mlp pet feature importance

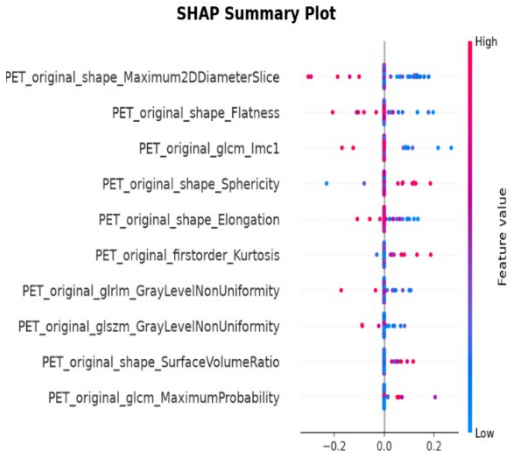


Figure 84 mlp pet shap summary

The strongest SHAP feature was PET_original_shape_Maximum2DDiameterSlice. Other important SHAP features included PET_original_shape_Flatness, PET_original_glcm_Imc1, PET_original_shape_Sphericity, PET_original_shape_Elongation, PET_original_firstorder_Kurtosis, PET_original_glrlm_GrayLevelNonUniformity, PET_original_glszm_GrayLevelNonUniformity, PET_original_shape_SurfaceVolumeRatio, and PET_original_glcm_MaximumProbability.

The automatically selected dependence feature was PET_original_shape_Maximum2DDiameterSlice. This plot shows that this tumor shape/size-related feature has a visible effect on the model output. The relationship is not perfectly linear, which is expected for an MLP.

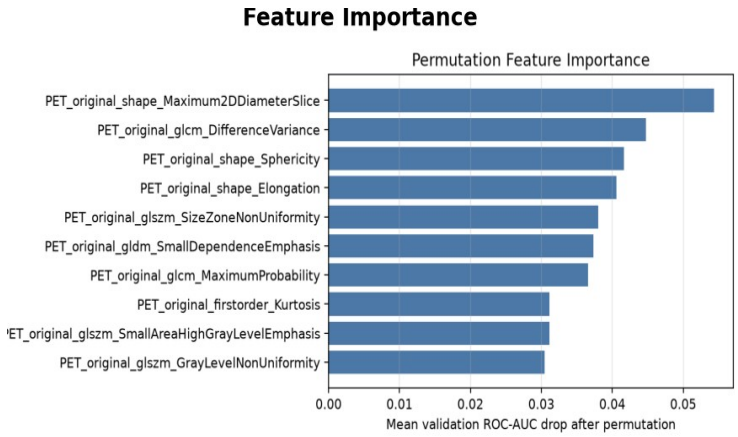


Figure 85 mlp pet shap dependence

Figure 86 mlp pet train loss

The training loss decreased from 0.665 to 0.073, while validation loss ended around 0.463. Training ROC-AUC became very high, but validation ROC-AUC was lower. This indicates overfitting: the PET model learns the training data strongly, but the validation and test results are weaker.

Before vs After Scaling Metrics

Experiment	Split	ROC-AUC	Bal Acc	Sens	Spec	F1	Brier	ECE
Before scaling	Validation	0.500	0.500	1.000	0.000	0.945	0.213	0.346
Before scaling	Test	0.500	0.500	1.000	0.000	0.939	0.214	0.335
fter StandardSca	Validation	0.976	0.743	0.986	0.500	0.965	0.041	0.056
fter StandardSca	Test	0.813	0.756	0.957	0.556	0.950	0.087	0.092

Figure 87 mlp pet shap heatmap

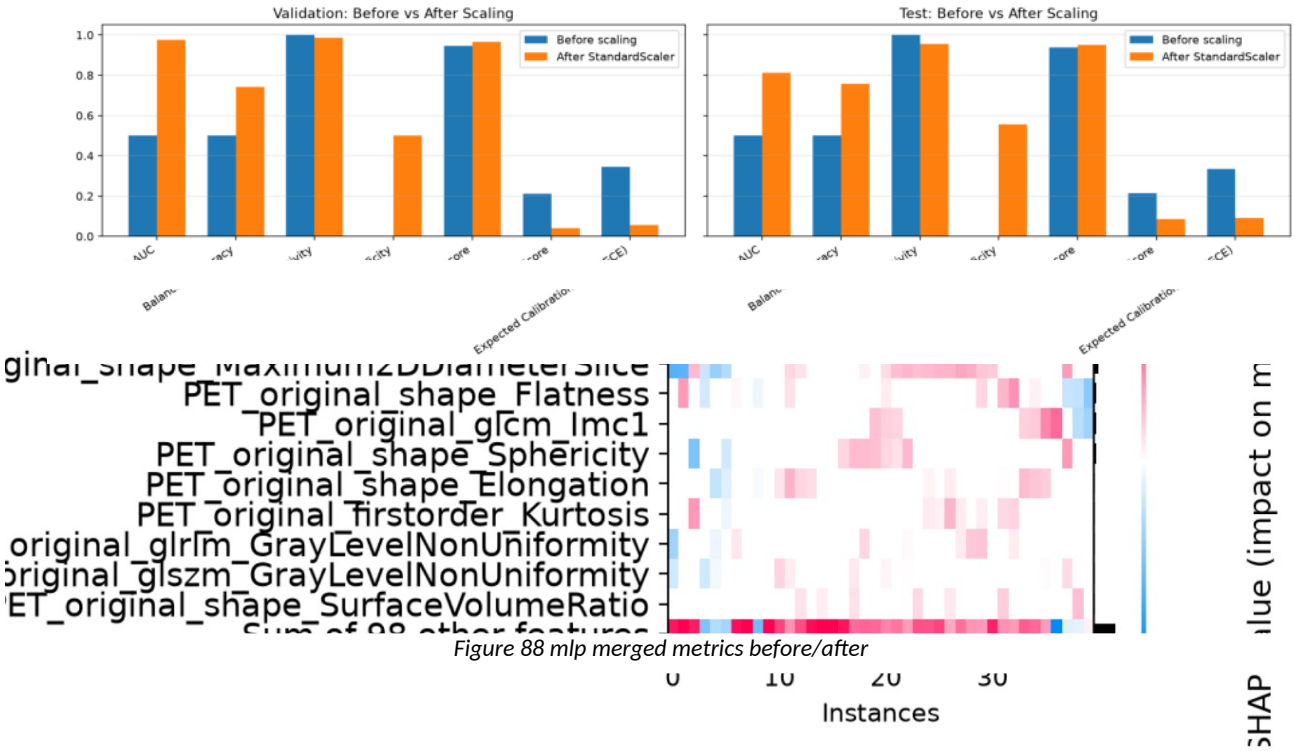


Figure 88 mlp merged metrics before/after

The SHAP heatmap shows that PET_original_shape_Maximum2DDiameterSlice repeatedly influences many validation patients. Other PET shape and texture features also contribute, but the heatmap suggests the PET model is less stable and less clinically clean than the clinical-only model.

MERGED ONLY :

- **_Before PET scaling:**
  - High accuracy (0.896 validation, 0.885 test) was misleading because the model predicted every patient as HPV-positive.
  - Balanced Accuracy was only **0.500** on both validation and test.
  - Specificity was **0.000**, meaning no HPV-negative patients were correctly identified.
- **After PET-column scaling:**

Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.019	0.000	0.019
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	1	0.312	1.000	0.688
[0.4, 0.5)	0			
[0.5, 0.6)	1	0.512	0.000	0.512
[0.6, 0.7)	2	0.664	1.000	0.336
[0.7, 0.8)	2	0.756	0.500	0.256
[0.8, 0.9)	2	0.859	0.000	0.859
[0.9, 1.0]	65	0.997	1.000	0.003

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	5	0.008	0.600	0.592
[0.1, 0.2)	2	0.152	0.000	0.152
[0.2, 0.3)	0			
[0.3, 0.4)	1	0.303	0.000	0.303
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	0			
[0.7, 0.8)	0			
[0.8, 0.9)	2	0.878	0.500	0.378
[0.9, 1.0]	68	0.997	0.956	0.042

- ROC-AUC improved from **0.500 → 0.976** (validation) and **0.500 → 0.813** (test).
- Accuracy increased to **0.935** (validation) and **0.910** (test).
- Balanced Accuracy improved to **0.743** (validation) and **0.756** (test).
- High sensitivity was maintained (**0.986** validation, **0.957** test).
- Specificity increased to **0.500** (validation) and **0.556** (test).
- Calibration improved, with the test **Brier Score** decreasing from **0.214 to 0.087** and **ECE** from **0.335 to 0.092**.

- **Overall:** Scaling only the PET features resolved the scale issue, allowing the merged model to effectively use both clinical and PET information, making it balanced, well-calibrated, and competitive with the clinical-only model.

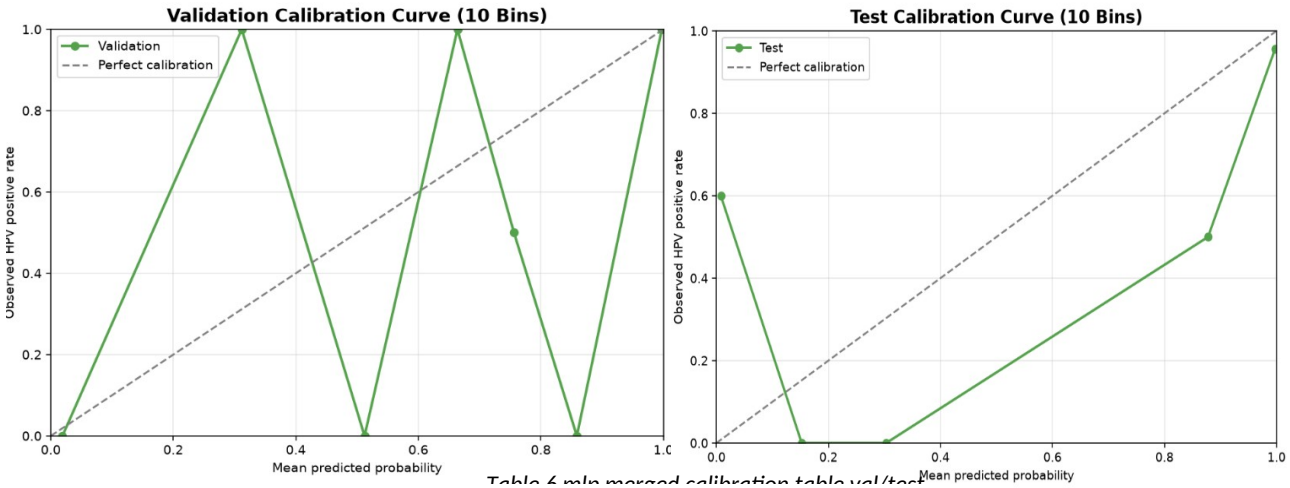


Table 4 mlp merged calibration table val/test

Figure 90 mlp merged calibration curve val/test

Calibration improved strongly after PET-column scaling. Validation ECE improved from 0.346 to 0.056, and test ECE improved from 0.335 to 0.092. Brier Score improved from 0.213 to 0.041 on validation and from 0.214 to 0.087 on test. This means the scaled merged model gives much more reliable probabilities than the unscaled merged model.

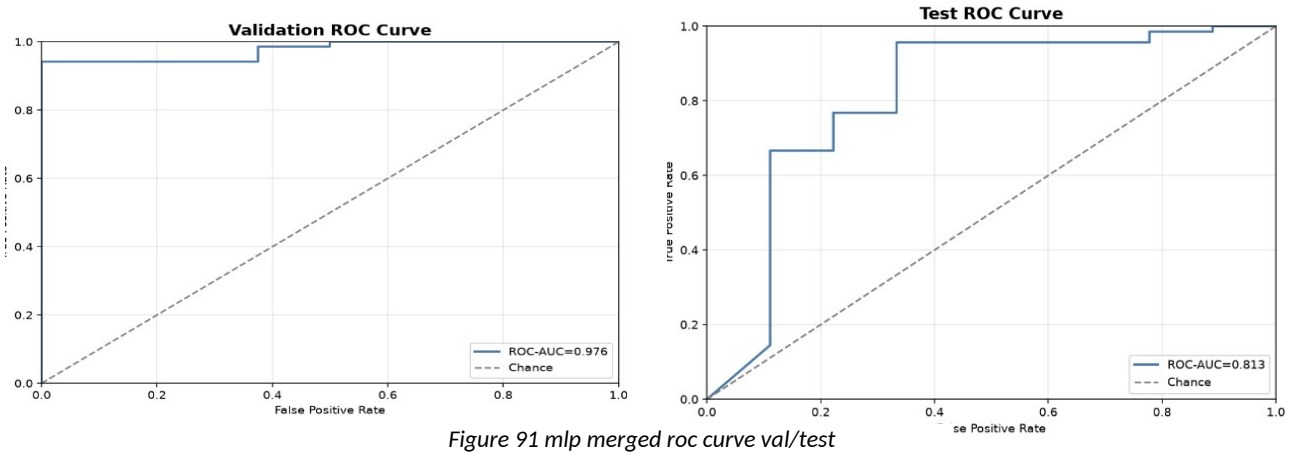


Figure 91 mlp merged roc curve val/test

After PET-column scaling, validation ROC-AUC is 0.976 and test ROC-AUC is 0.813. The validation curve is very strong and the test curve is still clearly better than chance. The model now ranks HPV-positive patients.

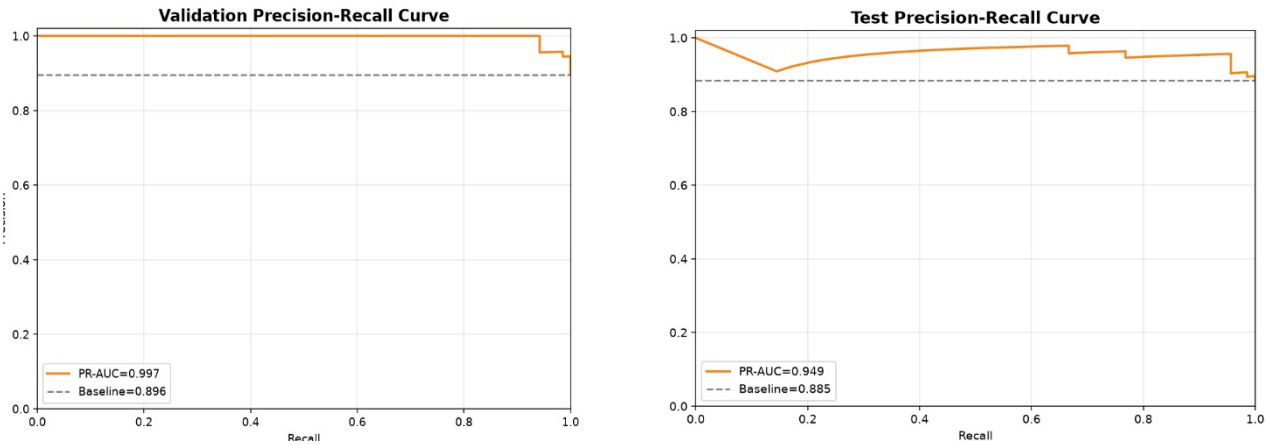
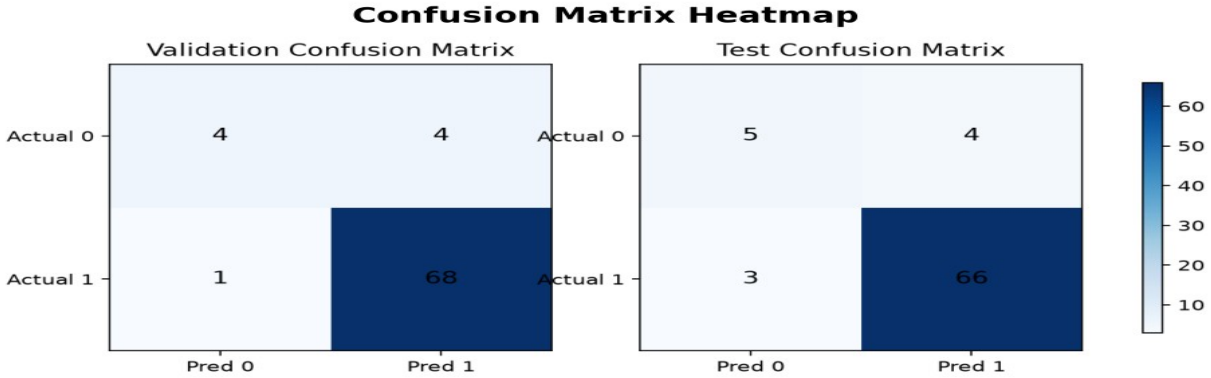


Figure 92 mlp merged pr-auc val/test

After scaling, PR-AUC is 0.997 on validation and 0.949 on test. This shows strong positive-class performance. Because the dataset is imbalanced toward HPV-positive patients, PR-AUC should be interpreted together with specificity. The merged model has better specificity than PET-only, so the strong PR-AUC is more meaningful here.nts above HPV-negative patients much better than before scaling.



Before scaling, the merged model detected no HPV-negative patients: validation was [[0, 8], [0, 69]] and test was [[0, 9], [0, 69]].

After scaling, validation improved to [[4, 4], [1, 68]], meaning it correctly detected 4 of 8 HPV-negative patients and 68 of 69 HPV-positive patients. Test improved to [[5, 4], [3, 66]], meaning it correctly detected 5 of 9 HPV-negative patients and 66 of 69 HPV-positive patients.

This is a strong practical improvement. The model keeps high sensitivity while gaining real specificity.

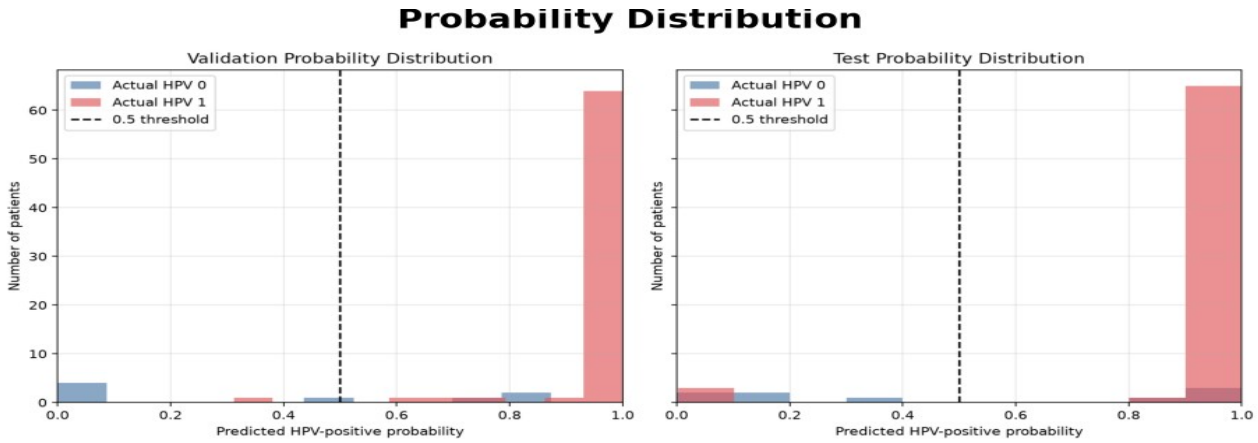


Figure 94 mlp merged probability distribution val/test

The probability distribution plot after scaling shows better separation than the unscaled merged model. Many HPV-positive patients still receive high predicted probabilities, but more HPV-negative patients now receive lower probabilities. This matches the improvement in specificity from 0.000 to 0.500 on validation and from 0.000 to 0.556 on test.

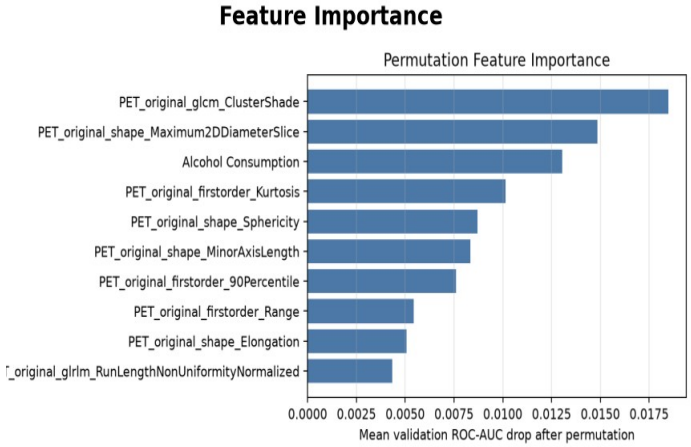


Figure 95 mlp merged features importance

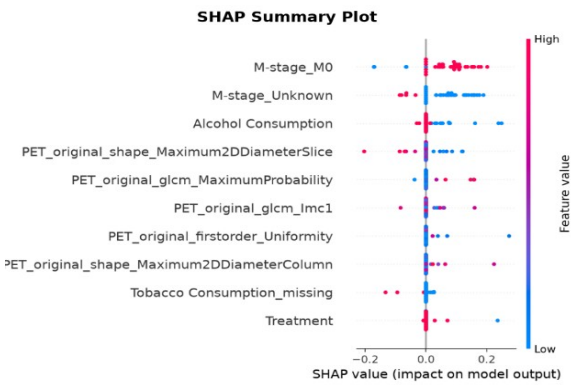


Figure 96 mlp merged shap summary

This is the best sign from the merged model: both clinical and PET features contribute. The model is not relying only on radiomics or only on clinical variables.

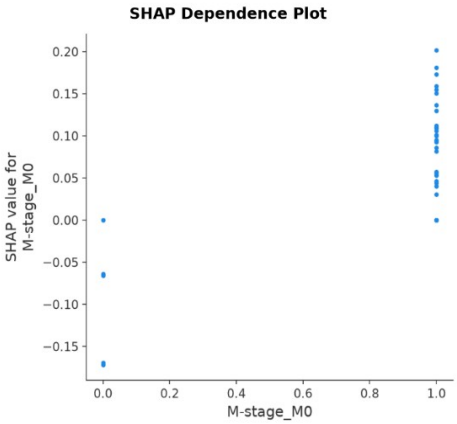


Figure 97 mlp merged shap dependence



Figure 98 mlp merged training loss

The automatically selected dependence feature was M-stage_M0. This means that, even in the merged model, clinical tumor-stage information remains one of the most influential drivers of the MLP output. The PET features improve the model, but M-stage still carries major explanatory power.

The training loss decreased from 0.704 to 0.017, while validation loss ended around 0.113. Training ROC-AUC became almost perfect, and validation ROC-AUC ended around 0.976 in the training history. This suggests the model learned a strong signal. The large gap between training loss and validation loss suggests some overfitting risk, but the test ROC-AUC of 0.813 and test Balanced Accuracy of 0.756 show that the model still generalizes much better than the unscaled merged model.

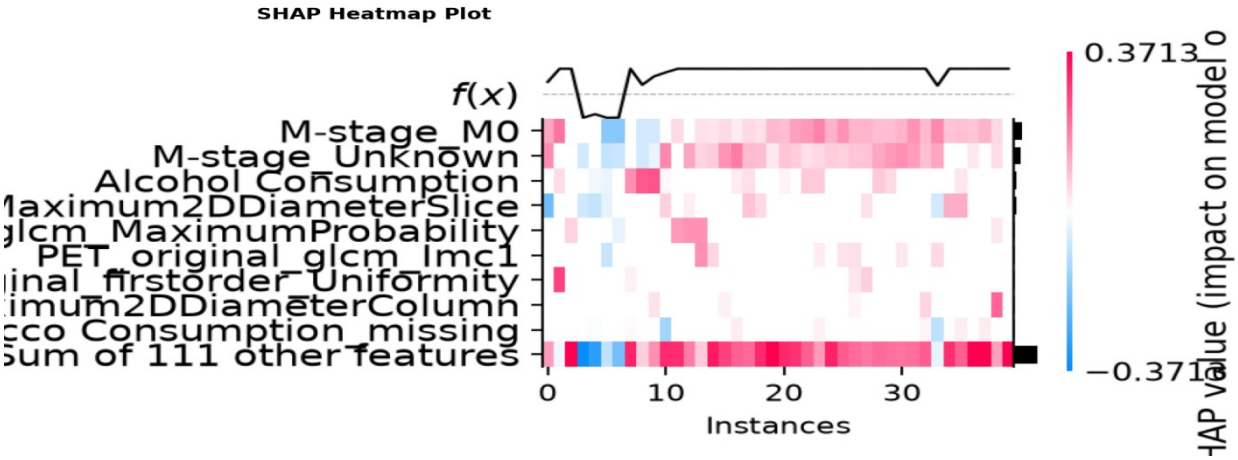


Figure 99 mlp merged shap heatmap

The SHAP heatmap shows patient-level contributions from both clinical and PET variables. M-stage_M0 and M-stage_Unknown remain important across many validation samples. PET shape and texture features also appear in the heatmap, showing that the scaled PET features add useful information instead of overwhelming the clinical signal.

FINAL INTERPRETATION:

Three MLP classification experiments were performed to evaluate the ability of deep learning to classify the HPV status of head and neck cancer tumors using **clinical features**, **PET radiomic features**, and a **combination of clinical and PET features**. To address the class imbalance caused by the limited number of HPV-negative patients, **SMOTE** and **SMOTENC** were applied to the training data. Since the PET radiomic features were not on a comparable numerical scale, **StandardScaler** was additionally applied to the PET-only and combined experiments to improve the learning process.

The **clinical-only model** achieved the best overall performance among the three experiments. It produced excellent discrimination (test ROC-AUC = **0.862**), high accuracy (**0.872**), and strong sensitivity (**0.899**), while maintaining acceptable specificity (**0.667**). Feature importance and SHAP analyses consistently identified **M-stage**, along with tobacco and alcohol consumption variables, as the main contributors to HPV status prediction. The training history also showed stable convergence with no evidence of severe overfitting, indicating that clinical variables provide the strongest predictive signal for HPV classification.

The **PET-only model** showed the weakest performance. Although feature scaling substantially improved the model compared with the initial unscaled version, the test ROC-AUC reached only **0.621**, and specificity remained very low (**0.111**). The model continued to favor HPV-positive predictions, resulting in poor identification of HPV-negative patients. Furthermore, the large gap between training and validation performance suggested overfitting, indicating that PET radiomic features alone are insufficient for reliable HPV status classification.

The **combined clinical and PET model** demonstrated that integrating PET radiomic features with clinical information improves overall predictive performance compared with PET alone. After scaling the PET features, the model achieved a test ROC-AUC of **0.813**, balanced accuracy of **0.756**, and specificity of **0.556**, while maintaining very high sensitivity (**0.957**). Calibration also improved considerably, and SHAP analysis showed that both clinical and PET features contributed to the predictions. Nevertheless, **M-stage** remained the most influential feature, confirming that clinical information continued to drive most of the model's decisions, while PET radiomic features provided complementary information that enhanced classification performance.

Overall, the MLP experiments demonstrate that **clinical features remain the most informative predictors of HPV status**, whereas PET radiomic features alone have limited predictive capability. However, when PET features are appropriately scaled and combined with clinical variables, they contribute additional complementary information that improves model balance, calibration, and generalization. These findings confirm that the combination of clinical and PET radiomic data provides a more robust framework for HPV status classification than PET radiomics alone, although the clinical information remains the dominant source of predictive power.

TRANSFORMER CLASSIFICATION:

Following the evaluation of the MLP model, the internship progressed to its fourth phase, which focused on **transformer-based classification**. In this stage, an **FT-Transformer (Feature Tokenizer Transformer)** model was implemented to classify the **HPV status of head and neck cancer tumors**. The objective was to investigate whether a transformer architecture could better capture complex relationships between clinical and PET radiomic features, leading to improved predictive performance.

To ensure a fair comparison with the previous machine learning and deep learning approaches, the same experimental protocol was maintained. The same **training, validation, and test datasets**, preprocessing pipeline, and evaluation metrics were used throughout the experiments. The same three classification experiments were performed:

- **Clinical features only**
- **PET radiomic features only**
- **Combined Clinical + PET radiomic features**

The FT-Transformer model was selected because it is specifically designed for **tabular data**, where each feature is represented as a token and processed through the transformer's self-attention mechanism. This enables the model to learn complex interactions between features while effectively handling both numerical and categorical variables.

The performance of the FT-Transformer was evaluated using the same metrics as the previous models, including **ROC-AUC, Accuracy, Balanced Accuracy, Sensitivity, Specificity, Precision, F1-score, PR-AUC, Brier Score, Expected Calibration Error (ECE), confusion matrices, calibration curves, feature importance, and SHAP analysis**. This consistent evaluation framework allowed a direct comparison between the **LightGBM**, **MLP**, and **FT-Transformer** models to determine the most effective approach for HPV status classification.

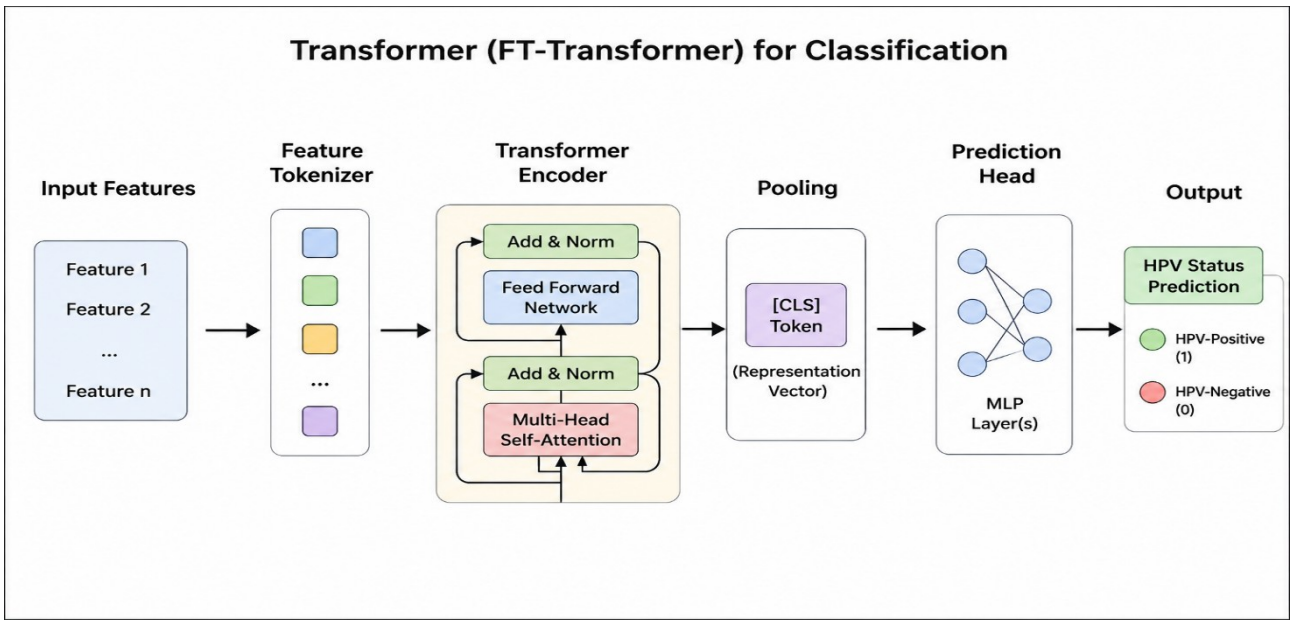


Figure 100 ft transformer workflow

How the FT-Transformer Works:

The **FT-Transformer** is a transformer architecture designed specifically for **tabular data classification**. Instead of processing words like traditional transformers, it processes each clinical or radiomic feature as an individual input token. The model then applies a **self-attention mechanism** to learn how these features interact with one another. After extracting the most relevant information, the learned feature representation is passed to a **Multi-Layer Perceptron (MLP) prediction head**, which classifies the tumor as **HPV-positive** or **HPV-negative**.

Workflow:

1. **Input Features** (Clinical, PET, or Combined data)
2. **Feature Tokenization** (Convert each feature into a token representation)
3. **Transformer Encoder** (Self-attention learns relationships between features)
4. **Pooling** (Combine the learned information into a single feature representation)
5. **MLP Prediction Head** (Perform the final classification)
6. **Output:** HPV-positive or HPV-negative prediction.

This architecture enables the model to capture complex interactions between clinical and PET radiomic features, making it well suited for tabular medical datasets.

WORK APPLIED:

CLINICAL:

Validation and Test Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity	Specificity	Precision	F1-score	Confusion Matrix	Brier Score	PR-AUC	Expected Calibration Error (E
Validation	0.951	0.935	0.743	0.986	0.500	0.944	0.965	TN=4, FP=4, FN=1, TP=6	0.045	0.994	0.043
Test	0.820	0.936	0.771	0.986	0.556	0.944	0.965	TN=5, FP=4, FN=1, TP=6	0.063	0.961	0.062

Table 7 ft transformer clinical metrics results

The clinical-only FT-Transformer performs strongly overall. Validation ROC-AUC is 0.951 and test ROC-AUC is 0.820. This means the clinical features contain useful information for ranking HPV-positive patients above HPV-negative patients. The test ROC-AUC is lower than validation, so the model generalizes less perfectly on unseen patients, but the test result remains clearly better than chance.

Accuracy is high on both splits. Validation accuracy is 0.935, meaning 72 of 77 validation patients were classified correctly. Test accuracy is 0.936, meaning 73 of 78 test patients were classified correctly. Because the dataset is imbalanced toward HPV-positive patients, accuracy should not be interpreted alone.

Balanced Accuracy gives a more realistic picture. Validation Balanced Accuracy is 0.743 and test Balanced Accuracy is 0.771. These values are lower than raw accuracy because the HPV-negative class is harder for the model. Still, the model does detect some negative cases instead of predicting every patient as HPV positive.

Sensitivity is very high on both validation and test. Validation sensitivity is 0.986 and test sensitivity is 0.986. On both splits, the model missed only 1 HPV-positive patient. This means the model is very strong at identifying HPV-positive patients.

Specificity is moderate. Validation specificity is 0.500, meaning 4 of 8 HPV-negative validation patients were correctly identified. Test specificity is 0.556, meaning 5 of 9 HPV-negative test patients were correctly identified. The main weakness is still false positives among HPV-negative patients.

Precision and F1-score are strong. Test precision is 0.944 and test F1-score is 0.965. This means that when the model predicts HPV positive, it is usually correct, and the balance between precision and sensitivity is strong.

Calibration is good for this run. Validation Brier Score is 0.045 and test Brier Score is 0.063. Validation ECE is 0.043 and test ECE is 0.062. These values show that the predicted probabilities are reasonably reliable, especially compared with weaker PET-only classification.

Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	3	0.026	0.000	0.026
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	1	0.313	0.000	0.313
[0.4, 0.5)	1	0.479	1.000	0.521
[0.5, 0.6)	0			
[0.6, 0.7)	0			
[0.7, 0.8)	1	0.779	1.000	0.221
[0.8, 0.9)	12	0.845	0.667	0.178
[0.9, 1.0]	59	1.000	1.000	0.000

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	6	0.019	0.167	0.147
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	0			
[0.7, 0.8)	2	0.787	1.000	0.213
[0.8, 0.9)	9	0.834	0.889	0.055
[0.9, 1.0]	61	1.000	0.951	0.049

Table 8 ft transformer clinical calibration table val/test

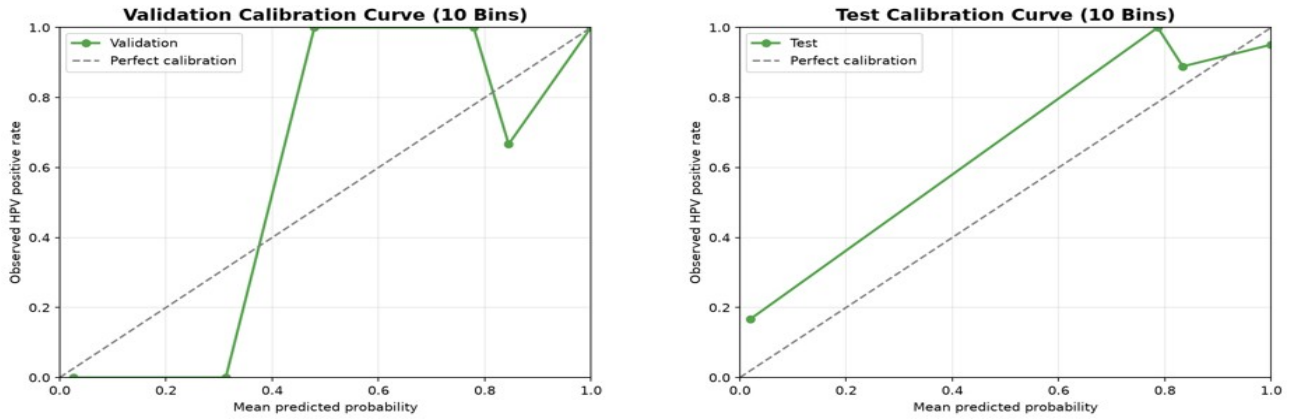


Figure 101 ft transnformer clinical calibration curve val/test

The 10-bin calibration tables show low calibration error. Test ECE is 0.062 and test Brier Score is 0.063, meaning the model's predicted probabilities are fairly close to the observed HPV-positive rates. Validation calibration is slightly better than test calibration, but both are acceptable for this experiment.

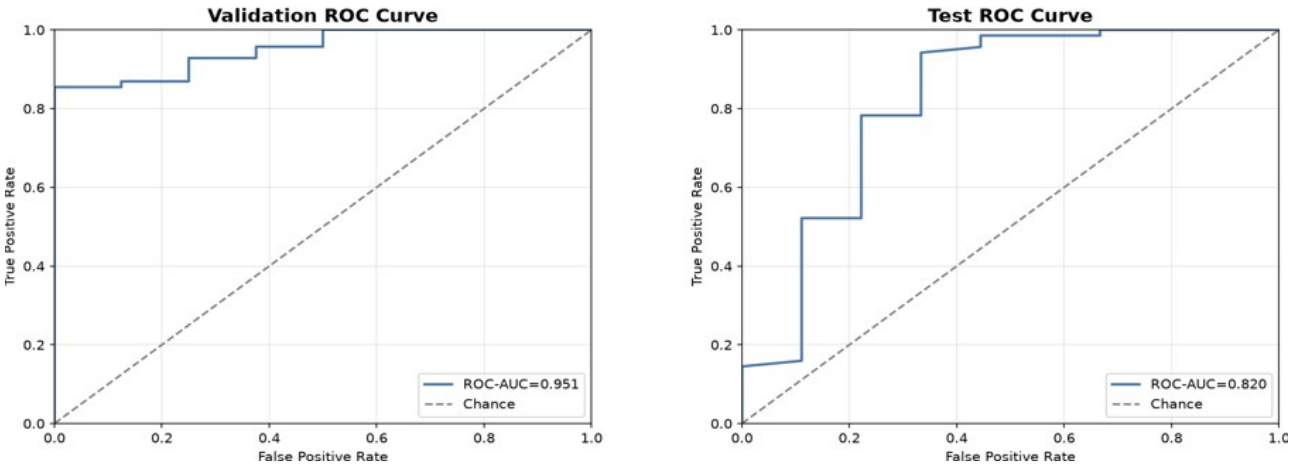


Figure 102 ft transformer clinical roc curve val/test

The ROC curve shows strong discrimination for the clinical-only FT-Transformer. Validation ROC-AUC is 0.951 and test ROC-AUC is 0.820. The validation curve is very strong, while the test curve shows a noticeable but acceptable drop. The model still ranks patients meaningfully on the test set.

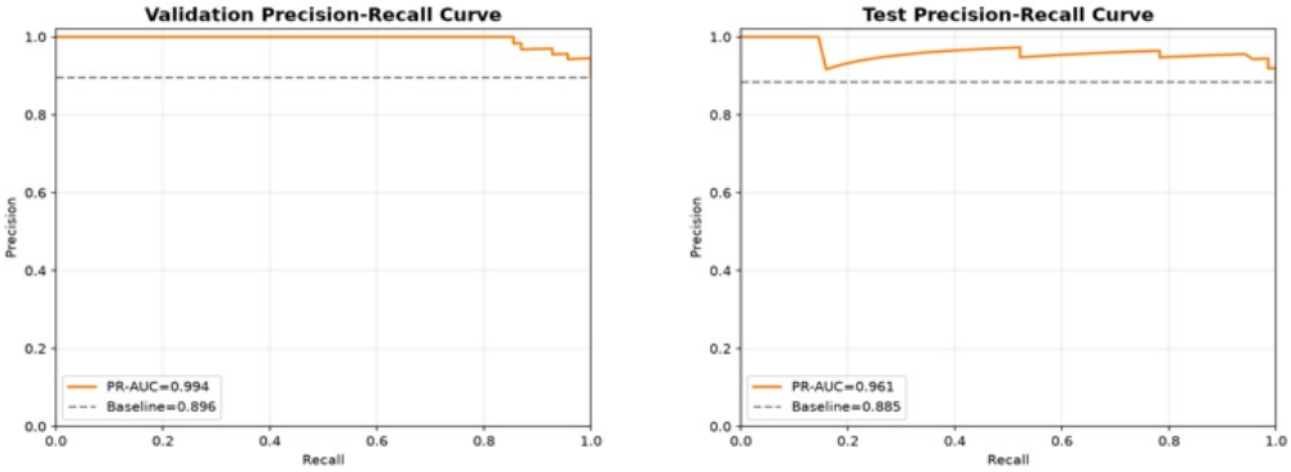


Figure 103 ft transformer clinical pr-auccurve val/test

The PR-AUC is 0.994 on validation and 0.961 on test. These are high values, but the dataset contains many more HPV-positive patients than HPV-negative patients, so PR-AUC should be read together with Balanced Accuracy and specificity. The clinical model has useful positive-class performance and still detects some negative cases.

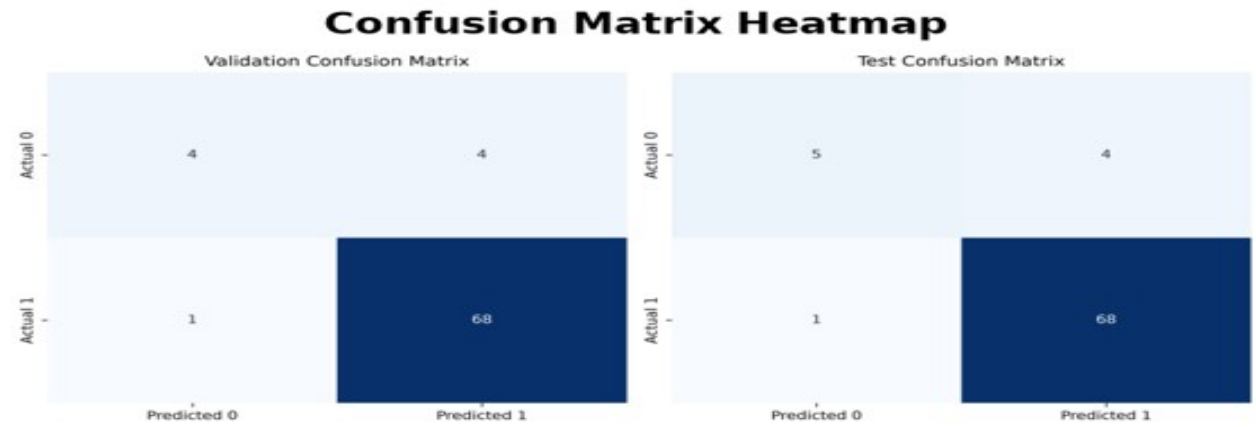


Figure 104 ft transformer clinical confusion matrix val/test

On validation, the model correctly identified 4 of 8 HPV-negative patients and 68 of 69 HPV-positive patients. It made 4 false positives and 1 false negative. On test, it correctly identified 5 of 9 HPV-negative patients and 68 of 69 HPV-positive patients. It made 4 false positives and 1 false negative.

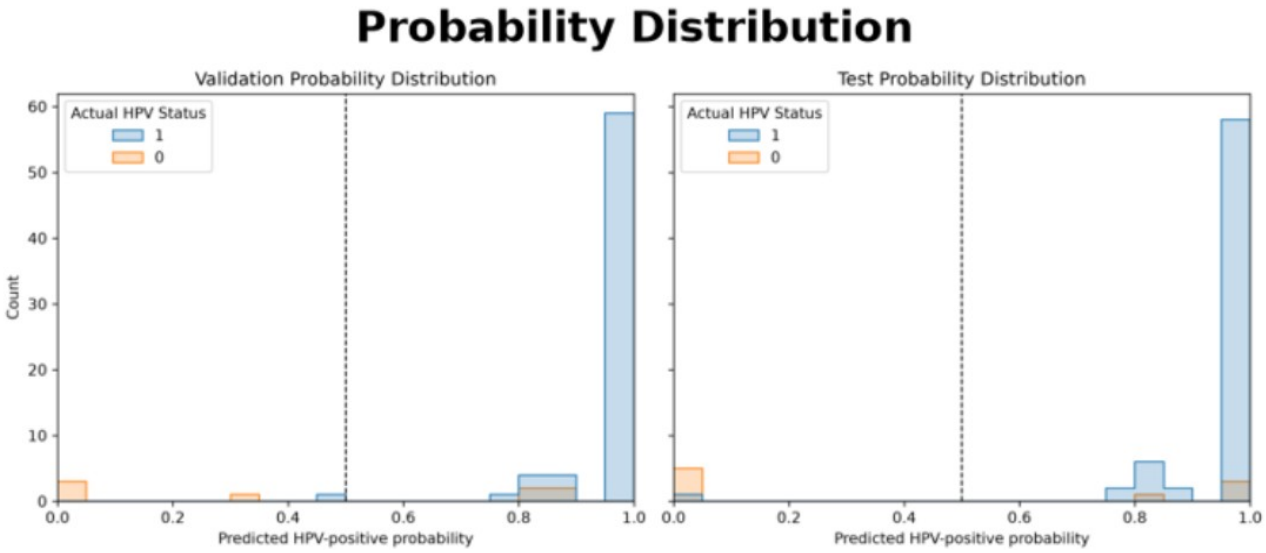


Figure 105 ft transformer clinical probability distributio val/test

The probability distribution plot should show most HPV-positive patients receiving high predicted HPV-positive probabilities. Some HPV-negative patients overlap with the positive group, which explains why specificity is not perfect. The threshold of 0.5 was fixed by rule, so this script does not tune the threshold to improve specificity.

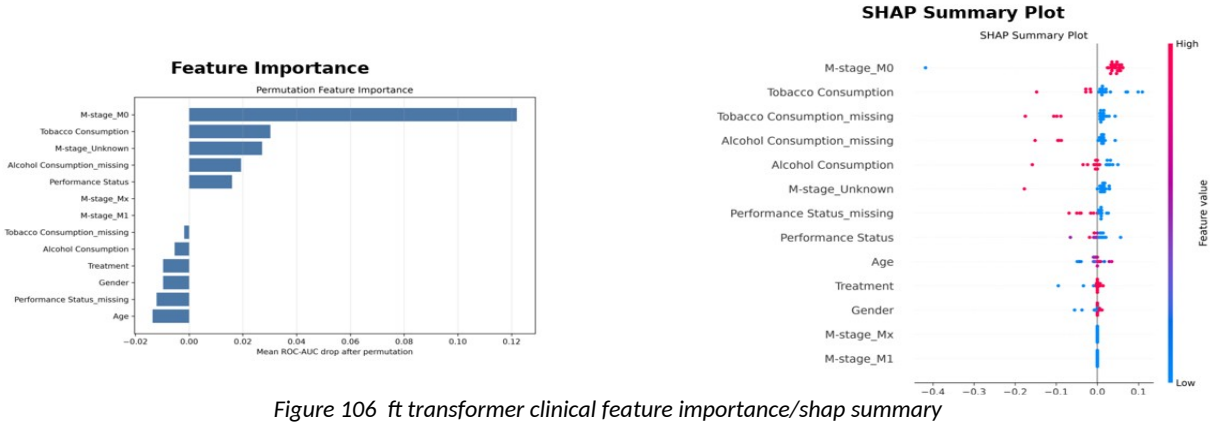


Figure 106 ft transformer clinical feature importance/shap summary

The strongest permutation feature was M-stage_M0 with a mean ROC-AUC drop of 0.122. Other useful features included Tobacco Consumption, M-stage_Unknown, Alcohol Consumption_missing, and Performance Status. This means the clinical FT-Transformer relies mainly on tumor-stage information and clinical lifestyle/status variables.

Some permutation importance values were near zero or negative. This does not mean the features were removed; the rule was to keep all features. It only means that shuffling those features did not reduce validation ROC-AUC in this trained model.

The strongest SHAP features were M-stage_M0, Tobacco Consumption, Tobacco Consumption_missing, Alcohol Consumption_missing, Alcohol Consumption, M-stage_Unknown, Performance Status_missing, Performance Status, Age, and Treatment. This agrees with the permutation feature-importance plot: stage and smoking/alcohol-related clinical variables are the strongest explanatory signals.

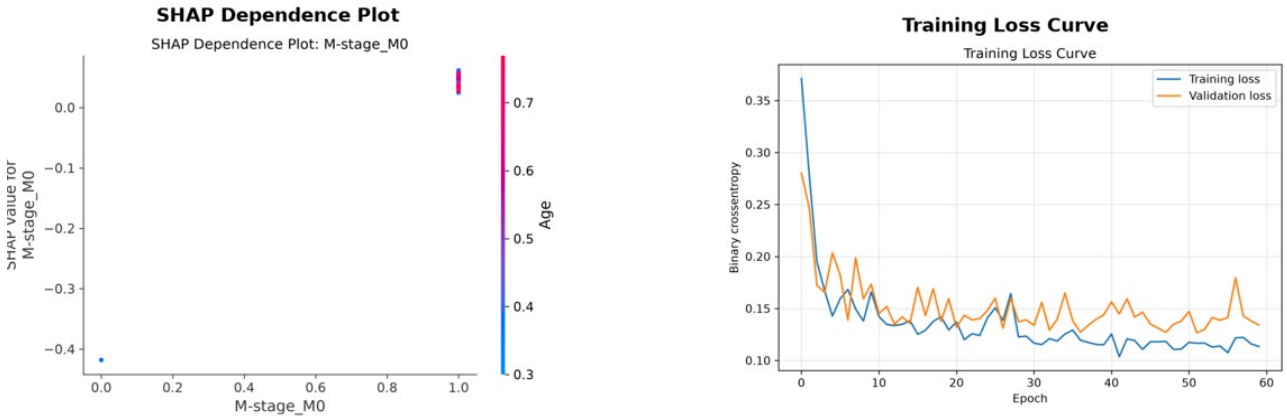


Figure 107 ft transformer clinical shap dependence/training loss

The automatically selected dependence feature was M-stage_M0. Because this is a one-hot encoded feature, the dependence plot mainly compares patients with M-stage_M0 = 0 against patients with M-stage_M0 = 1. The feature has the largest average SHAP contribution in this model.

Training loss decreased from 0.371 at epoch 1 to 0.113 at epoch 60. Validation loss decreased from 0.280 to 0.134. Training ROC-AUC increased from 0.596 to 0.974, while validation ROC-AUC ended at 0.952. This suggests the FT-Transformer learned a stable clinical signal without an extreme validation collapse.

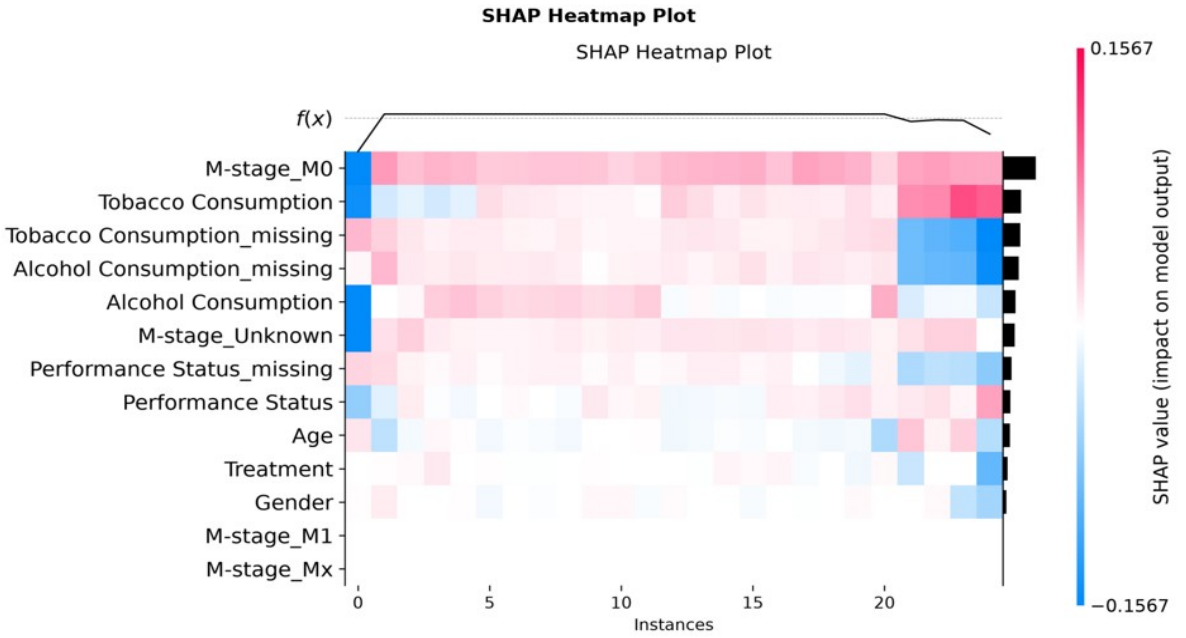


Figure 108 ft transformer clinical shap heatmap

The SHAP heatmap shows patient-level explanations across validation samples. M-stage_M0 appears as the dominant clinical feature, with tobacco and alcohol variables also contributing across multiple patients. The model is not controlled by one feature only, but M-stage_M0 is the clearest signal.

PET:

Validation and Test Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity	Specificity	Precision	F1-score	Confusion Matrix	Brier Score	PR-AUC	Expected Calibration Error (E
Validation	0.884	0.896	0.500	1.000	0.000	0.896	0.945	TN=0, FP=8, FN=0, TP=6	0.073	0.983	0.050
Test	0.638	0.885	0.500	1.000	0.000	0.885	0.939	TN=0, FP=9, FN=0, TP=6	0.099	0.915	0.036

Table 9 ft transformer pet metrics results

The PET-only FT-Transformer has a mixed result. The validation ROC-AUC is 0.884, which means the predicted probabilities have some ranking ability on validation. However, test ROC-AUC drops to 0.638, showing much weaker generalization on unseen test patients.

Raw accuracy looks high, but it is misleading. Validation accuracy is 0.896 and test accuracy is 0.885, but this happens because the dataset is mostly HPV positive and the model predicts every patient as HPV positive at the fixed 0.5 threshold.

Balanced Accuracy reveals the real classification problem. Validation Balanced Accuracy is 0.500 and test Balanced Accuracy is 0.500, which is chance-level balanced performance. This happens because sensitivity is perfect but specificity is zero.

Sensitivity is 1.000 on validation and test. The model detected all 69 HPV-positive validation patients and all 69 HPV-positive test patients. This is good for the positive class, but it is not enough for balanced classification.

Specificity is 0.000 on both splits. The model identified 0 of 8 HPV-negative validation patients and 0 of 9 HPV-negative test patients. Every negative patient was classified as HPV positive. This is the main weakness of the PET-only FT-Transformer under the no-scaling and fixed-threshold rules.

Precision and F1-score remain high because HPV-positive patients dominate the dataset. Test precision is 0.885 and test F1-score is 0.939, but these values should be interpreted carefully because the model never predicts the negative class.

The Brier Score and ECE are not terrible, especially validation ECE of 0.050 and test ECE of 0.036. This means the probabilities are not wildly miscalibrated overall, but calibration does not solve the fixed-threshold classification issue. A model can have reasonable probability calibration and still fail to produce negative predictions at threshold 0.5.

Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	0			
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	7	0.685	0.429	0.256
[0.7, 0.8)	9	0.762	0.778	0.016
[0.8, 0.9)	14	0.851	0.929	0.077
[0.9, 1.0]	47	0.961	0.979	0.018

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	0			
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	3	0.681	0.667	0.014
[0.7, 0.8)	13	0.752	0.769	0.017
[0.8, 0.9)	12	0.861	0.917	0.056
[0.9, 1.0]	50	0.958	0.920	0.038

Figure 109 ft transformer pet calibration table val/test

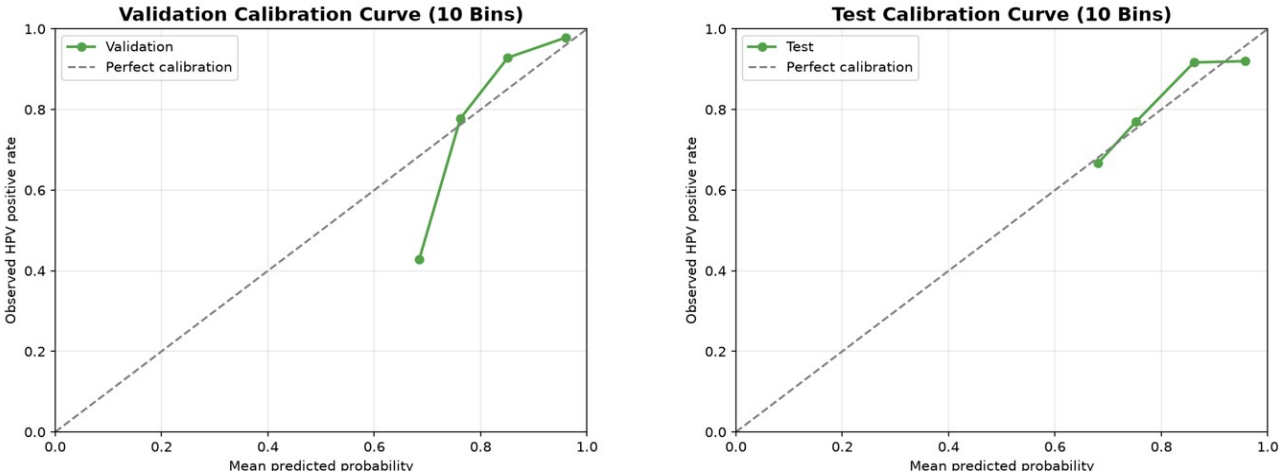


Figure 110 ft transformer pet calibration curve val/test

The calibration results are better than the confusion matrix might suggest. Validation ECE is 0.050 and test ECE is 0.036. Test Brier Score is 0.099. This means the probability estimates are somewhat reasonable, but most predicted probabilities still fall on the HPV-positive side of the 0.5 threshold.

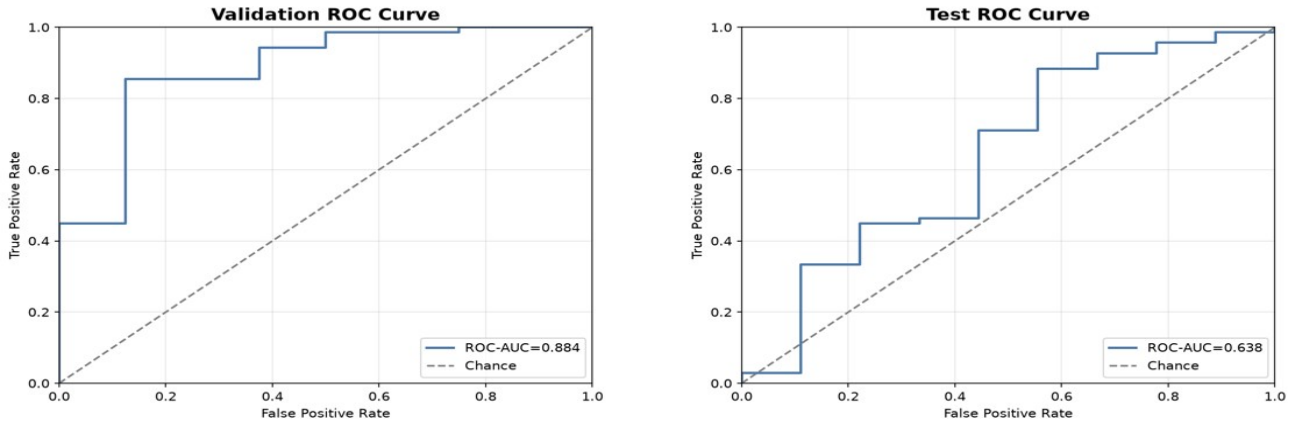


Figure 111 ft transformer pet roc curve val/test

The validation ROC-AUC is 0.884, so the PET model has some ability to rank HPV-positive patients above HPV-negative patients on validation. The test ROC-AUC is only 0.638, which is much weaker. This validation-test gap suggests the PET-only FT-Transformer is less stable than the clinical and merged models.

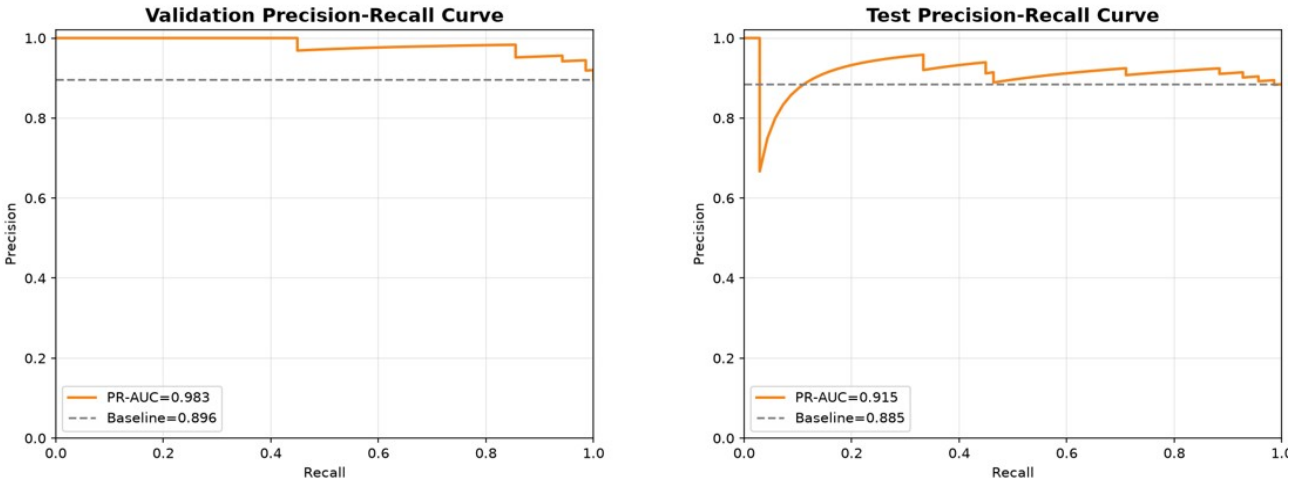


Figure 112 ft transformer pet pr-auc val/test

The PR-AUC is 0.983 on validation and 0.915 on test. These values are high, but the positive class is very common. Because specificity is 0.000, the PR curve should not be read as proof of a clinically balanced classifier. The PET model is strongly biased toward the HPV-positive class at the fixed threshold.

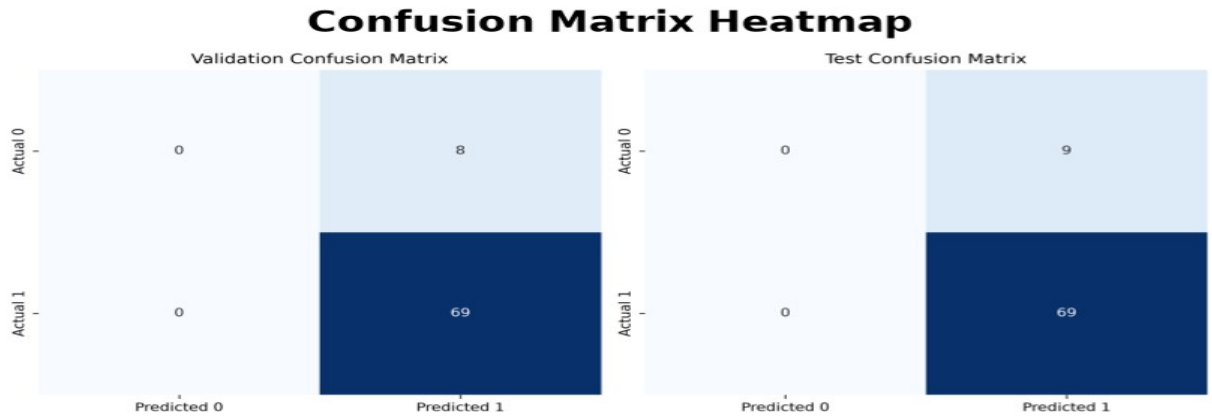
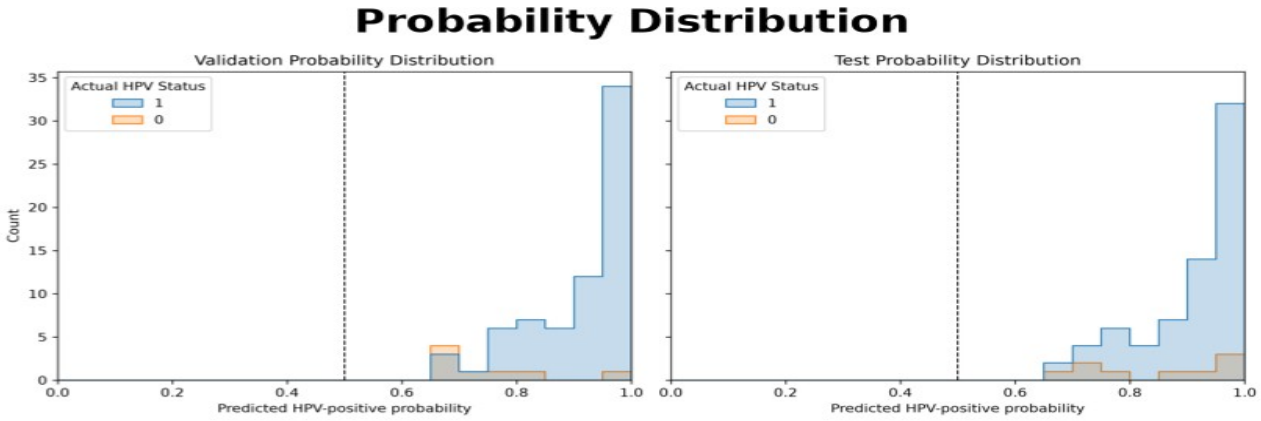


Figure 113 ft transformer pet confusion matrix val/test

The validation confusion matrix is [[0, 8], [0, 69]]. The test confusion matrix is [[0, 9], [0, 69]]. This means the model predicts every validation and test patient as HPV positive. It misses no HPV-positive patients, but it also detects no HPV-negative patients.



The probability distribution plot should show most patients receiving probabilities above the 0.5 threshold. HPV-negative and HPV-positive patients overlap heavily, and the negative patients are not pushed below 0.5. This explains the perfect sensitivity and zero specificity.

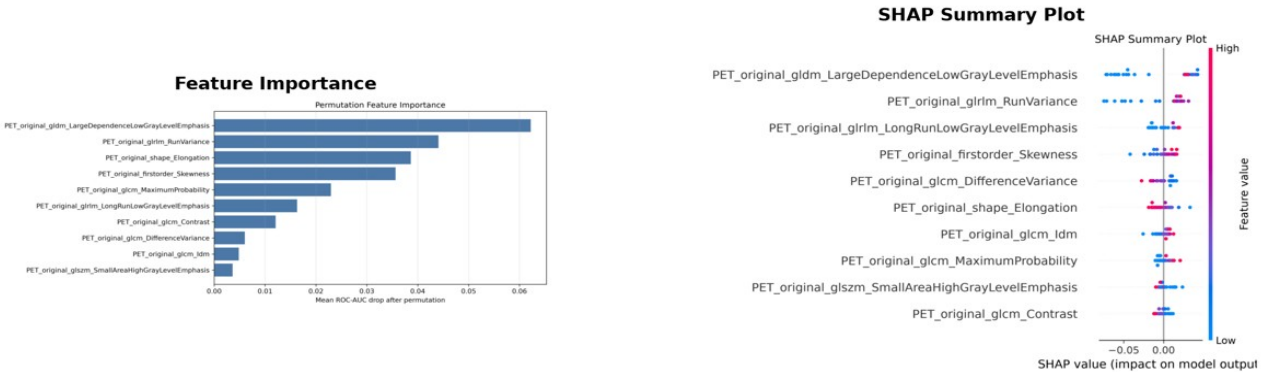


Figure 115 ft transformer pet feature importance/shap summary val/test

The top permutation importance features were PET_original_gldm_LargeDependenceLowGrayLevelEmphasis, PET_original_glrlm_RunVariance, PET_original_shape_Elongation, PET_original_firstorder_Skewness, PET_original_glcm_MaximumProbability, PET_original_glrlm_LongRunLowGrayLevelEmphasis, PET_original_glcm_Contrast, PET_original_glcm_DifferenceVariance, PET_original_glcm_Idm, and PET_original_glszm_SmallAreaHighGrayLevelEmphasis.

These are mainly PET radiomic texture, gray-level, run-length, and shape features. This means the PET-only model is learning from radiomic patterns, but those patterns are not enough to separate HPV-negative patients at the fixed 0.5 threshold.

The strongest SHAP features were PET_original_gldm_LargeDependenceLowGrayLevelEmphasis, PET_original_glrlm_RunVariance, PET_original_glrlm_LongRunLowGrayLevelEmphasis, PET_original_firstorder_Skewness, PET_original_glcm_DifferenceVariance, PET_original_shape_Elongation, PET_original_glcm_Idm, PET_original_glcm_MaximumProbability, PET_original_glszm_SmallAreaHighGrayLevelEmphasis, and PET_original_glcm_Contrast.

The SHAP summary agrees with permutation importance: PET texture and gray-level features dominate this model.

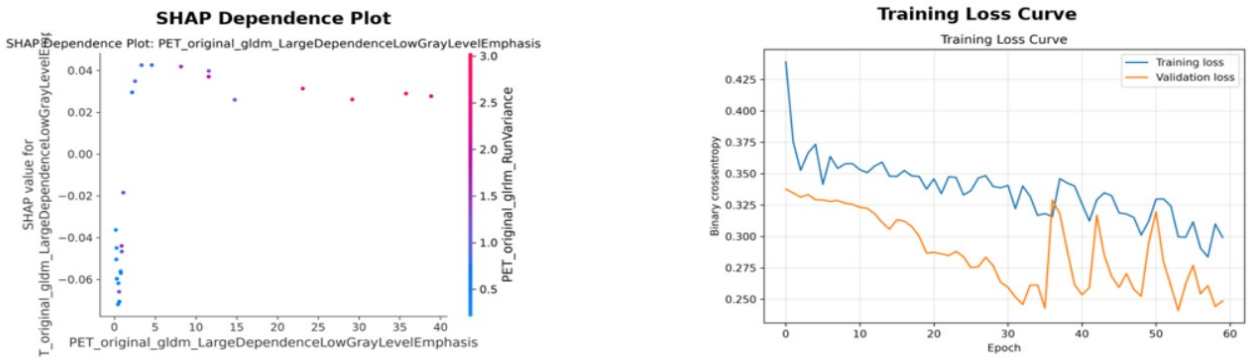


Figure 116 ft transformer pet shap dependence/training loss

The automatically selected dependence feature was PET_original_gldm_LargeDependenceLowGrayLevelEmphasis. This feature had the largest mean absolute SHAP contribution among the top-10 selected PET features. The dependence plot shows how this PET texture feature changes the predicted HPV-positive probability contribution.

Training loss decreased from 0.439 at epoch 1 to 0.299 at epoch 60. Validation loss decreased from 0.338 to 0.249. Training ROC-AUC increased from 0.460 to 0.782, and validation ROC-AUC ended at 0.886. The model learned something from PET features, but the final confusion matrix shows that the learned scores did not cross the threshold in a balanced way.

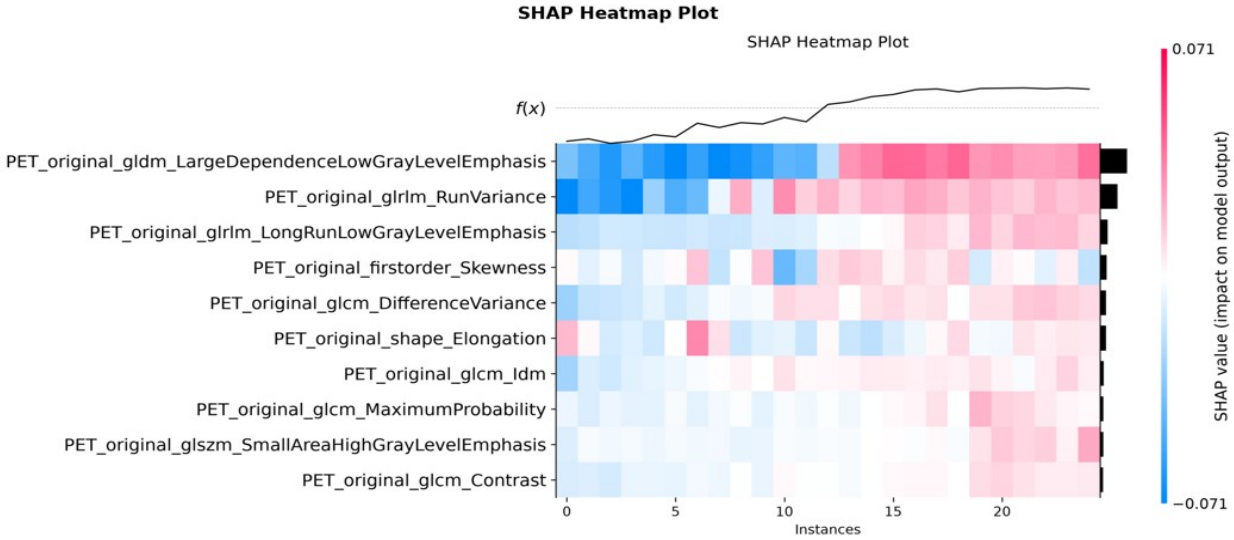


Figure 117 ft transformer pet shap heatmap

The SHAP heatmap shows that PET texture features repeatedly influence individual validation samples. However, the patient-level explanations do not translate into useful negative-class classification at threshold 0.5. The model has signal, but the thresholded classification remains all-positive.

MERGED:

Validation and Test Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity	Specificity	Precision	F1-score	Confusion Matrix	Brier Score	PR-AUC	Expected Calibration Error (E
Validation	0.962	0.935	0.743	0.986	0.500	0.944	0.965	TN=4, FP=4, FN=1, TP=6	0.056	0.996	0.056
Test	0.916	0.936	0.771	0.986	0.556	0.944	0.965	TN=5, FP=4, FN=1, TP=6	0.059	0.988	0.071

Table 10 ft transformer merged metrics results

The merged clinical + PET FT-Transformer is the strongest Week 12 model overall. Validation ROC-AUC is 0.962 and test ROC-AUC is 0.916. This means the merged model ranks HPV-positive patients above HPV-negative patients very well on both validation and test.

Accuracy is high on both splits. Validation accuracy is 0.935 and test accuracy is 0.936. In raw counts, the model correctly classified 72 of 77 validation patients and 73 of 78 test patients.

Balanced Accuracy is lower than raw accuracy but still useful. Validation Balanced Accuracy is 0.743 and test Balanced Accuracy is 0.771. This indicates that the model is not merely benefiting from the majority HPV-positive class; it also detects some HPV-negative patients.

Sensitivity is very high on both validation and test. The model correctly detected 68 of 69 HPV-positive patients on validation and 68 of 69 HPV-positive patients on test. This is excellent positive-class performance.

Specificity is moderate. The model correctly detected 4 of 8 HPV-negative validation patients and 5 of 9 HPV-negative test patients. The remaining weakness is the 4 false positives on each split.

Precision and F1-score are strong. Test precision is 0.944 and test F1-score is 0.965, showing that positive predictions are usually correct and that the model balances precision with very high sensitivity.

Calibration is acceptable. Validation Brier Score is 0.056 and test Brier Score is 0.059. Validation ECE is 0.056 and test ECE is 0.071. These values show that the predicted probabilities are reasonably reliable, although test ECE is slightly worse than the clinical-only model.

Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	3	0.043	0.000	0.043
[0.1, 0.2)	1	0.162	1.000	0.838
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	1	0.495	0.000	0.495
[0.5, 0.6)	0			
[0.6, 0.7)	3	0.676	0.667	0.009
[0.7, 0.8)	1	0.724	1.000	0.276
[0.8, 0.9)	1	0.870	0.000	0.870
[0.9, 1.0]	67	0.995	0.970	0.025

Test Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.055	0.000	0.055
[0.1, 0.2)	2	0.148	0.500	0.352
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	2	0.670	1.000	0.330
[0.7, 0.8)	1	0.795	1.000	0.205
[0.8, 0.9)	1	0.853	0.000	0.853
[0.9, 1.0]	68	0.998	0.956	0.042

Table 11 ft transformer merged calibration table val/test

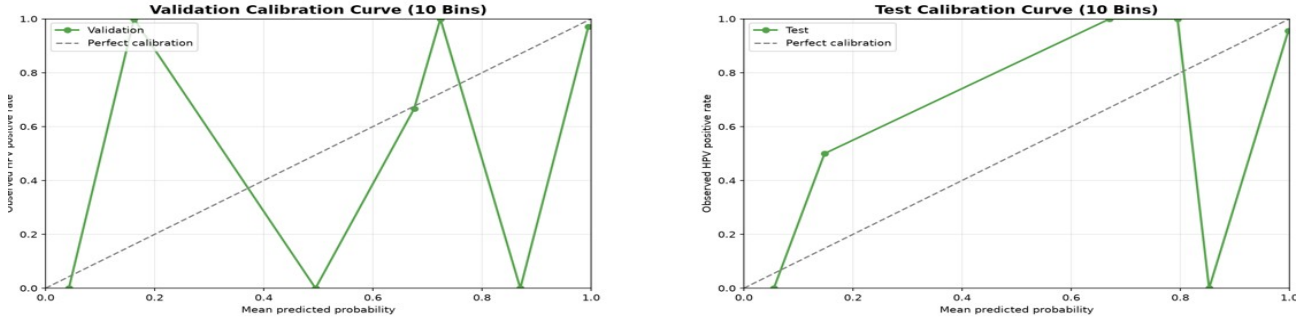
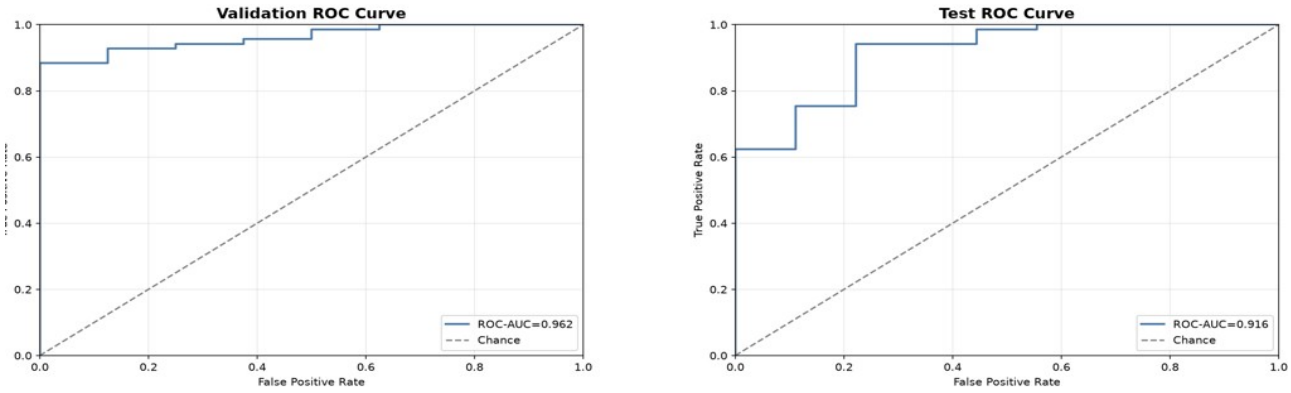


Figure 118 ft transformer merged calibration curve val/test

The 10-bin calibration tables show reasonable probability calibration. Test Brier Score is 0.059 and test ECE is 0.071. The model's probability estimates are useful, but the calibration curve should still be checked visually because the negative class has only 9 test patients.



The ROC curves show strong discrimination. Validation ROC-AUC is 0.962 and test ROC-AUC is 0.916. The small validation-test gap compared with the PET-only model suggests that adding clinical features makes the model more stable.

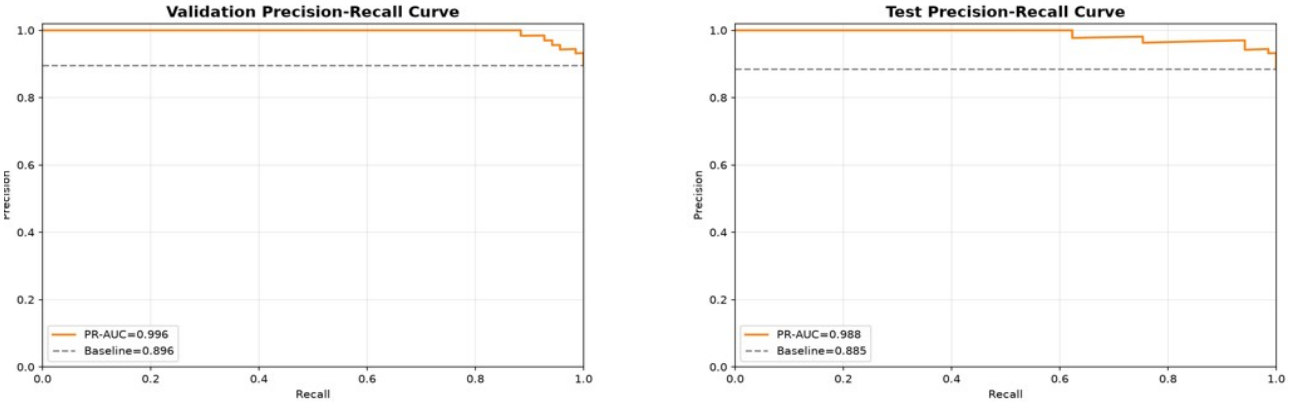


Figure 120 ft transformer merged pr-auc val/test

The PR-AUC is 0.996 on validation and 0.988 on test. These values are very high. Because the dataset is imbalanced toward HPV-positive patients, PR-AUC should still be read with specificity and Balanced Accuracy. Here, the merged model has strong PR-AUC and still detects several HPV-negative patients, so the result is more meaningful than PET-only.

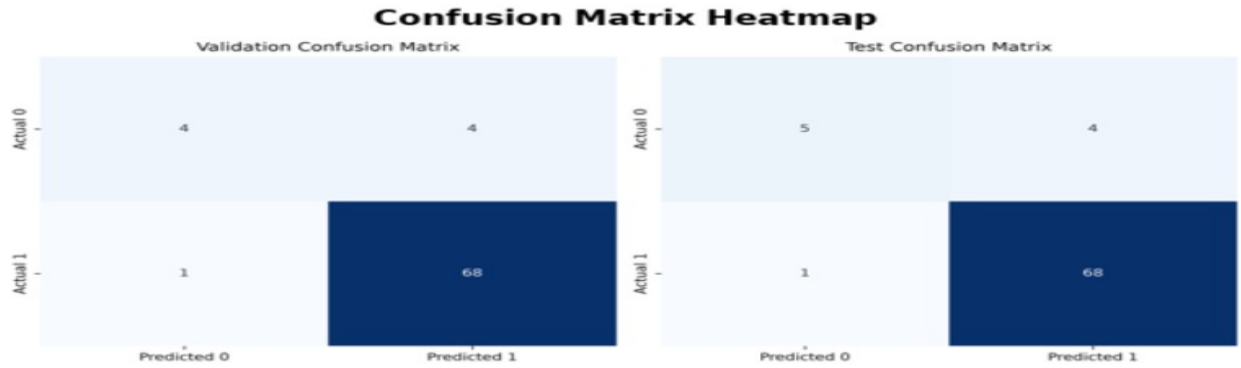


Figure 121 ft transformer merged confusiom matrix val/test

The validation confusion matrix is [[4, 4], [1, 68]]. The test confusion matrix is [[5, 4], [1, 68]]. This means the merged model keeps the same high sensitivity as the clinical-only model and identifies more than half of the HPV-negative test patients.

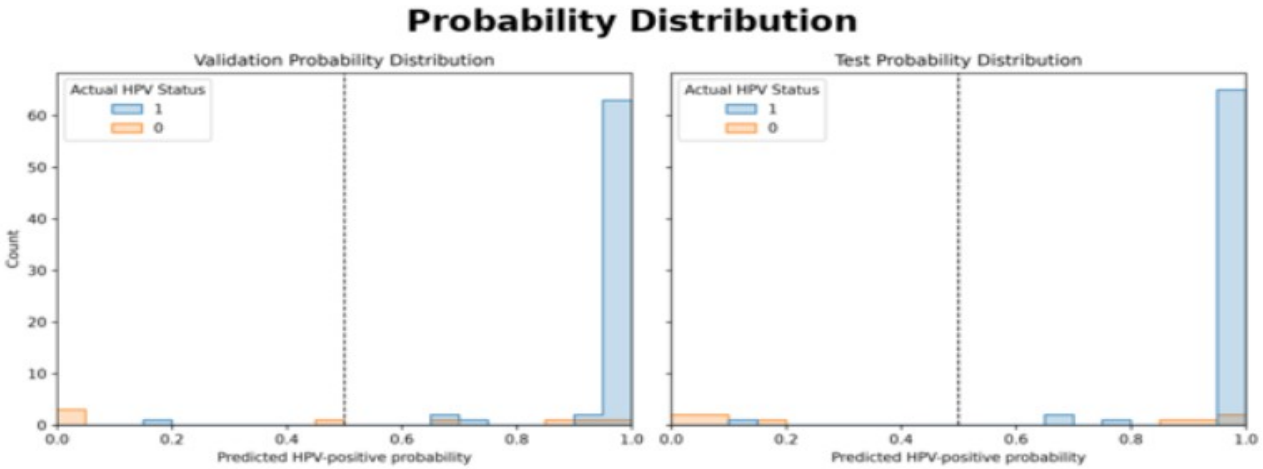


Figure 122 ft transformer merged probability distribution val/test

The probability distribution plot should show better separation than PET-only. Many HPV-positive patients should receive high probabilities, while some HPV-negative patients move lower. There is still overlap, which explains why specificity is moderate instead of high.

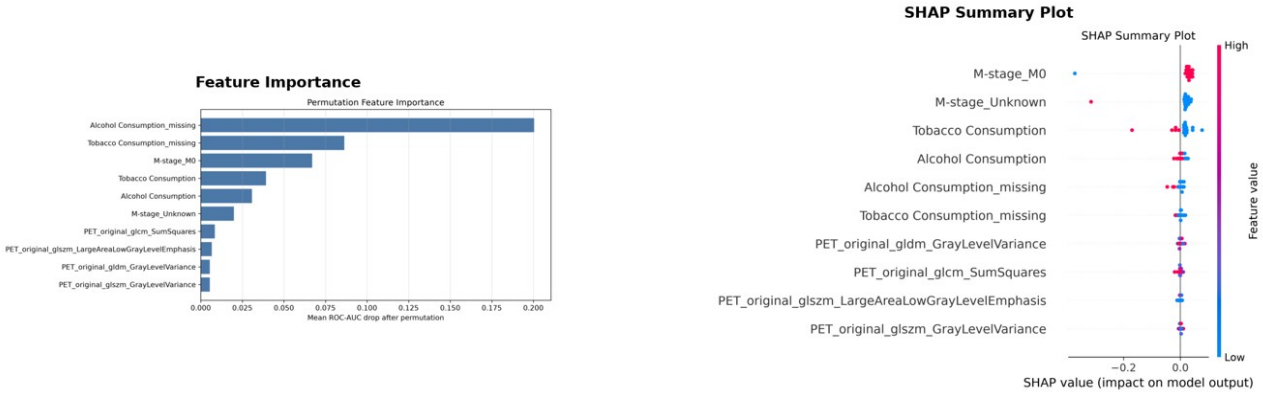


Figure 123 ft transformer merged features importances.shap summary

The top permutation importance features were Alcohol Consumption_missing, Tobacco Consumption_missing, M-stage_M0, Tobacco Consumption, Alcohol Consumption, M-stage_Unknown, PET_original_glcm_SumSquares, PET_original_glszm_LargeAreaLowGrayLevelEmphasis, PET_original_gldm_GrayLevelVariance, and PET_original_glszm_GrayLevelVariance.

This is an important result because both clinical and PET variables appear in the top 10. The strongest features are clinical/missingness and staging variables, while PET gray-level and texture features add extra information.

The strongest SHAP features were M-stage_M0, M-stage_Unknown, Tobacco Consumption, Alcohol Consumption, Alcohol Consumption_missing, Tobacco Consumption_missing, PET_original_gldm_GrayLevelVariance, PET_original_glcm_SumSquares, PET_original_glszm_LargeAreaLowGrayLevelEmphasis, and PET_original_glszm_GrayLevelVariance.

The SHAP result supports the permutation-importance result. The merged model is driven mainly by clinical staging and tobacco/alcohol variables, with PET texture/gray-level features contributing additional signal.

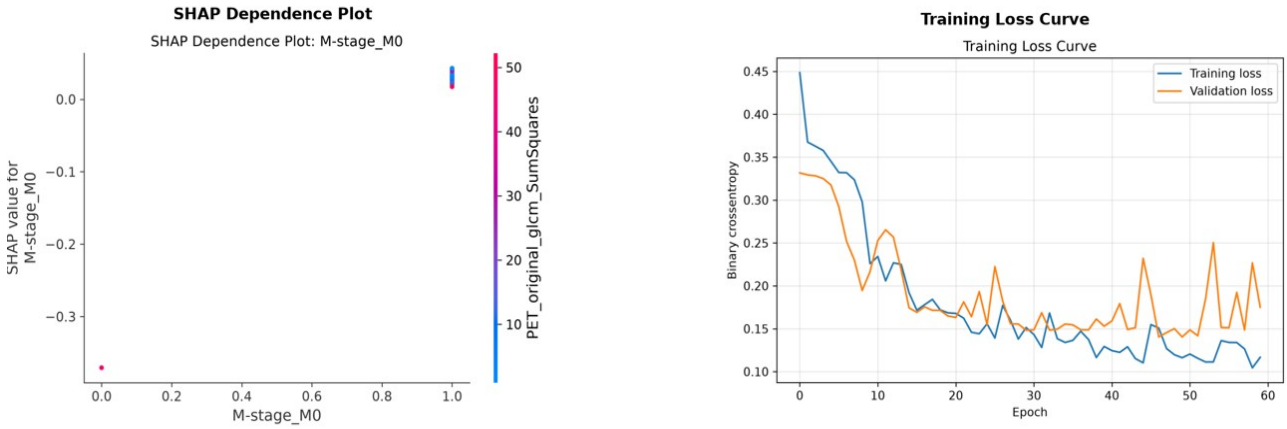


Figure 124 ft transformer merged shap dependence/training loss

The automatically selected dependence feature was M-stage_M0. Even in the merged model, M-stage_M0 has the largest mean absolute SHAP value among the selected top features. This means clinical staging remains the clearest explanatory driver of the FT-Transformer prediction.

Training loss decreased from 0.448 at epoch 1 to 0.117 at epoch 60. Validation loss decreased from 0.332 to 0.175. Training ROC-AUC increased from 0.387 to 0.976, and validation ROC-AUC ended at 0.962. The model learned a strong signal and kept strong validation performance.

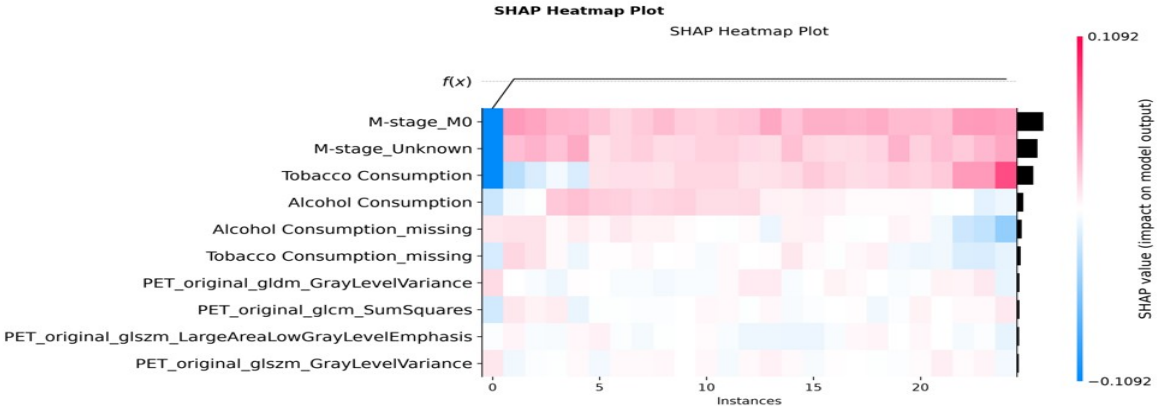


Figure 125 ft transformer merged shap heatmap

The SHAP heatmap shows patient-level contributions from both clinical and PET features. M-stage_M0 and M-stage_Unknown repeatedly influence predictions, while selected PET texture features add smaller patient-level contributions. This suggests the merged model is not simply a PET-only model; clinical information remains central.

FINAL INTERPRETATION:

Three FT-Transformer classification experiments were conducted to evaluate the ability of a transformer-based deep learning model to classify the **HPV status of head and neck cancer tumors** using **clinical features**, **PET radiomic features**, and a **combination of clinical and PET radiomic features**. All experiments were performed using the same training, validation, and test splits, without hyperparameter tuning, feature selection, feature engineering, scaling, or additional preprocessing, allowing a direct comparison between the three feature sets.

The **clinical-only FT-Transformer** demonstrated strong overall performance, achieving a **test ROC-AUC of 0.820**, high accuracy (**0.936**), excellent sensitivity (**0.986**), and acceptable specificity (**0.556**). The model consistently identified nearly all HPV-positive patients while correctly classifying more than half of the HPV-negative patients. Feature importance and SHAP analyses showed that **M-stage**, together with **tobacco consumption**, **alcohol consumption**, and **performance status**, were the main contributors to the predictions. These findings indicate that clinical variables provide a strong and stable signal for HPV status classification.

The **PET-only FT-Transformer** produced the weakest performance among the three experiments. Although the validation ROC-AUC (**0.884**) suggested that PET radiomic features contained useful predictive information, the **test ROC-AUC decreased to 0.638**, indicating limited generalization to unseen data. More importantly, the model classified every patient as HPV-positive at the fixed decision threshold, resulting in **100% sensitivity but 0% specificity**. Both the feature importance and SHAP analyses confirmed that the model relied mainly on PET texture, gray-level, and shape features. However, these radiomic features alone were insufficient to reliably distinguish HPV-negative tumors, making the PET-only model unsuitable for balanced HPV classification.

The **combined clinical + PET FT-Transformer** achieved the best overall performance. It obtained the highest **test ROC-AUC (0.916)** and **PR-AUC (0.988)** while maintaining high accuracy (**0.936**), excellent sensitivity (**0.986**), and moderate specificity (**0.556**). Compared with the PET-only model, the combined model produced more stable probability estimates and better discrimination between HPV-positive and HPV-negative patients. The feature importance and SHAP analyses demonstrated that **clinical variables, particularly M-stage and tobacco/alcohol-related features, remained the dominant predictors**, while PET radiomic texture and gray-level features provided complementary information that enhanced the model's predictive ability.

Overall, the FT-Transformer experiments demonstrate that **clinical information remains the strongest predictor of HPV status**, whereas PET radiomic features alone do not provide sufficient discriminatory power for reliable classification. However, combining PET radiomic features with clinical data improves the model's probability ranking, robustness, and overall predictive performance. Among the three experiments, the **combined clinical + PET FT-Transformer** provided the best balance between discrimination, sensitivity, calibration, and generalization, making it the most effective transformer-based approach evaluated during this phase of the internship.

Conclusion

This internship focused on the classification of HPV status in Head and Neck Squamous Cell Carcinoma (HNSCC) patients using CT/PET radiomic features combined with clinical data. Three modeling approaches — LightGBM, Multi-Layer Perceptron (MLP), and FT-Transformer — were developed and evaluated across three feature configurations: clinical-only, PET radiomic-only, and combined clinical + PET, using a consistent training/validation/test protocol and a comprehensive set of metrics (ROC-AUC, Accuracy, Balanced Accuracy, Sensitivity, Specificity, Precision, F1-score, PR-AUC, Brier Score, and Expected Calibration Error), along with feature importance and SHAP interpretability analysis.

Across all three model architectures, the same pattern emerged consistently. Clinical features alone provided a strong and stable predictive signal, with **M-stage**, **tobacco consumption**, and **alcohol consumption** repeatedly identified as the dominant predictors in every experiment. PET radiomic features alone were consistently the weakest predictors: they achieved high sensitivity but very poor specificity, often collapsing toward predicting every patient as HPV-positive, and showed the largest gap between validation and test performance — indicating limited generalization when used in isolation. Combining clinical and PET radiomic features produced the most balanced and reliable models overall, improving specificity, calibration, and probability ranking relative to PET-only, while PET features consistently played a *complementary* rather than dominant role.

Among the three architectures, the **FT-Transformer with combined clinical + PET features** delivered the strongest overall result (test ROC-AUC = 0.916, PR-AUC = 0.988, accuracy = 0.936, sensitivity = 0.986), closely followed by the LightGBM and MLP combined models. This confirms that transformer-based architectures can effectively model the interaction between structured clinical variables and radiomic texture features when properly combined.

Overall, this work demonstrates that **clinical data remains the strongest and most reliable driver of HPV status prediction**, while **PET radiomics alone are insufficient for robust classification** but add meaningful complementary value when integrated with clinical information. The results support the broader conclusion that non-invasive, AI-assisted HPV classification is feasible and clinically promising, particularly when radiomic and clinical data are combined rather than used in isolation — offering a potential decision-support tool that could reduce reliance on invasive pathological testing in head and neck cancer management. Beyond the technical outcomes, this internship provided valuable hands-on experience in medical data preprocessing, model development, performance

evaluation, and interpretability techniques (SHAP, feature importance), strengthening both my technical skills and my understanding of applying AI to real-world healthcare problems.

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